SQANTI3 report

Unique Genes: 11849

Unique Isoforms: 25618

Gene Classification

Category	Genes, count
Annotated Genes	11840
Novel Genes	9

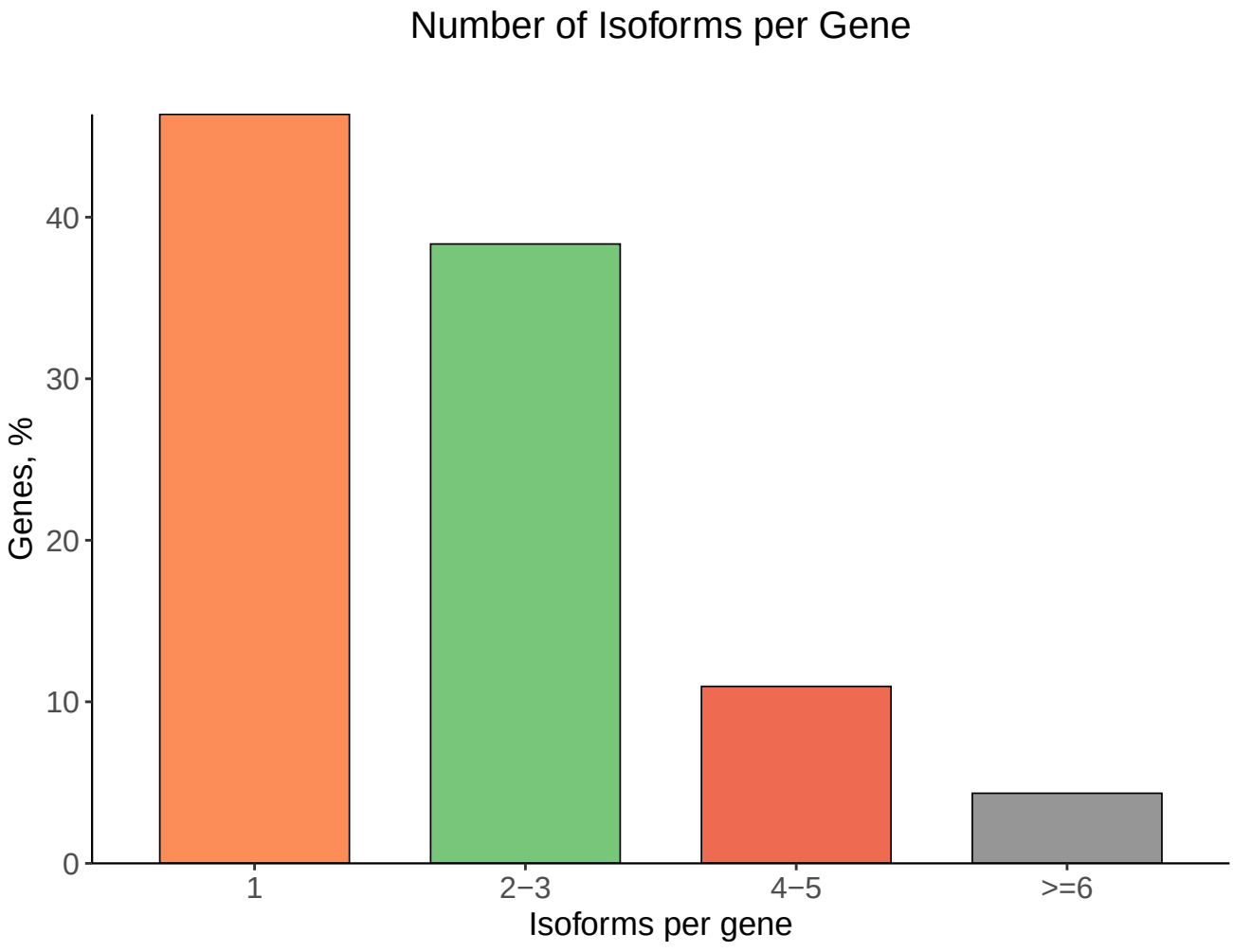
Transcript Classification

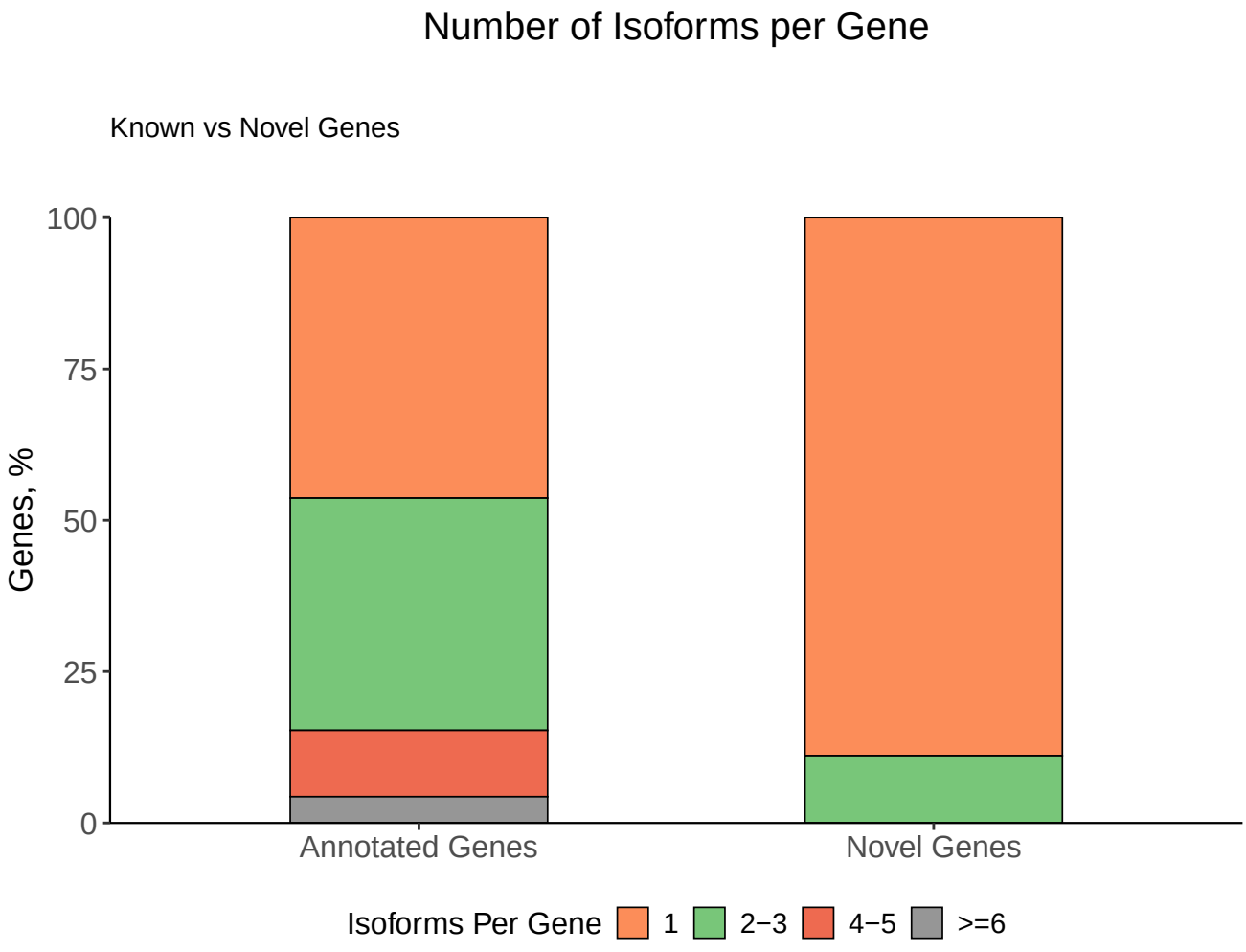
Category	Isoforms, count
FSM	24368
ISM	261
NIC	739
NNC	185
Genic Genomic	35
Antisense	9
Fusion	20
Intergenic	1
Genic Intron	0

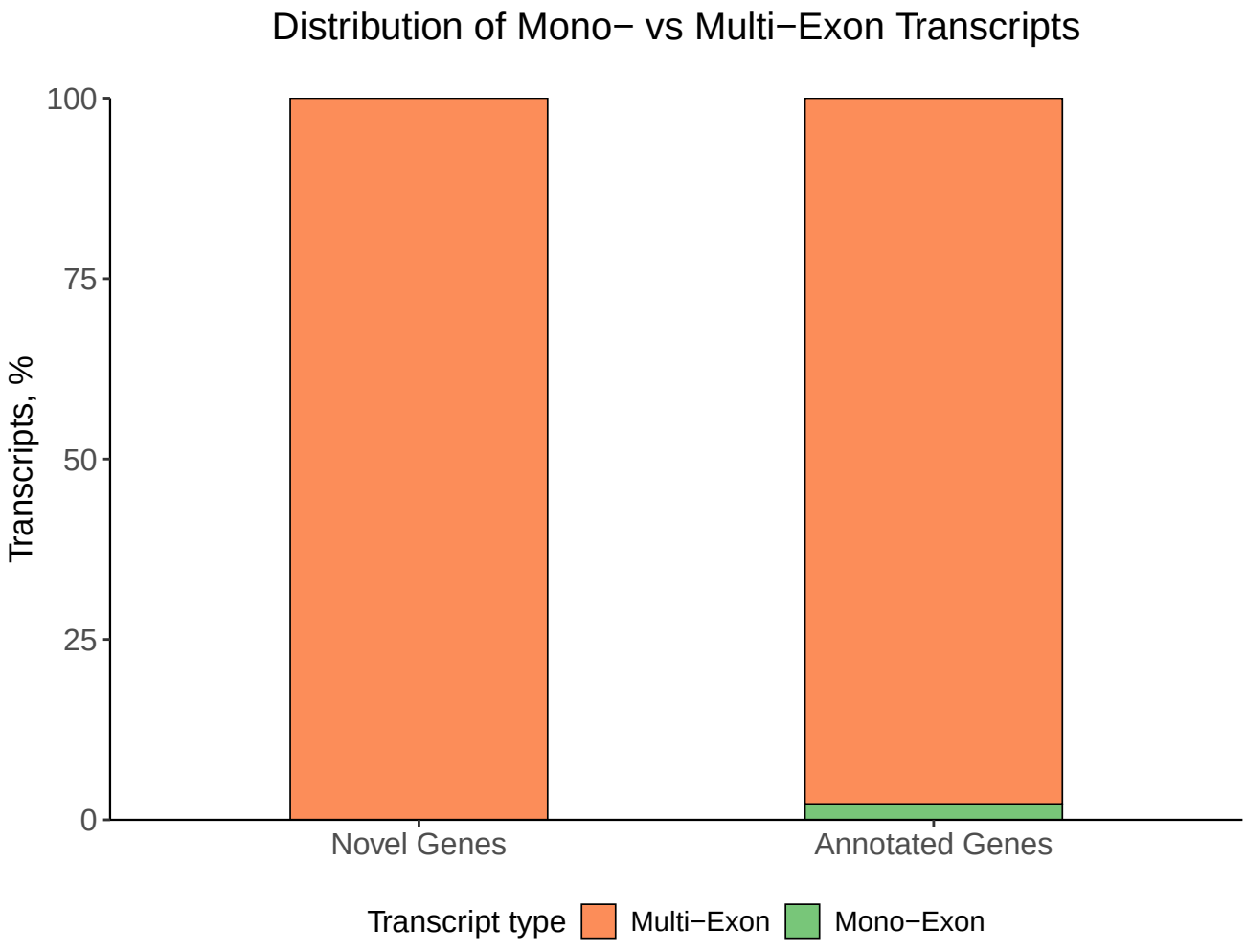
Splice Junction Classification

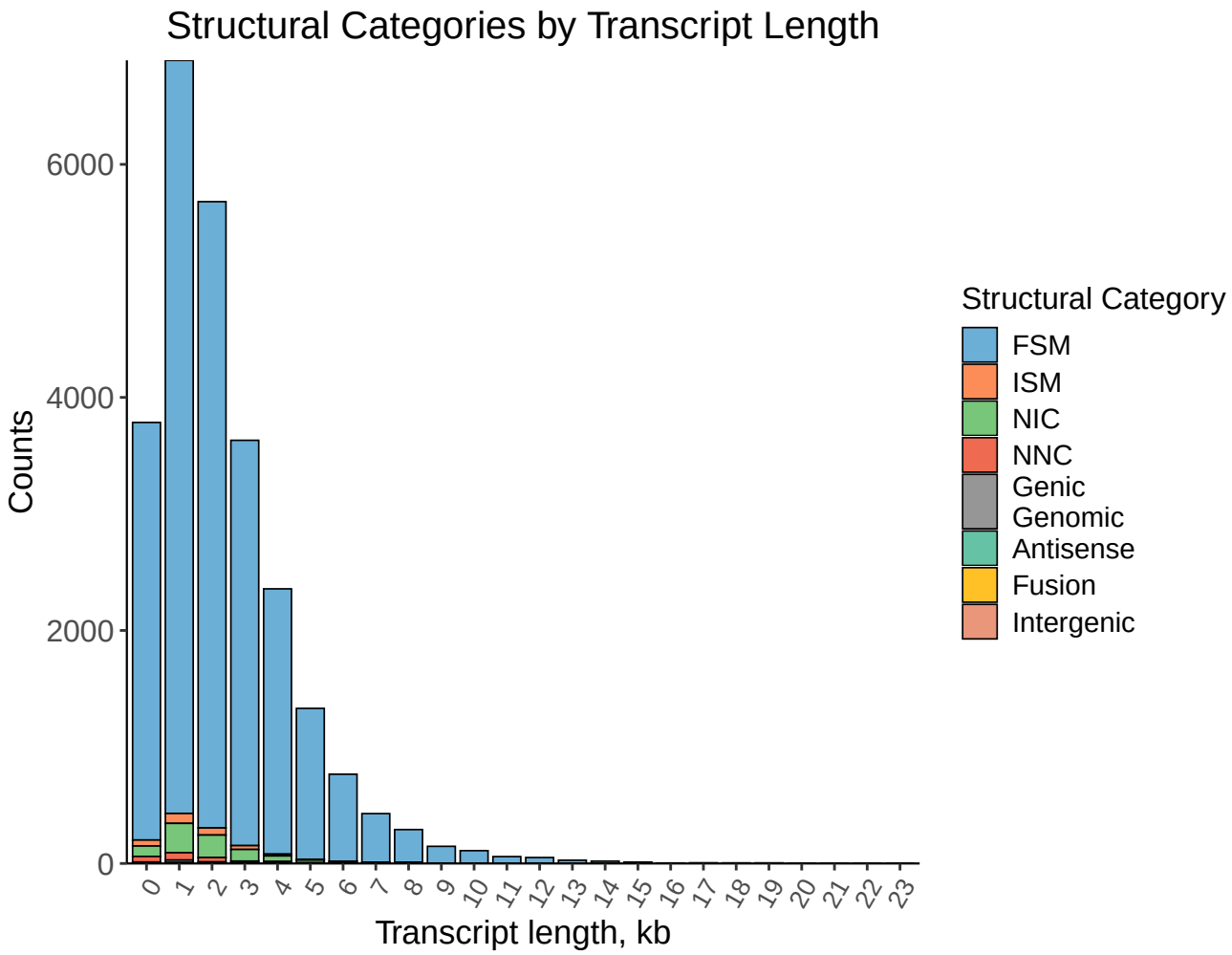
Category	SJs, count	Percent
Known canonical	120021	99.63
Known Non−canonical	63	0.05
Novel canonical	367	0.30
Novel Non−canonical	10	0.01

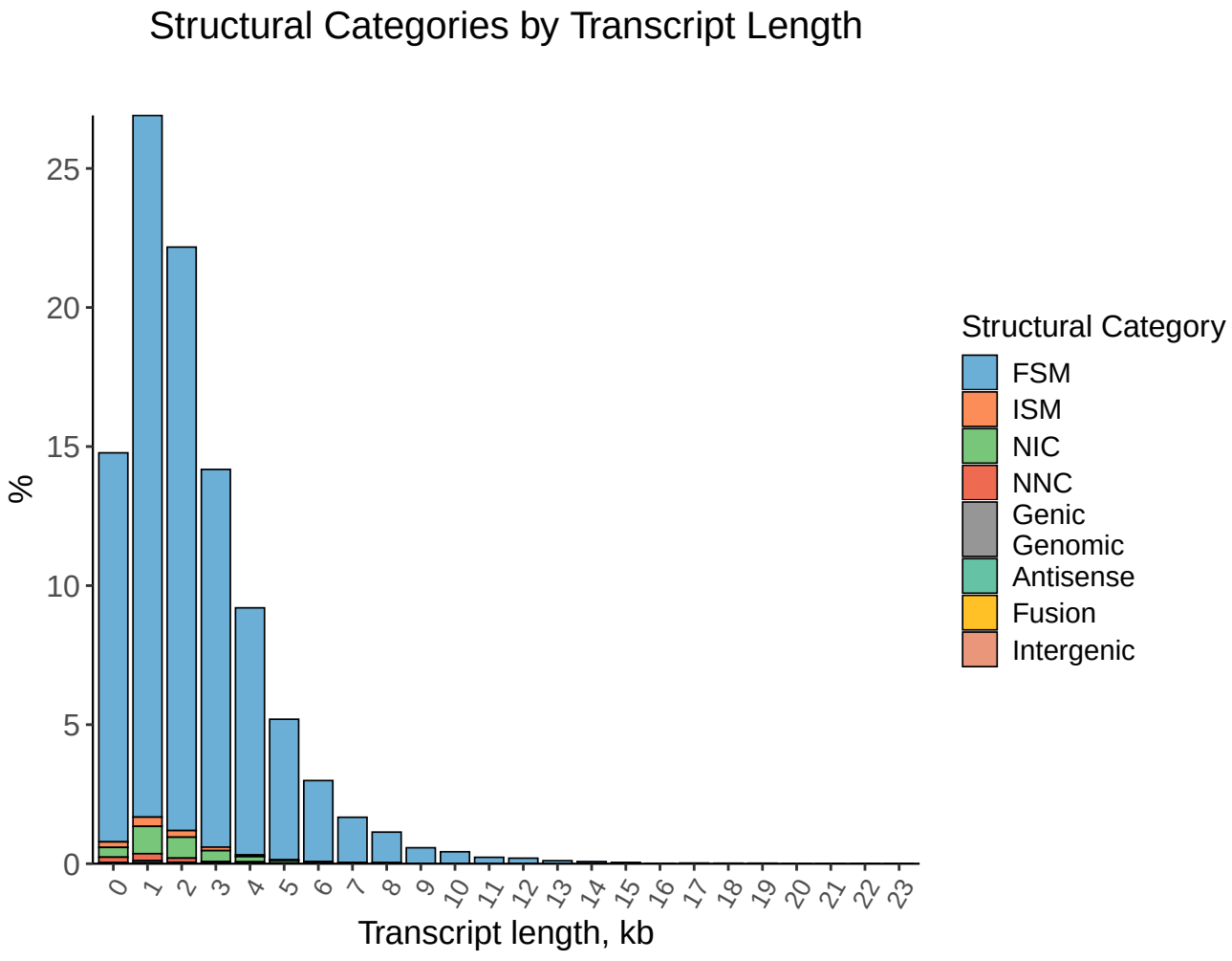
Gene Characterization

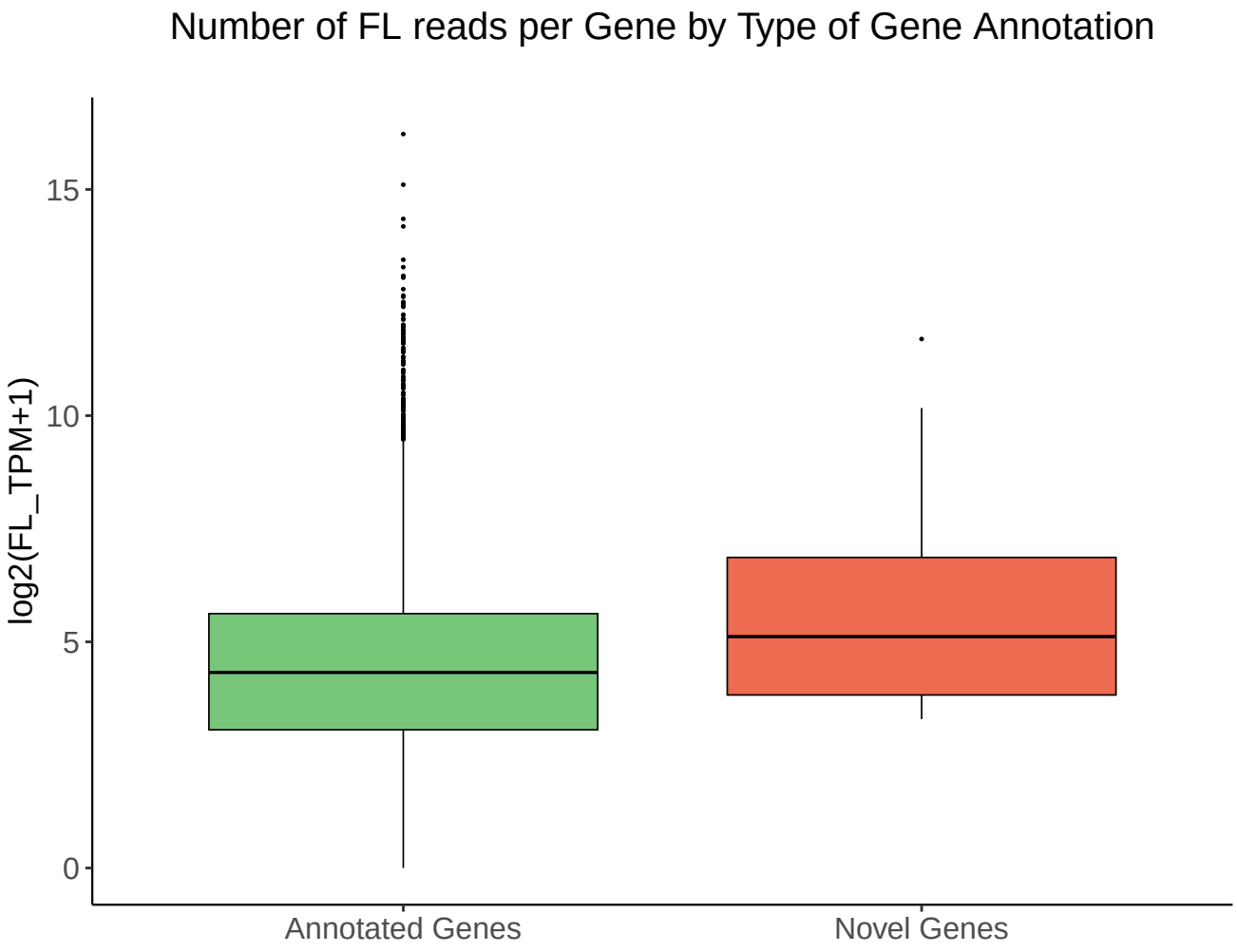

Number of Isoforms per Gene
Genes, %
0
10
20
30
40
1
2−3
4−5
>=6
Isoforms per gene

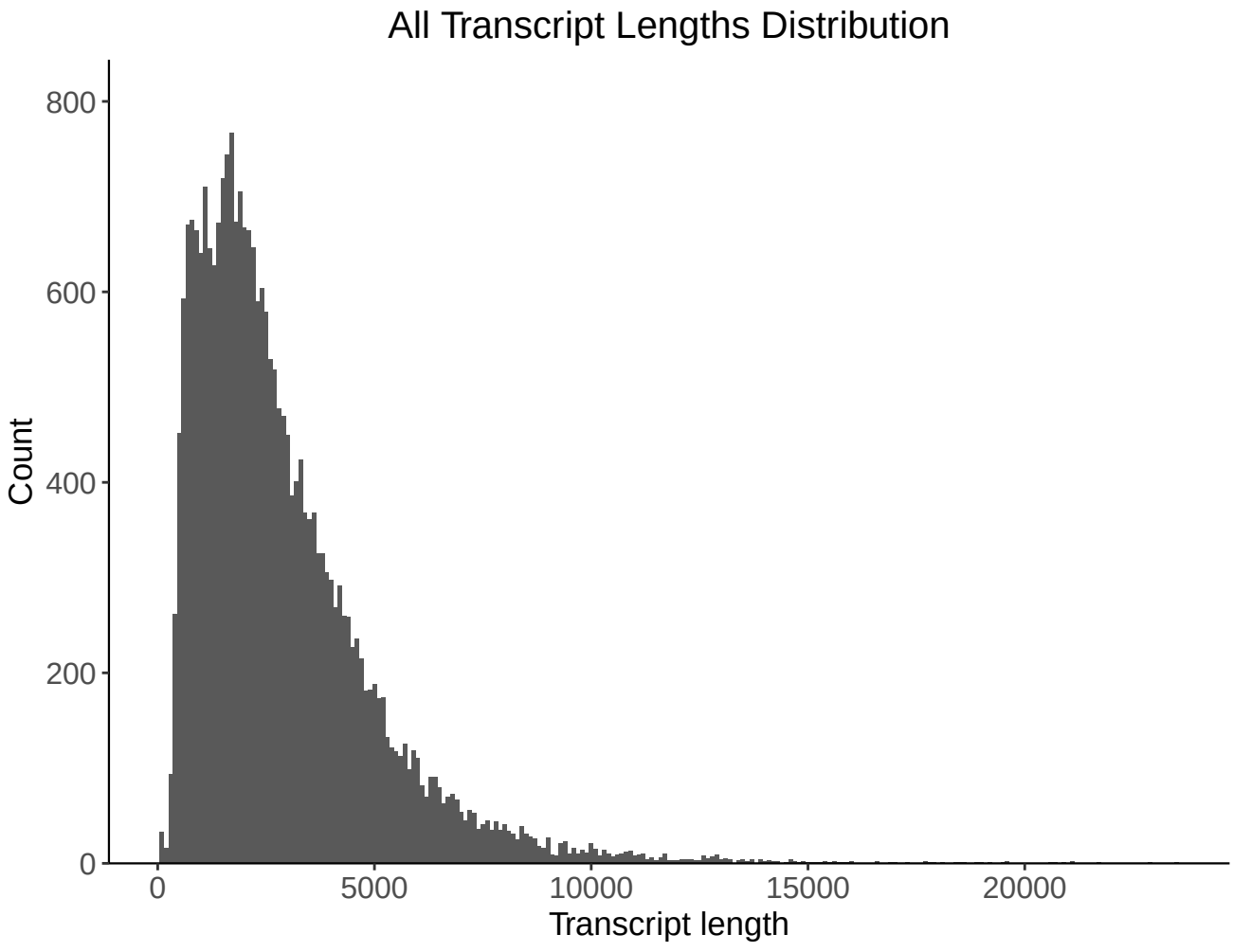

Number of Isoforms per Gene
Known vs Novel Genes
Genes, %
100
75
50
25
0
Annotated Genes
Novel Genes
Isoforms Per Gene
1
2−3
4−5
>=6

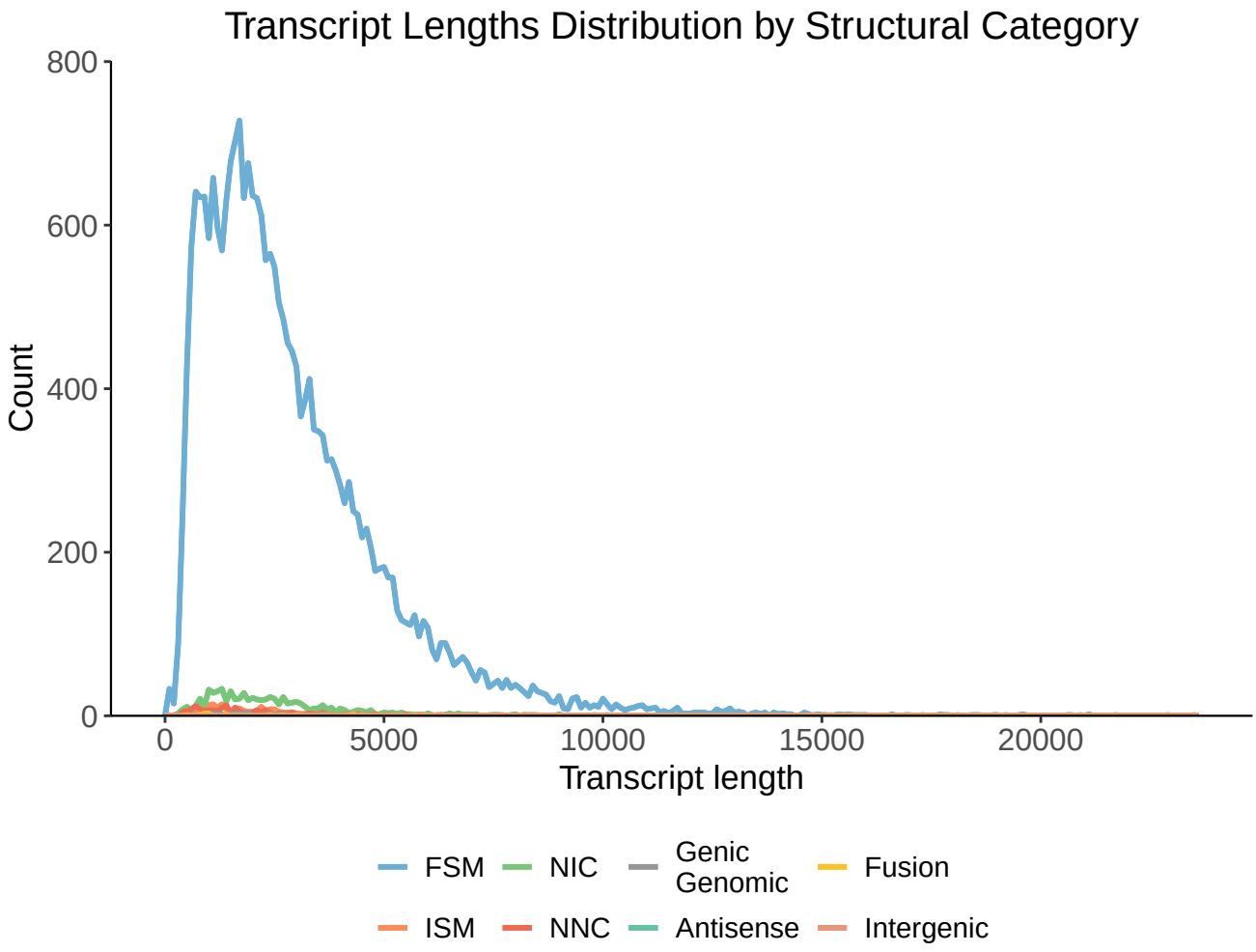

Distribution of Mono− vs Multi−Exon Transcripts
Transcripts, %
100
75
50
25
0
Novel Genes
Annotated Genes
Transcript type
Multi−Exon
Mono−Exon

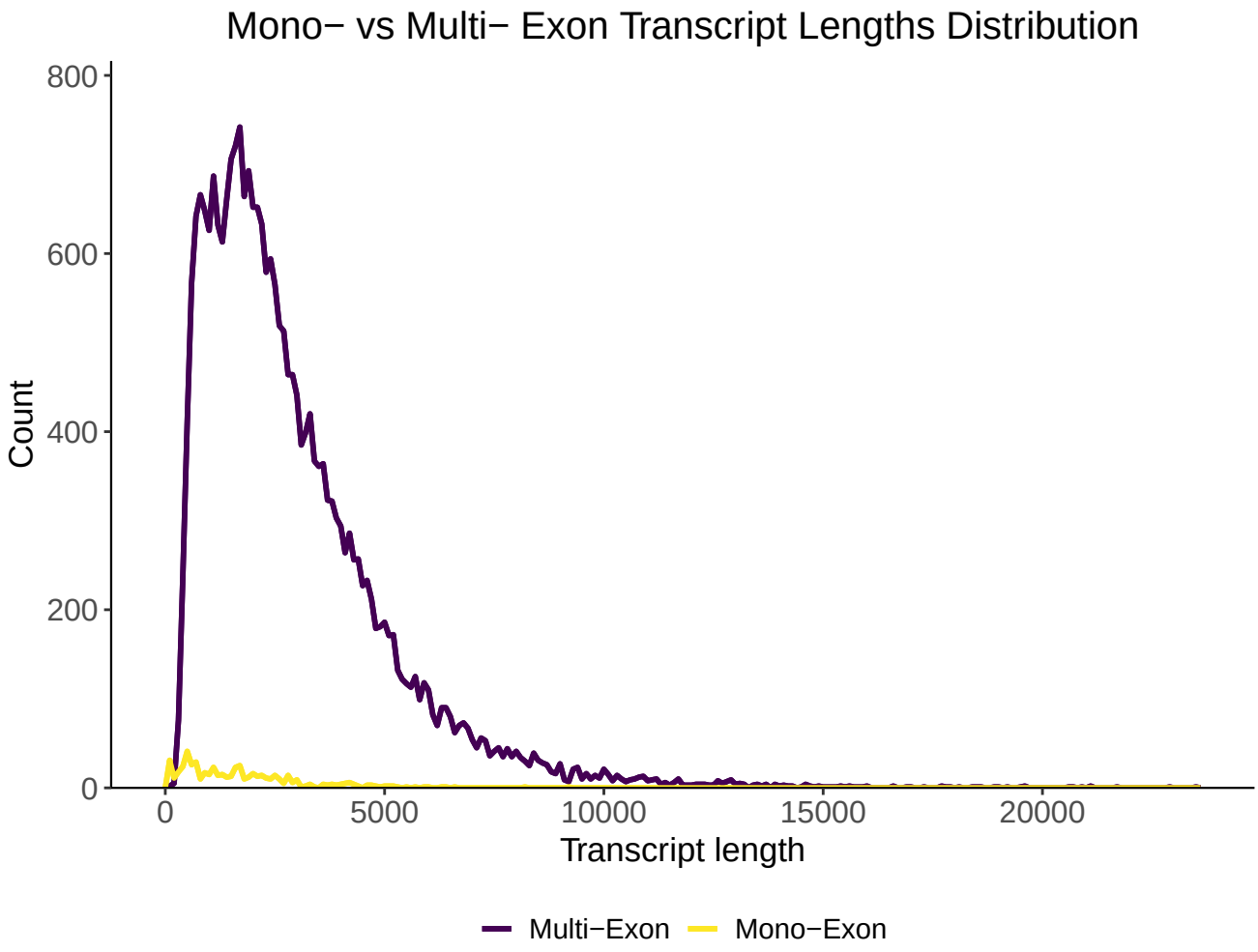

Structural Categories by Transcript Length
Counts
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
Transcript length, kb
Structural Category
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic


Structural Categories by Transcript Length
%
25
20
15
10
5
0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
Transcript length, kb
Structural Category
FSM
ISM
NIC
NNC
Genic
Genomic
Antisense
Fusion
Intergenic

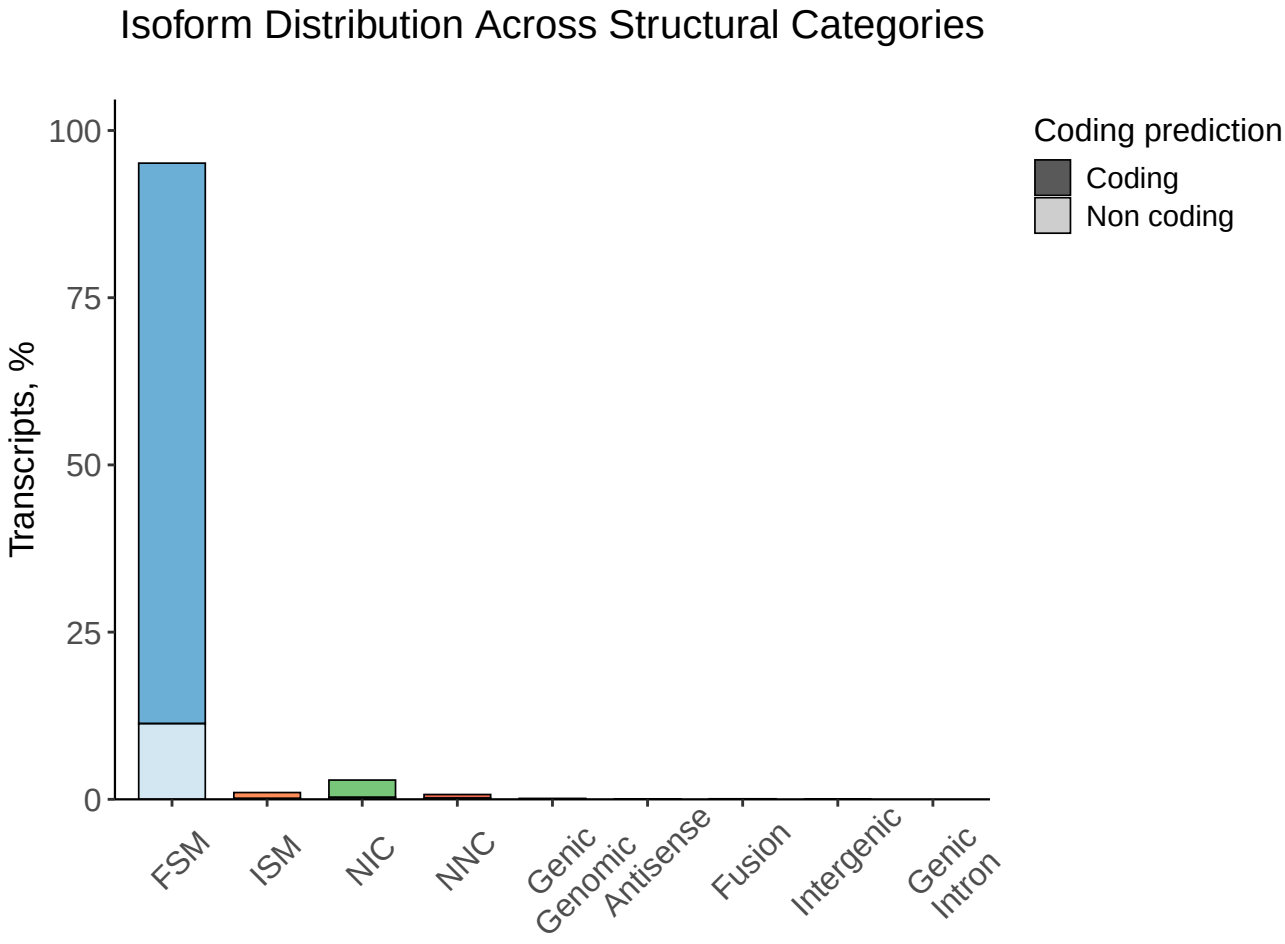

Number of FL reads per Gene by Type of Gene Annotation
log2(FL_TPM+1)
15
10
5
0
Annotated Genes
Novel Genes

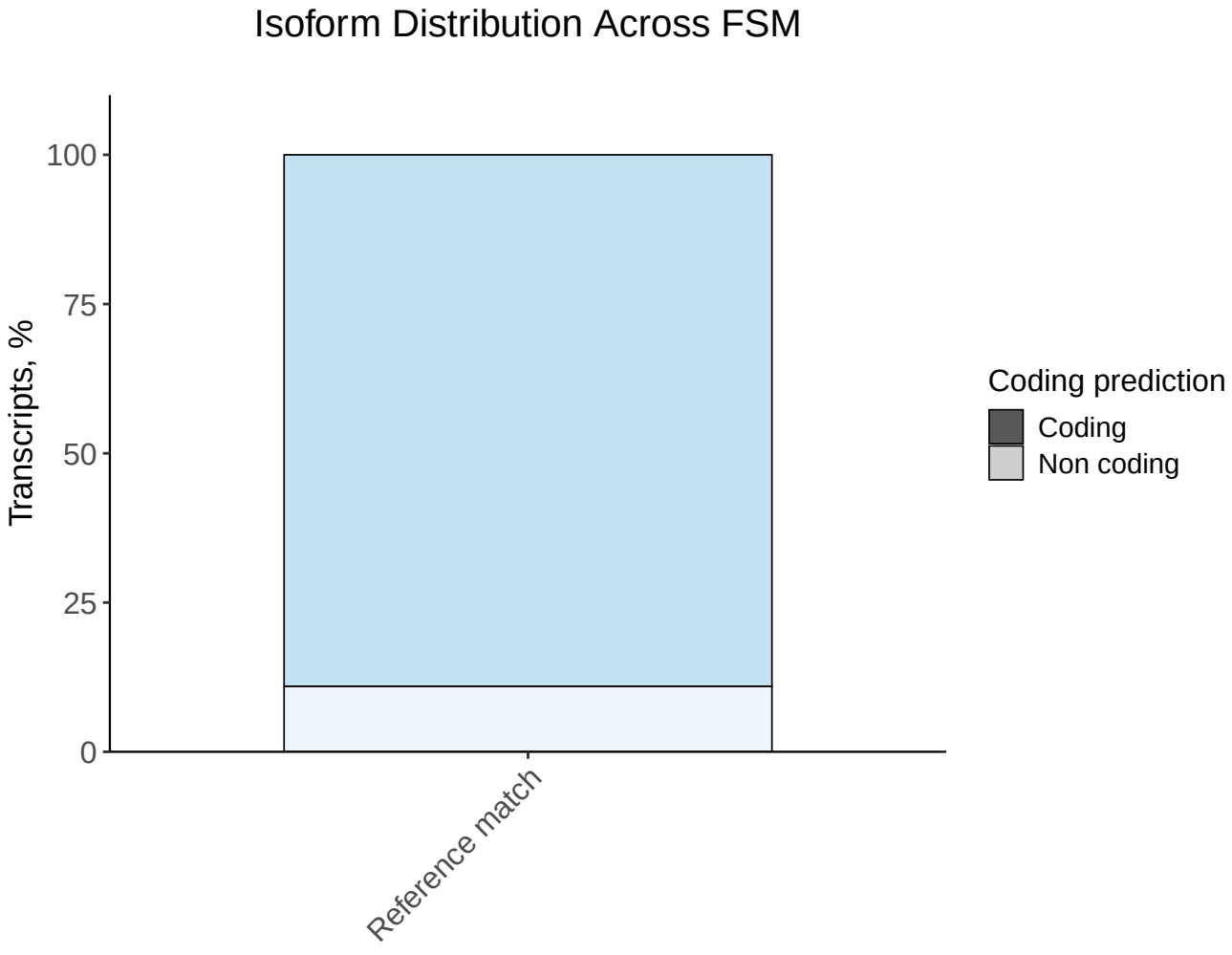

All Transcript Lengths Distribution
Count
800
600
400
200
0
0
5000
10000
15000
20000
Transcript length

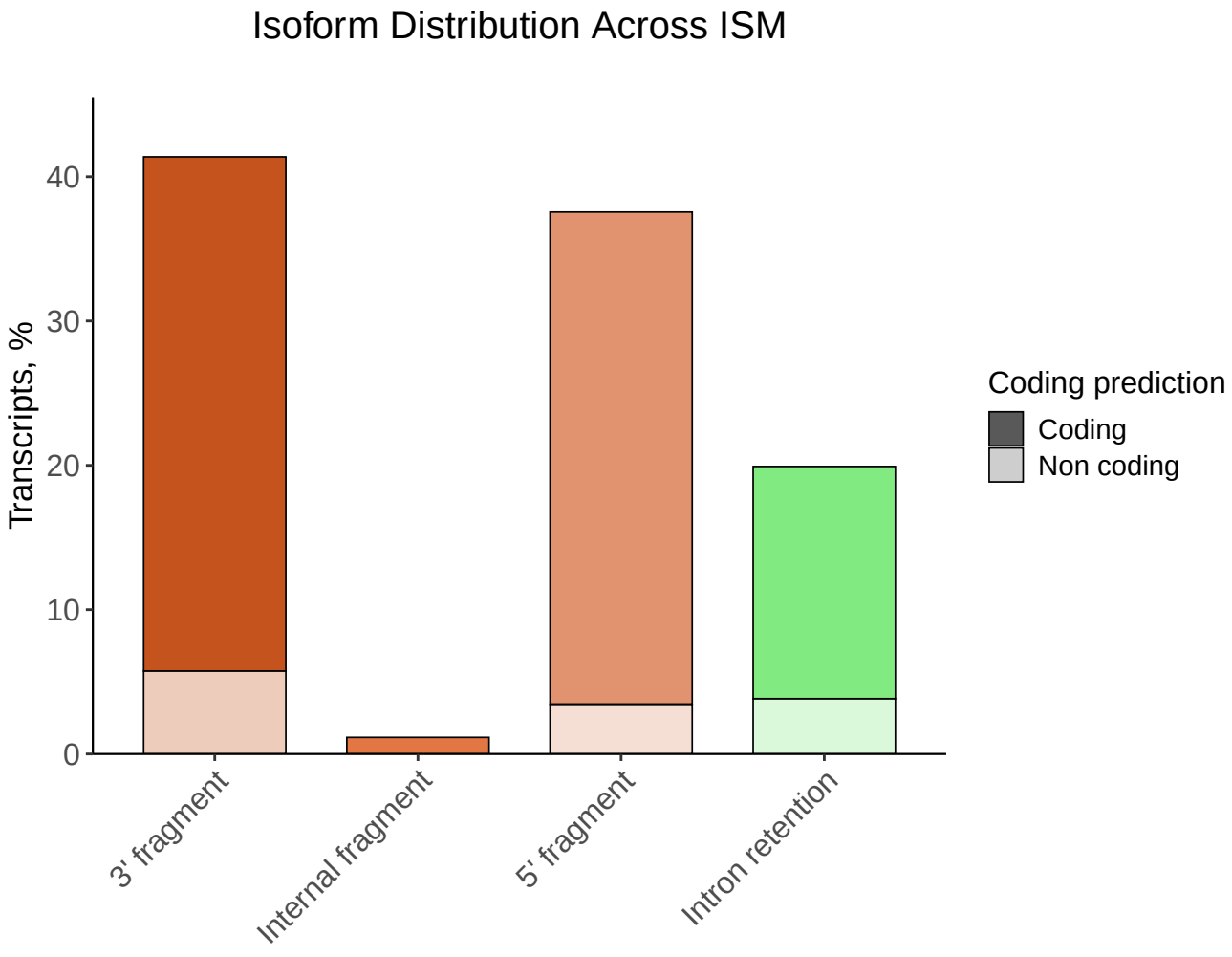

Transcript Lengths Distribution by Structural Category
Count
800
600
400
200
0
0
5000
10000
15000
20000
Transcript length
FSM
NIC
Genic Genomic
Fusion
ISM
NNC
Antisense
Intergenic

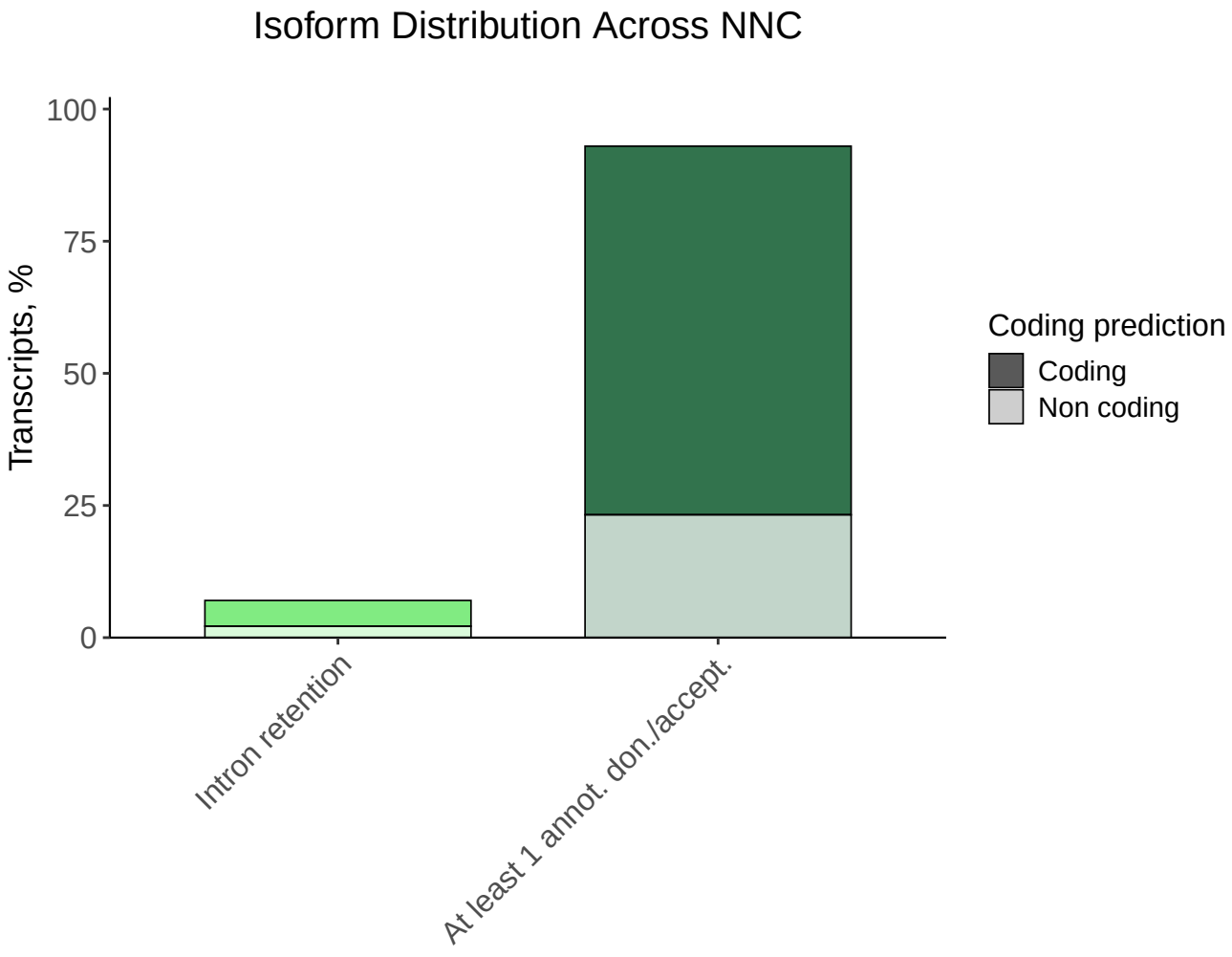

Mono− vs Multi− Exon Transcript Lengths Distribution
Count
800
600
400
200
0
0
5000
10000
15000
20000
Transcript length
Multi−Exon
Mono−Exon

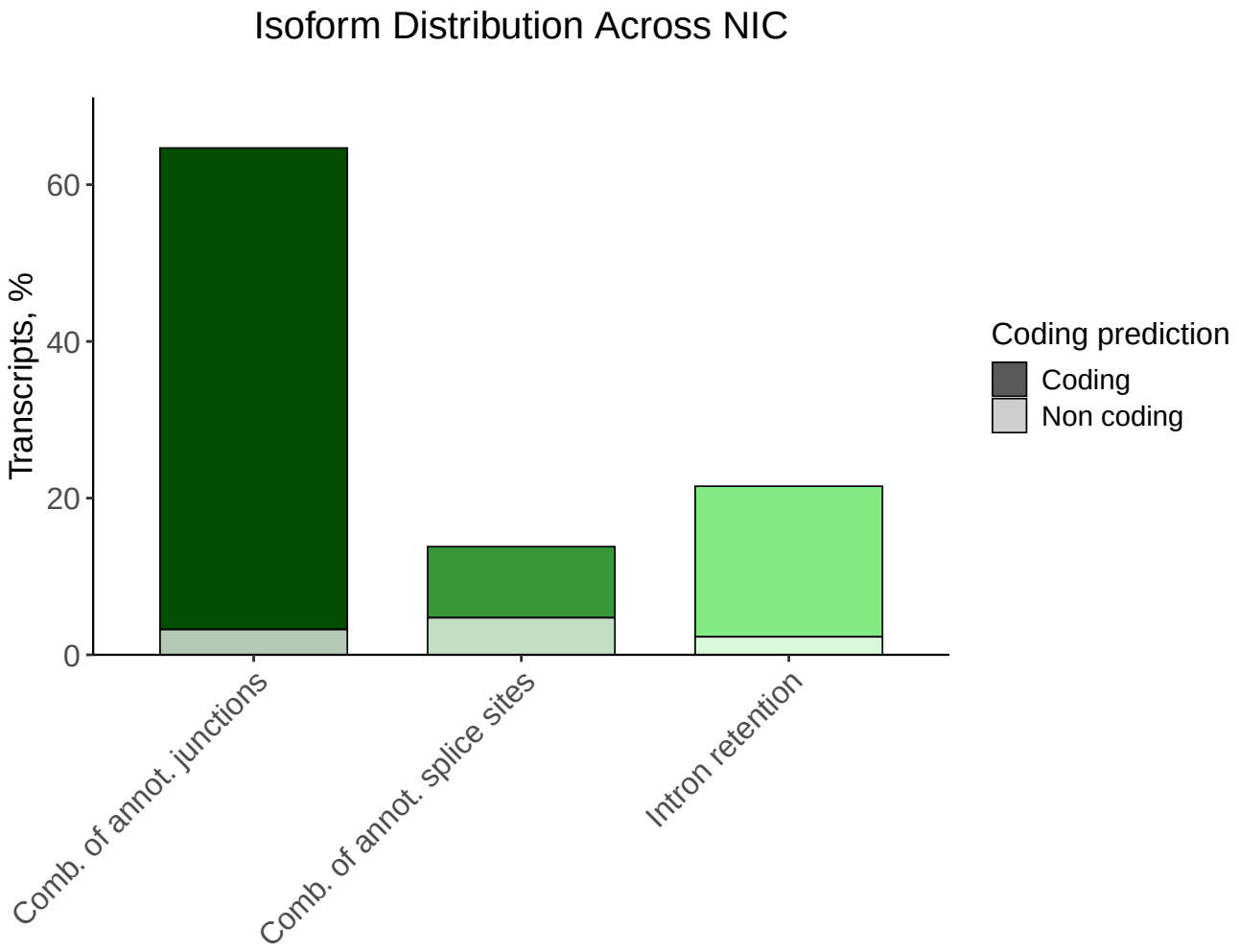
Structural Isoform Characterization

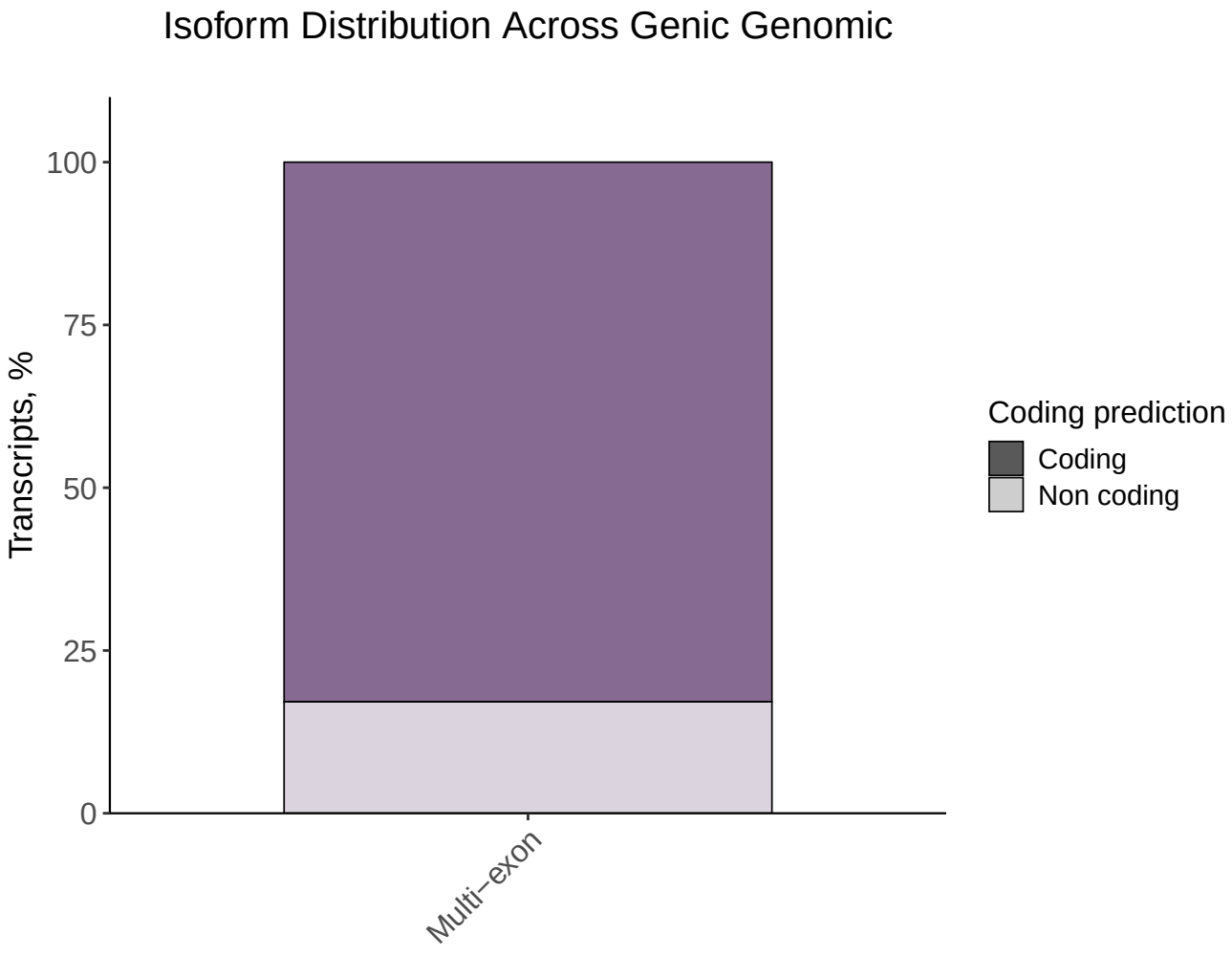

Isoform Distribution Across Structural Categories
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron
Coding prediction
Coding
Non coding

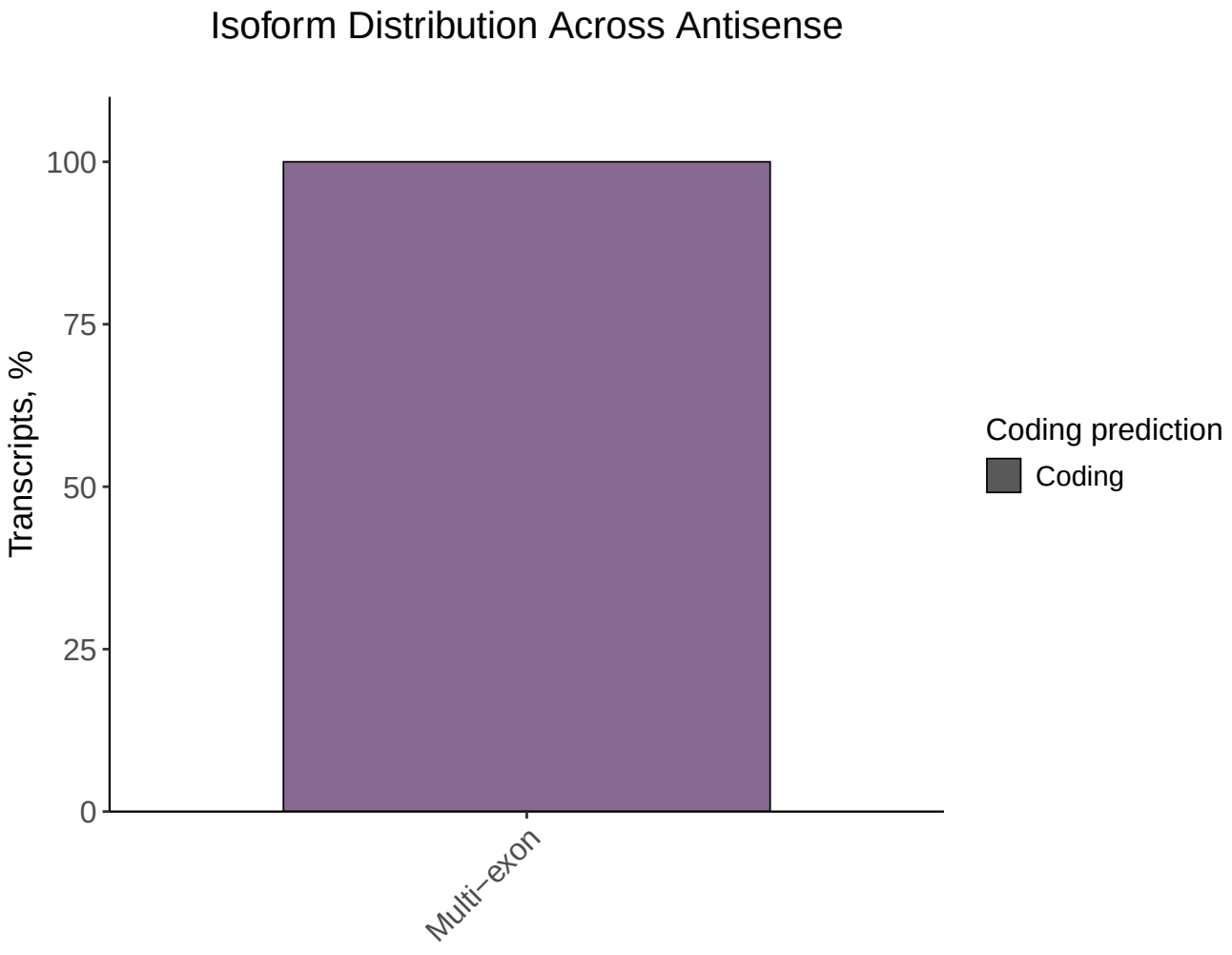

Isoform Distribution Across FSM
Transcripts, %
100
75
50
25
0
Reference match
Coding prediction
Coding
Non coding

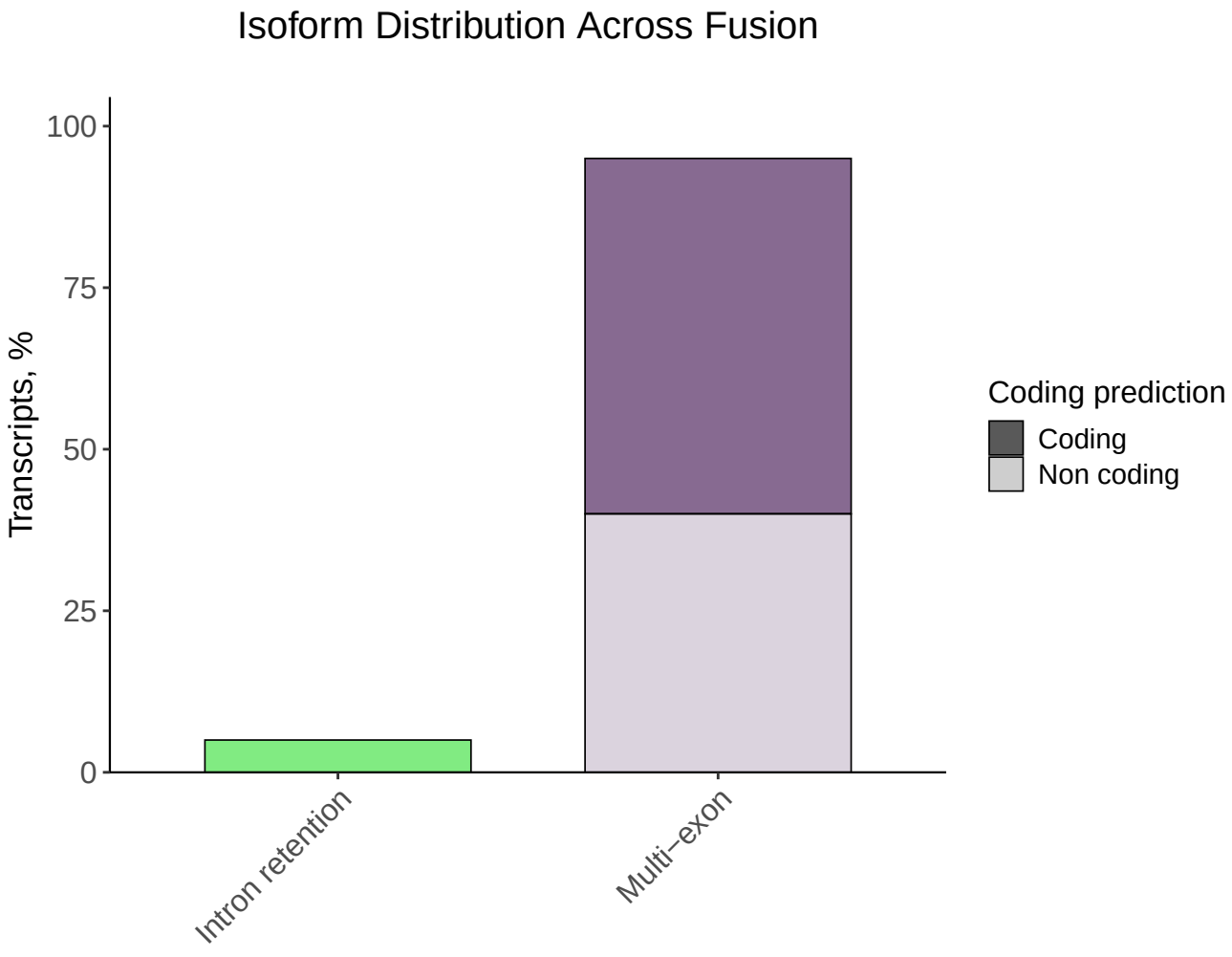

Isoform Distribution Across ISM
Transcripts, %
0
10
20
30
40
3' fragment
Internal fragment
5' fragment
Intron retention
Coding prediction
Coding
Non coding

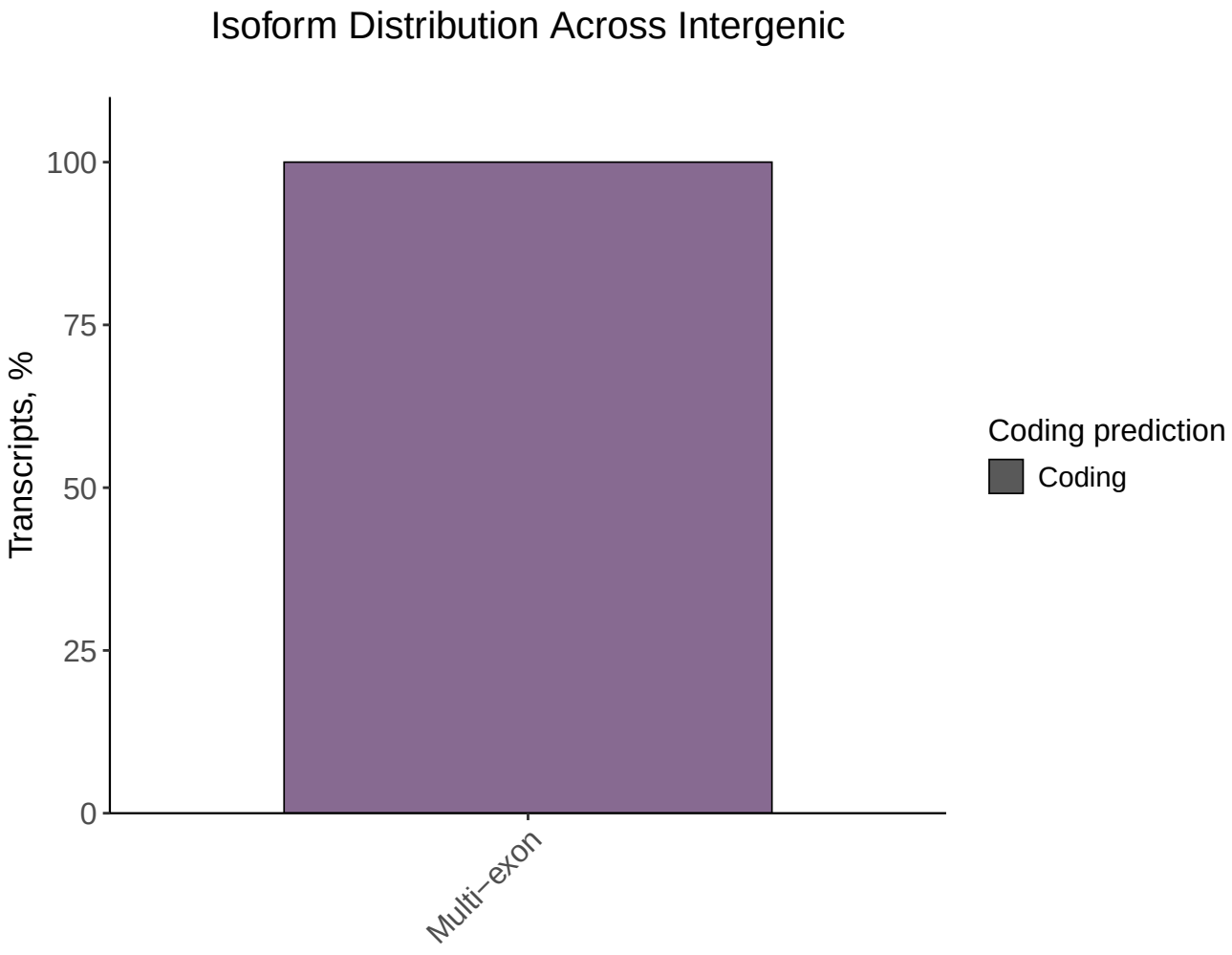

Isoform Distribution Across NNC
Transcripts, %
100
75
50
25
0
Intron retention
At least 1 annot. don./accept.
Coding prediction
Coding
Non coding

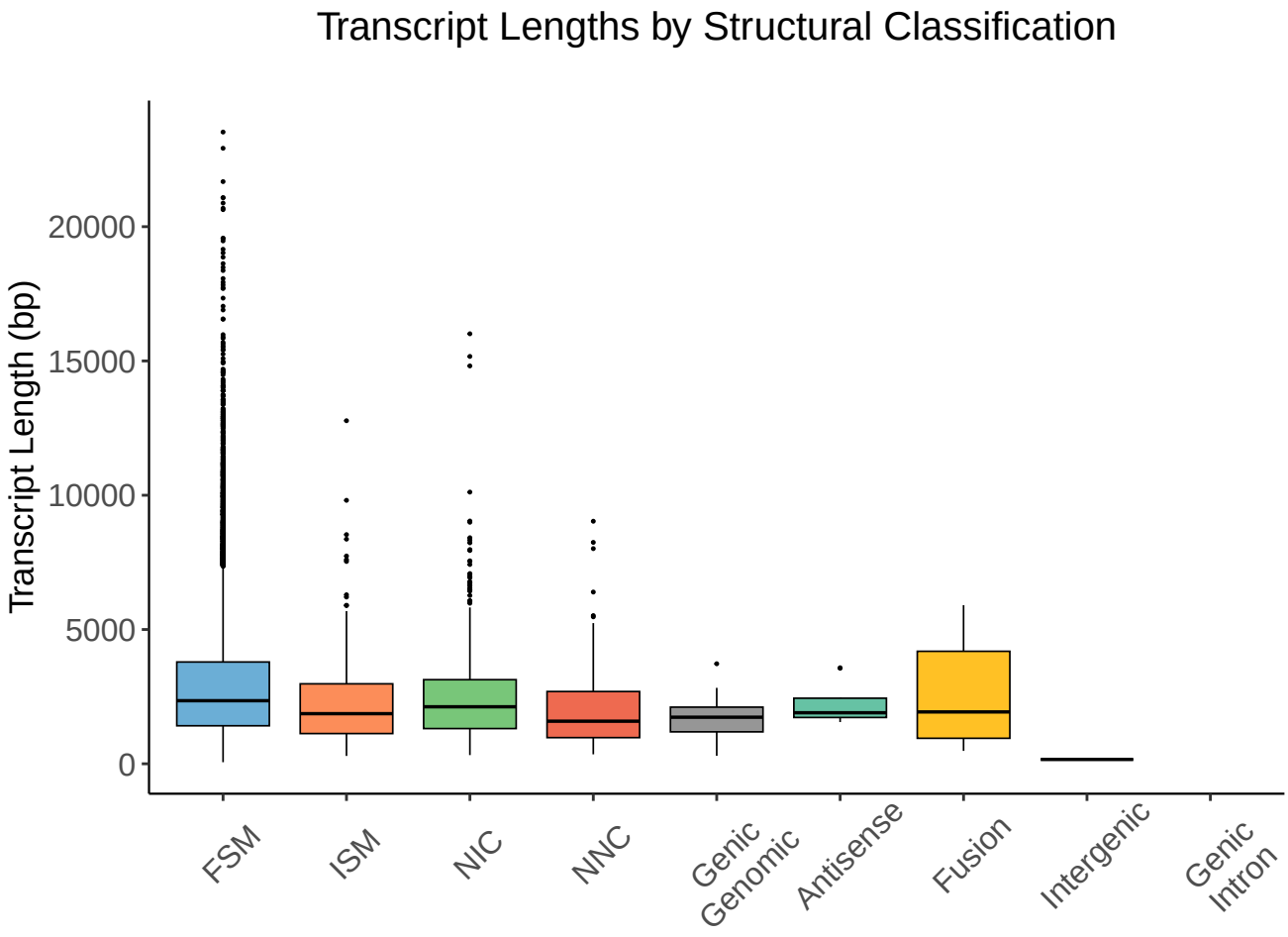

Isoform Distribution Across NIC
Transcripts, %
0
20
40
60
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Coding prediction
Coding
Non coding

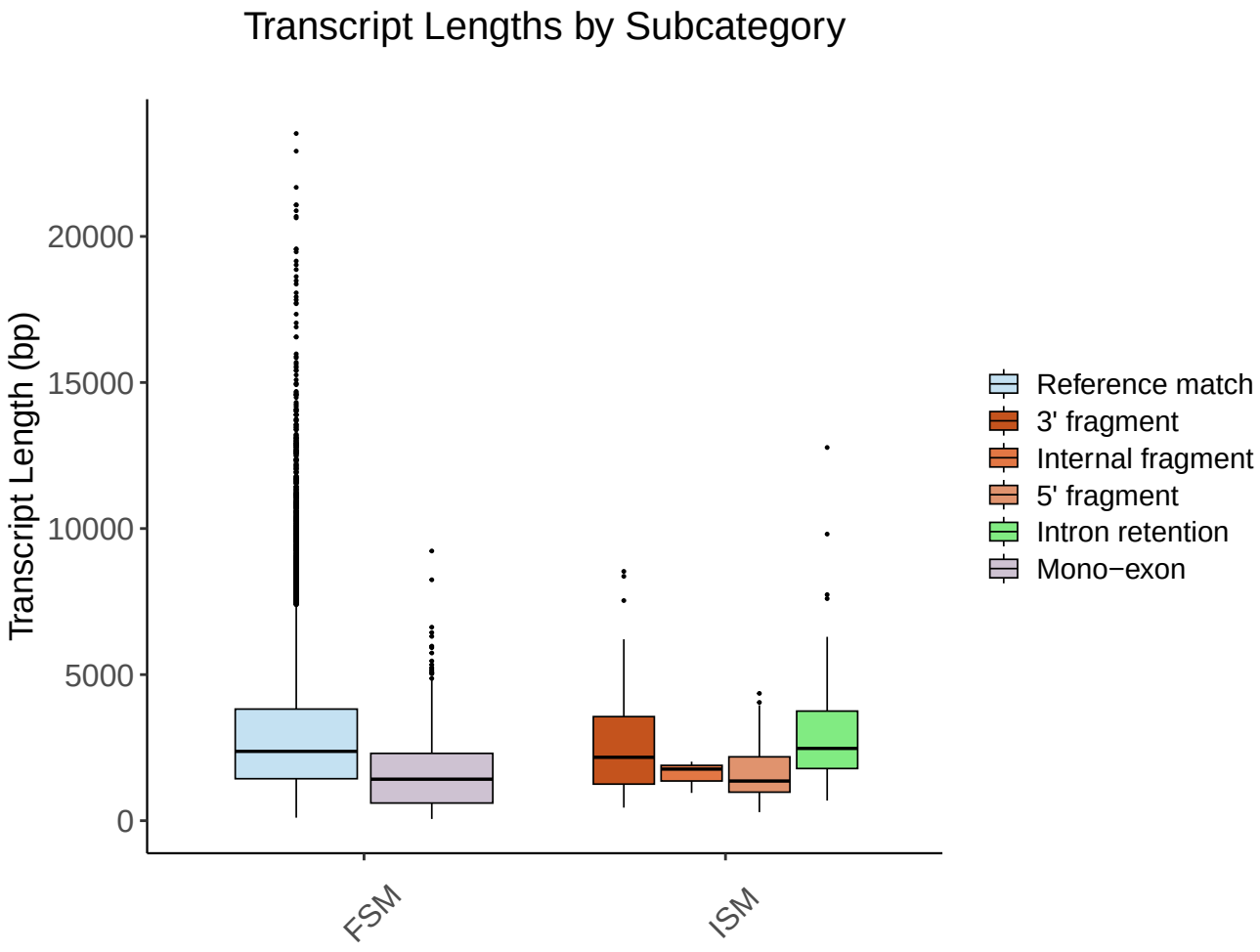

Isoform Distribution Across Genic Genomic
Transcripts, %
100
75
50
25
0
Multi−exon
Coding prediction
Coding
Non coding

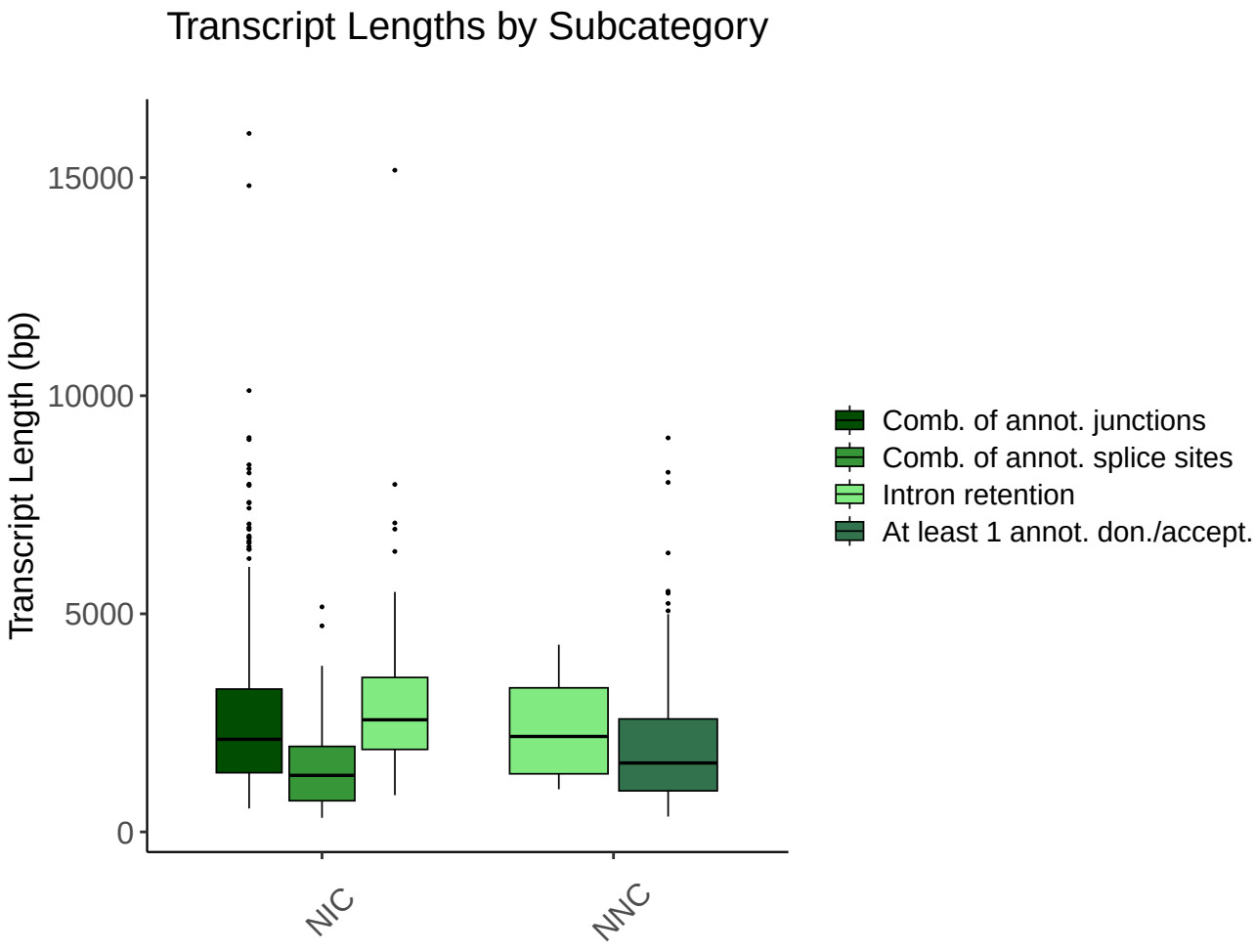

Isoform Distribution Across Antisense
Transcripts, %
100
75
50
25
0
Multi−exon
Coding prediction
Coding

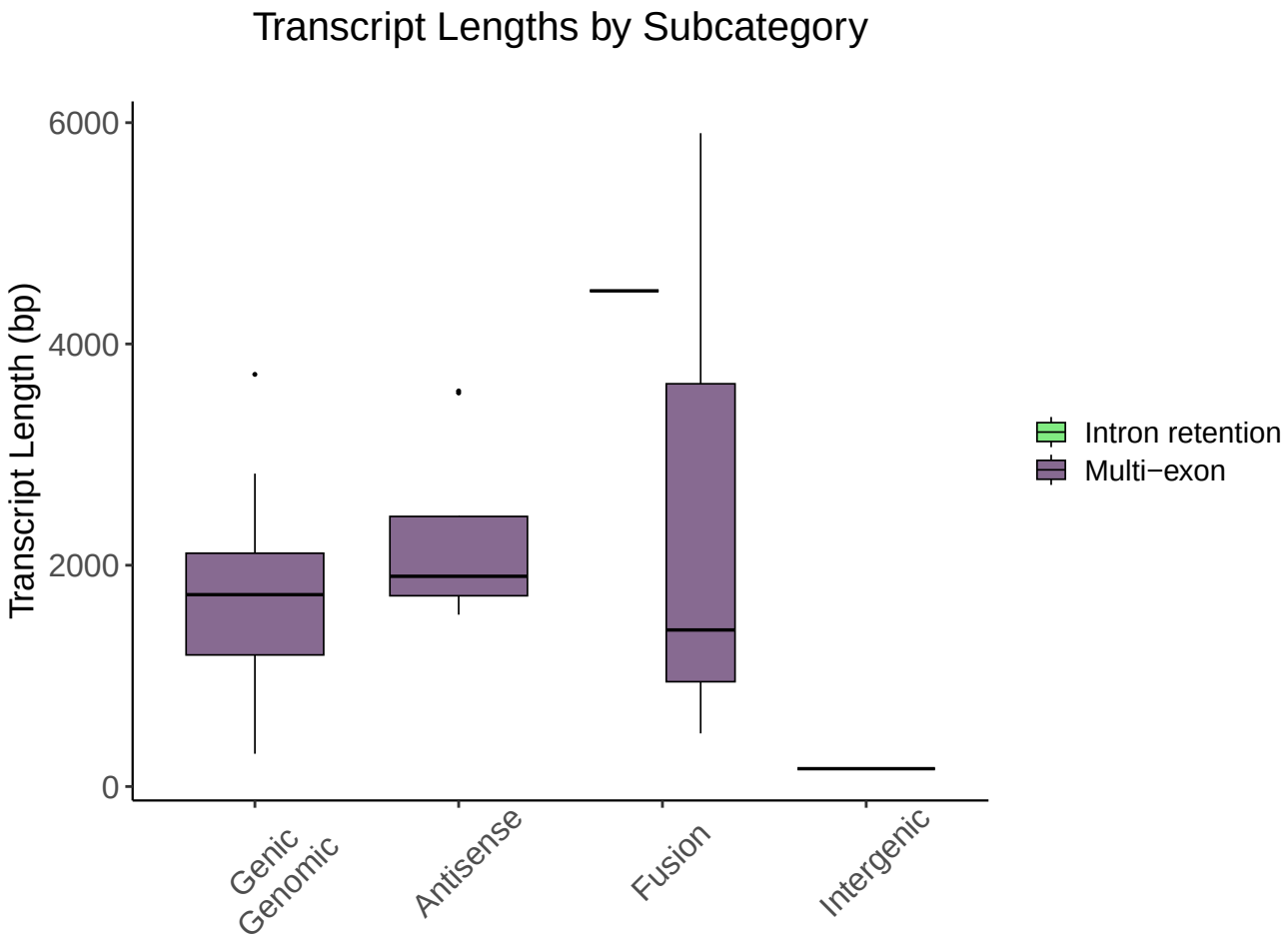

Isoform Distribution Across Fusion
Transcripts, %
100
75
50
25
0
Intron retention
Multi−exon
Coding prediction
Coding
Non coding

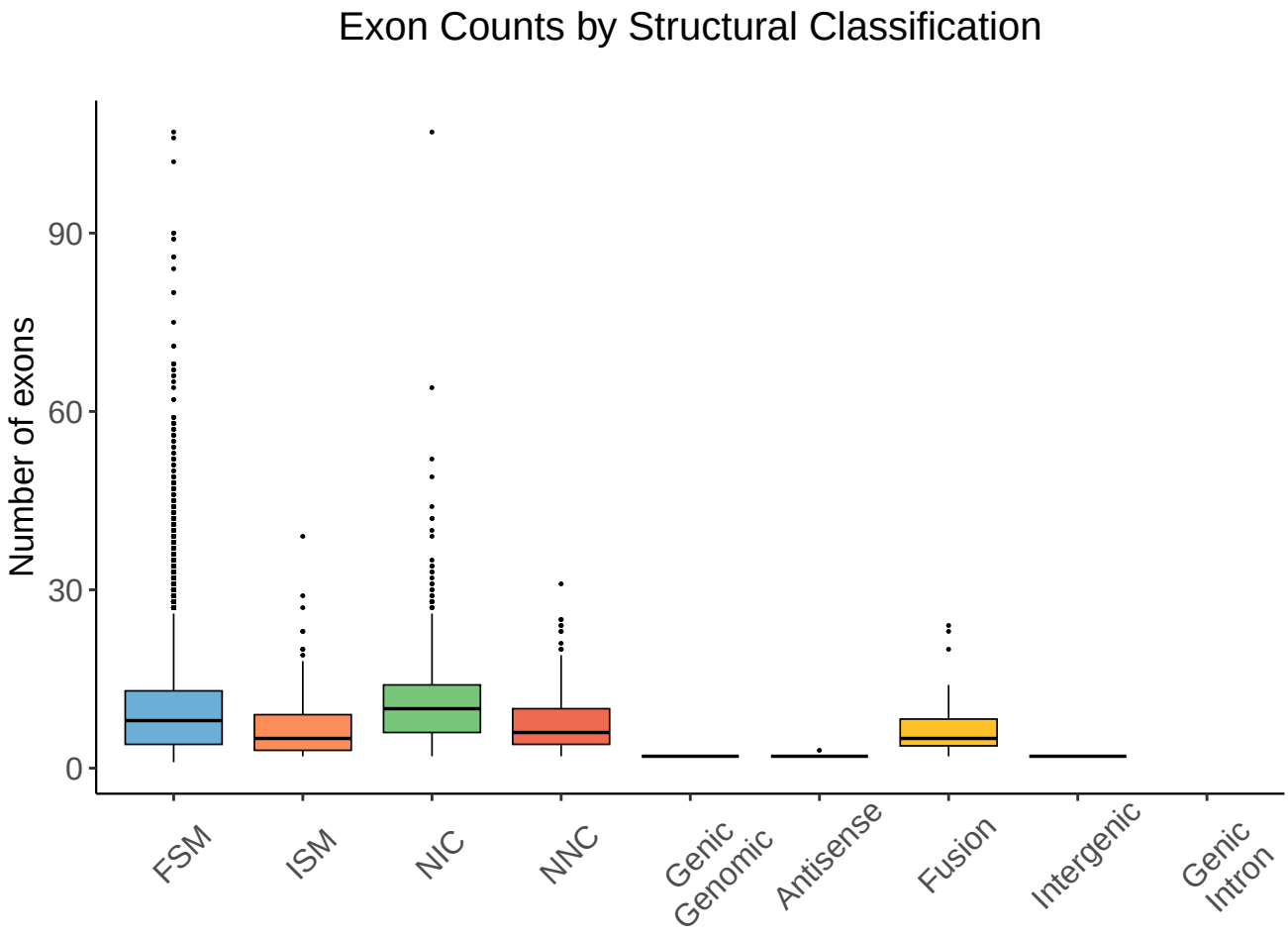

Isoform Distribution Across Intergenic
Transcripts, %
100
75
50
25
0
Multi−exon
Coding prediction
Coding

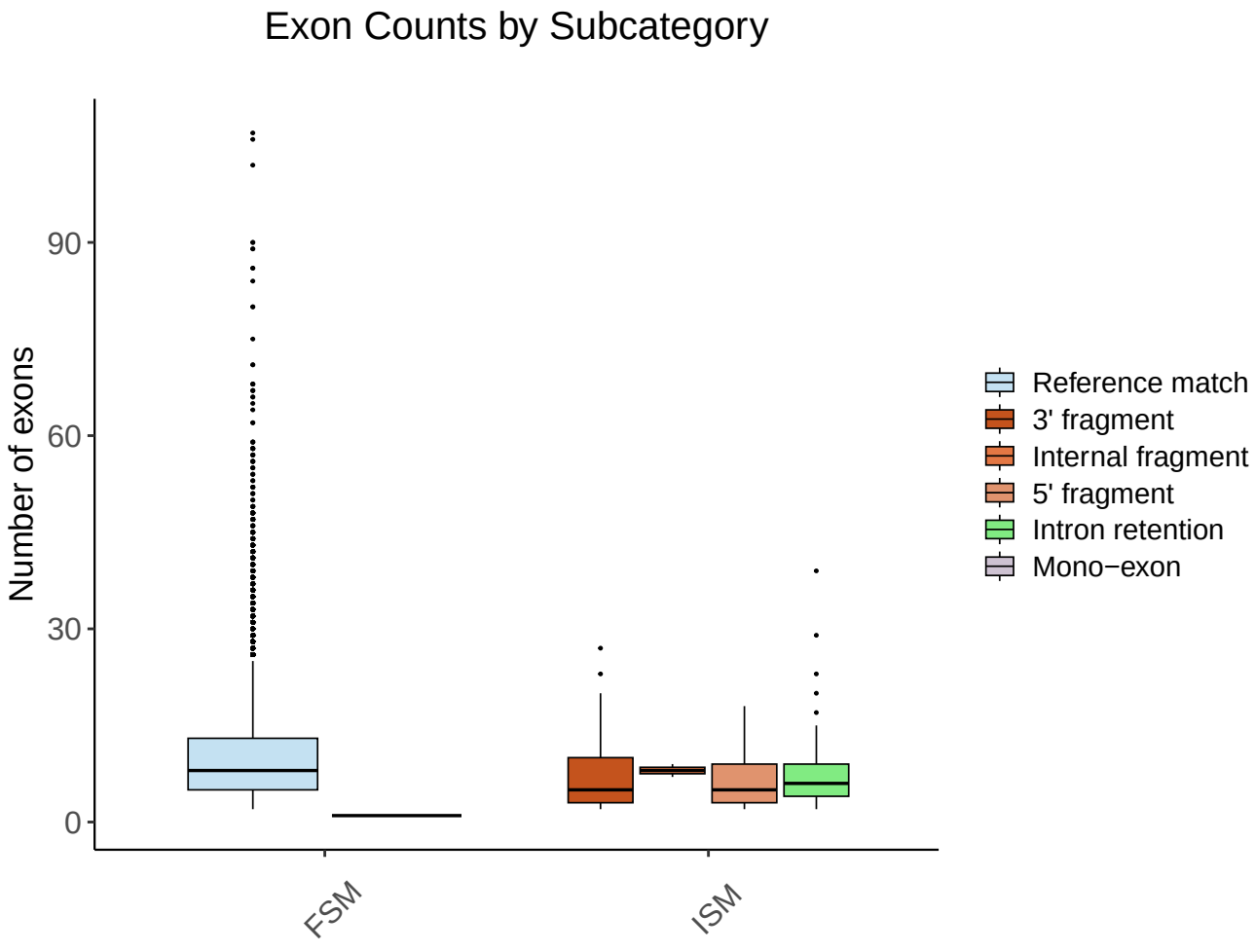

Transcript Lengths by Structural Classification
Transcript Length (bp)
20000
15000
10000
5000
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron

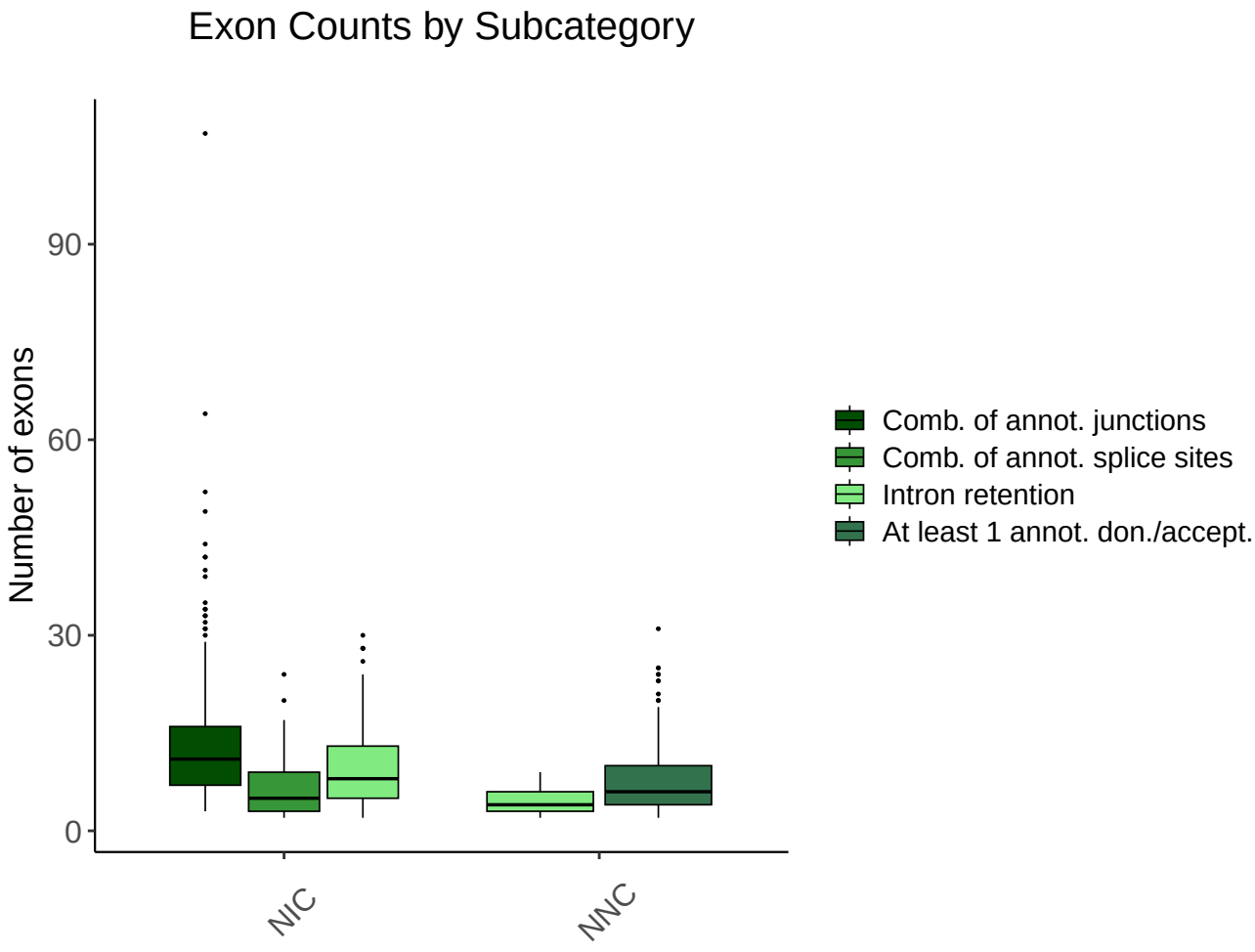

Transcript Lengths by Subcategory
Transcript Length (bp)
20000
15000
10000
5000
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon

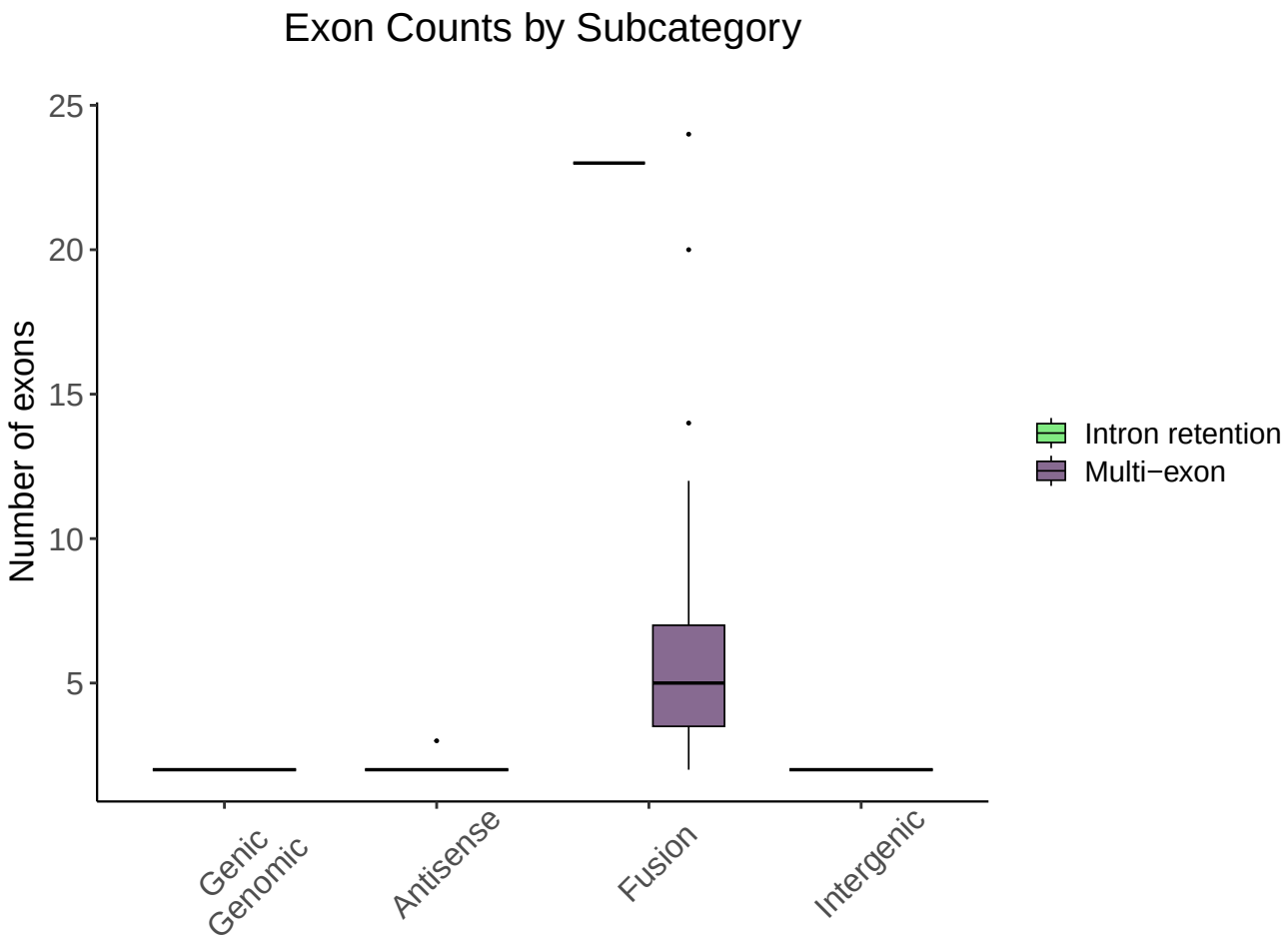

Transcript Lengths by Subcategory
Transcript Length (bp)
15000
10000
5000
0
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

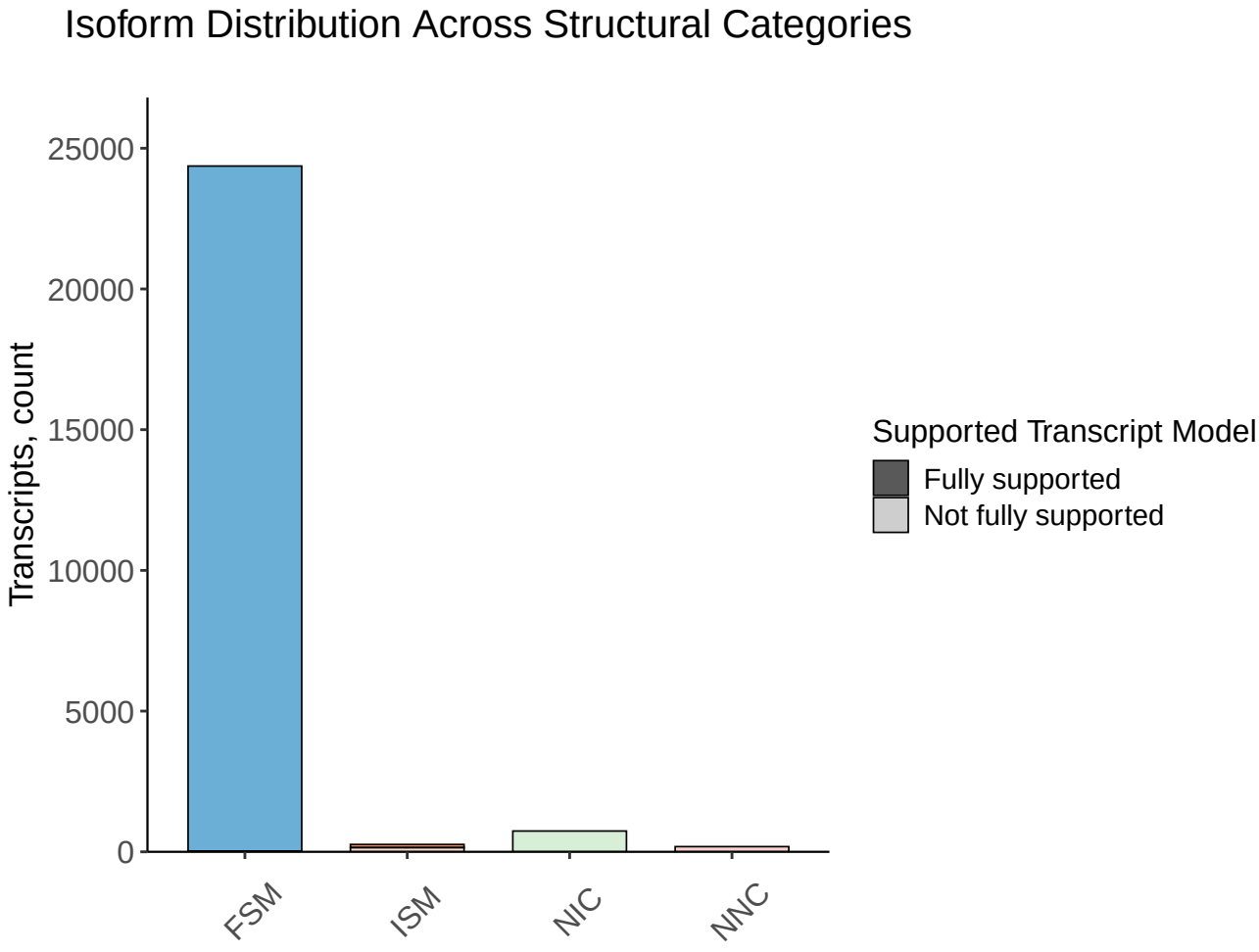

Transcript Lengths by Subcategory
Transcript Length (bp)
6000
4000
2000
0
Genic Genomic
Antisense
Fusion
Intergenic
Intron retention
Multi−exon

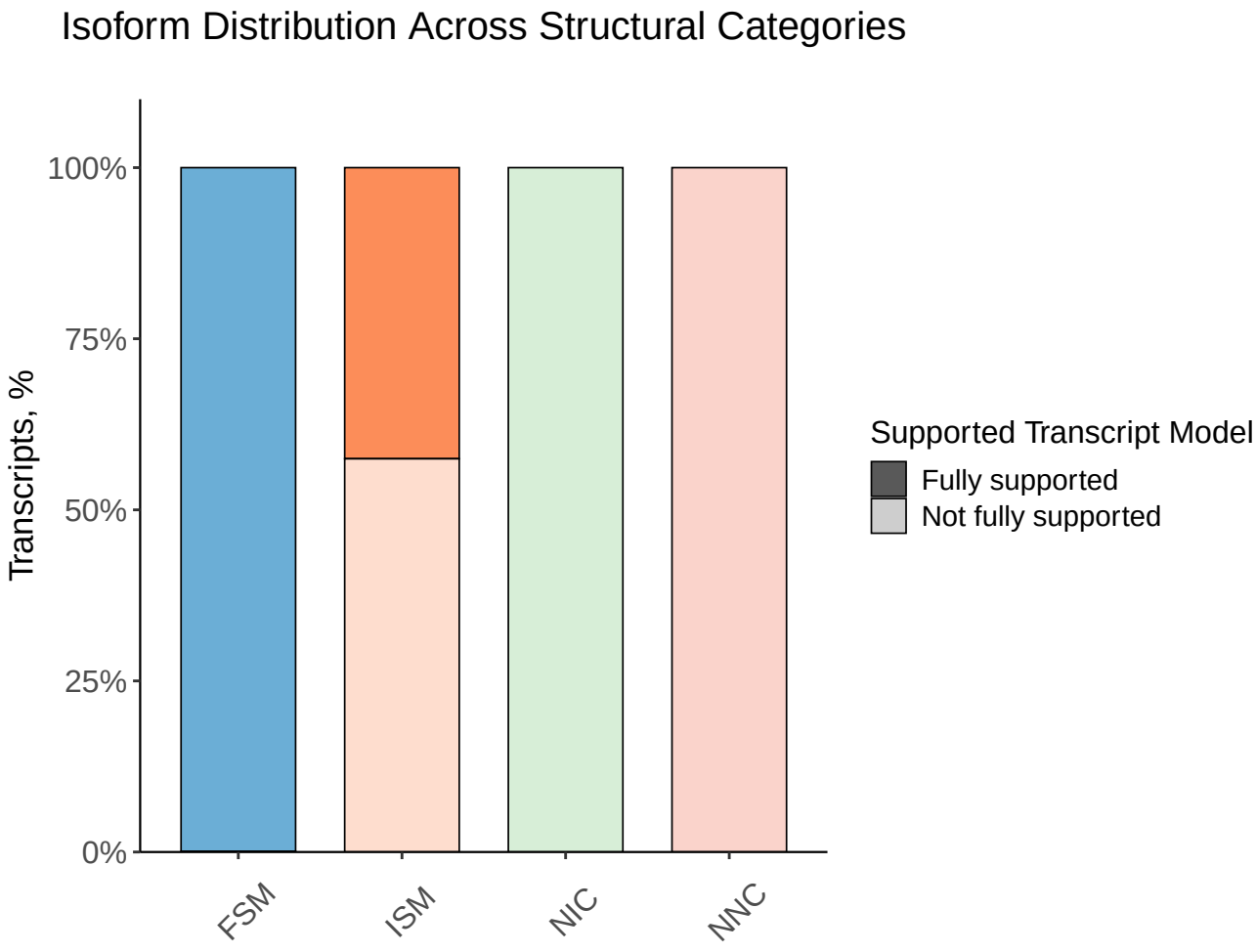

Exon Counts by Structural Classification
Number of exons
0
30
60
90
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron

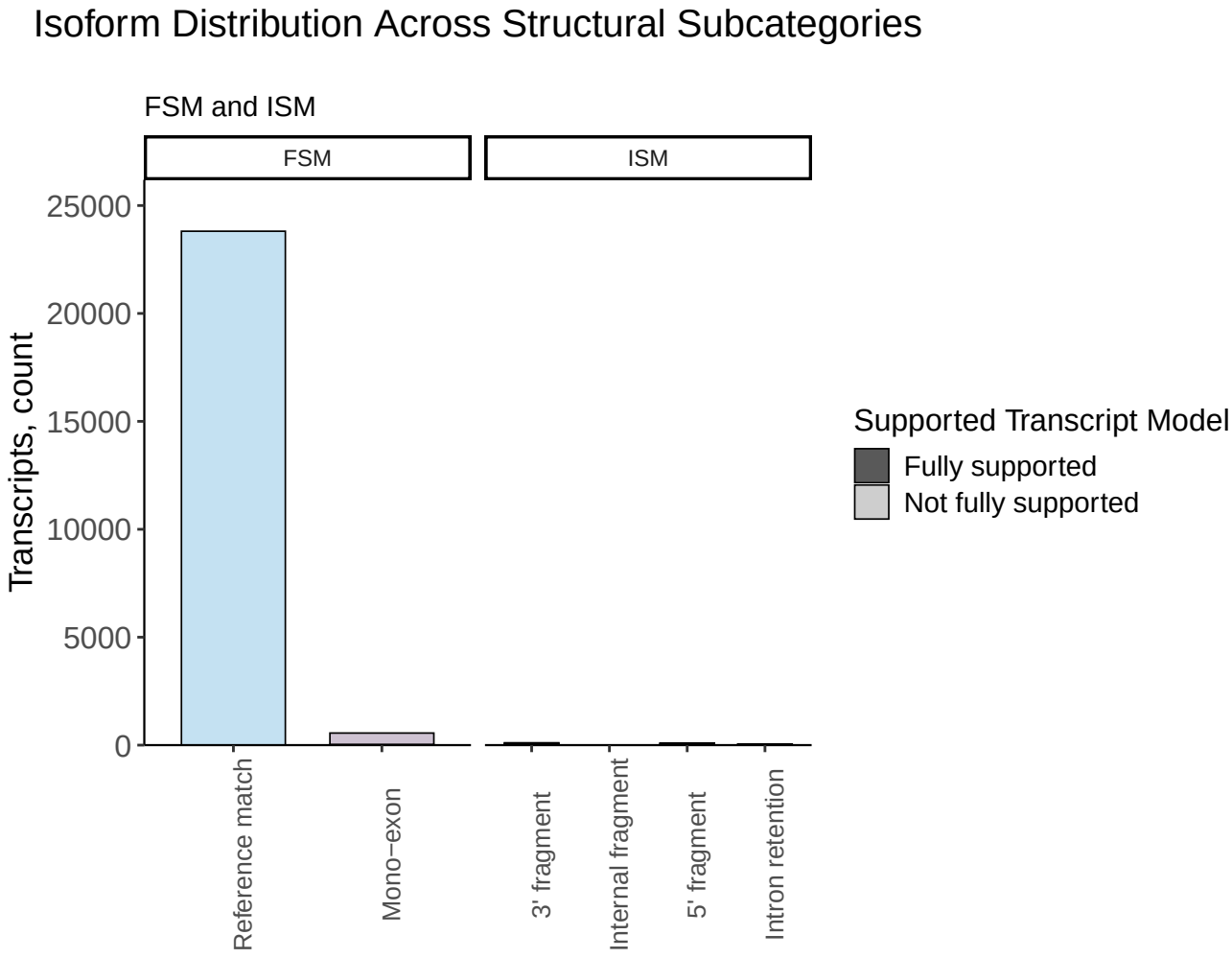

Exon Counts by Subcategory
Number of exons
90
60
30
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon

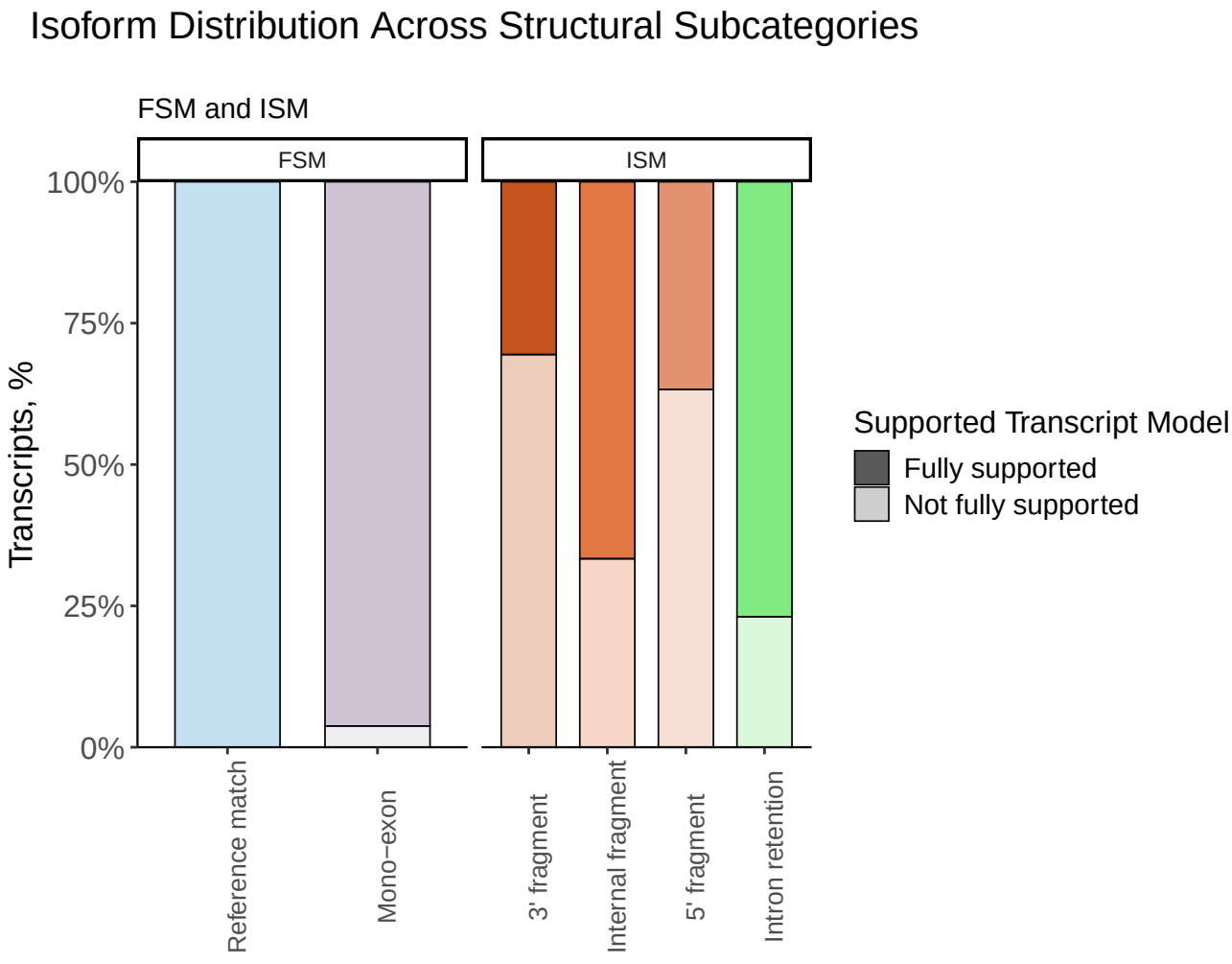

Exon Counts by Subcategory
Number of exons
90
60
30
0
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

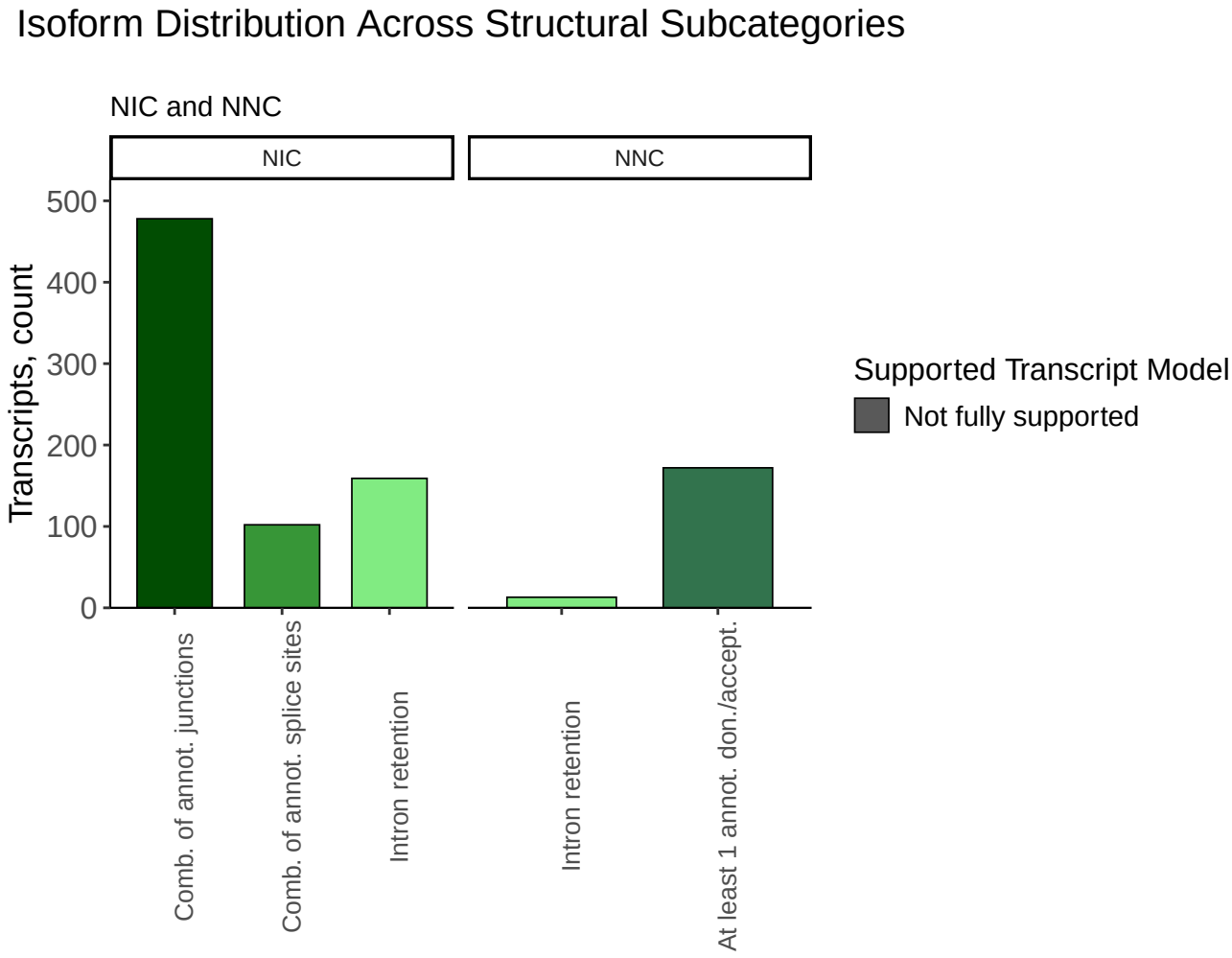

Exon Counts by Subcategory
Number of exons
25
20
15
10
5
Genic Genomic
Antisense
Fusion
Intergenic
Intron retention
Multi−exon

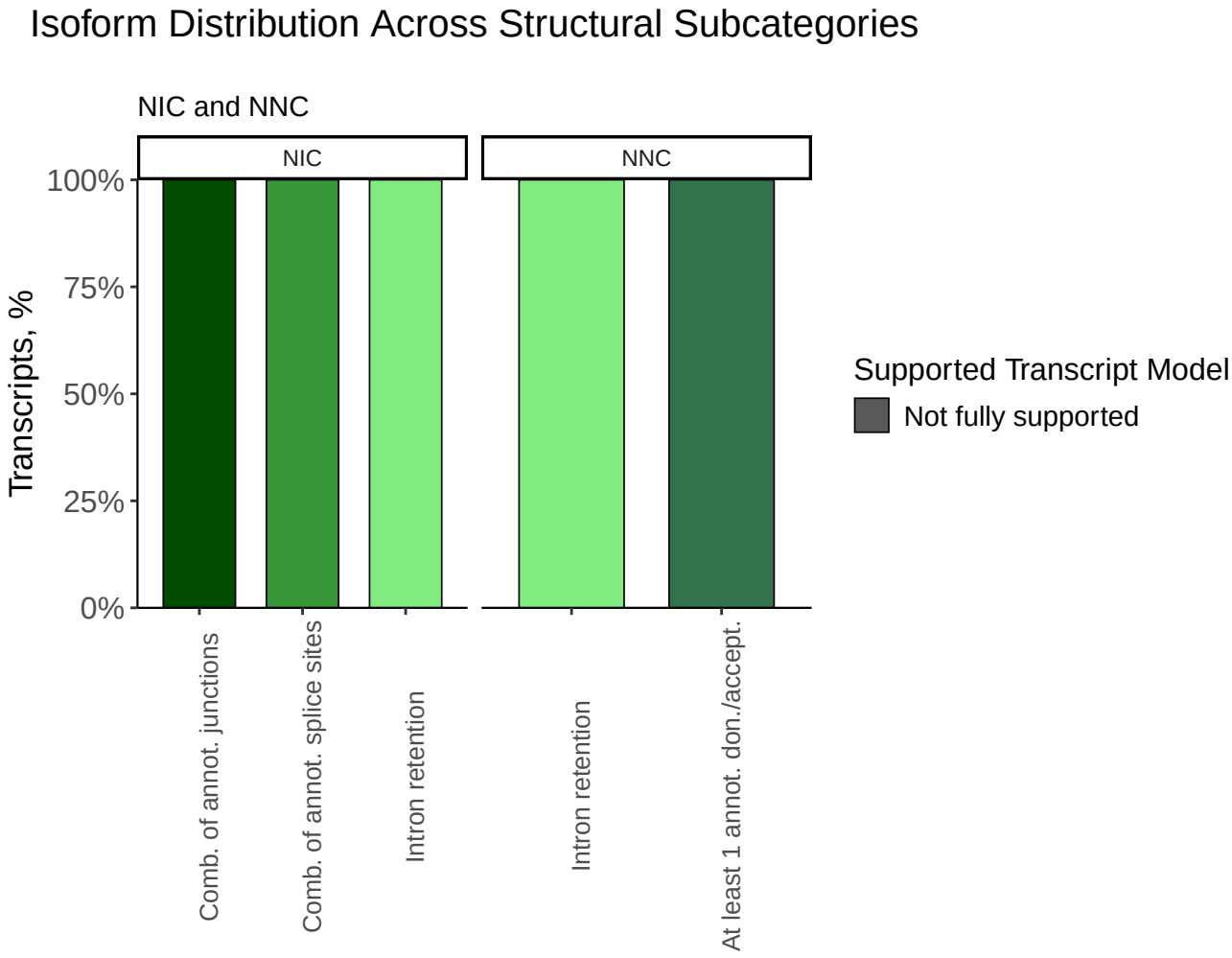

Isoform Distribution Across Structural Categories
Transcripts, count
25000
20000
15000
10000
5000
0
FSM
ISM
NIC
NNC
Supported Transcript Model
Fully supported
Not fully supported

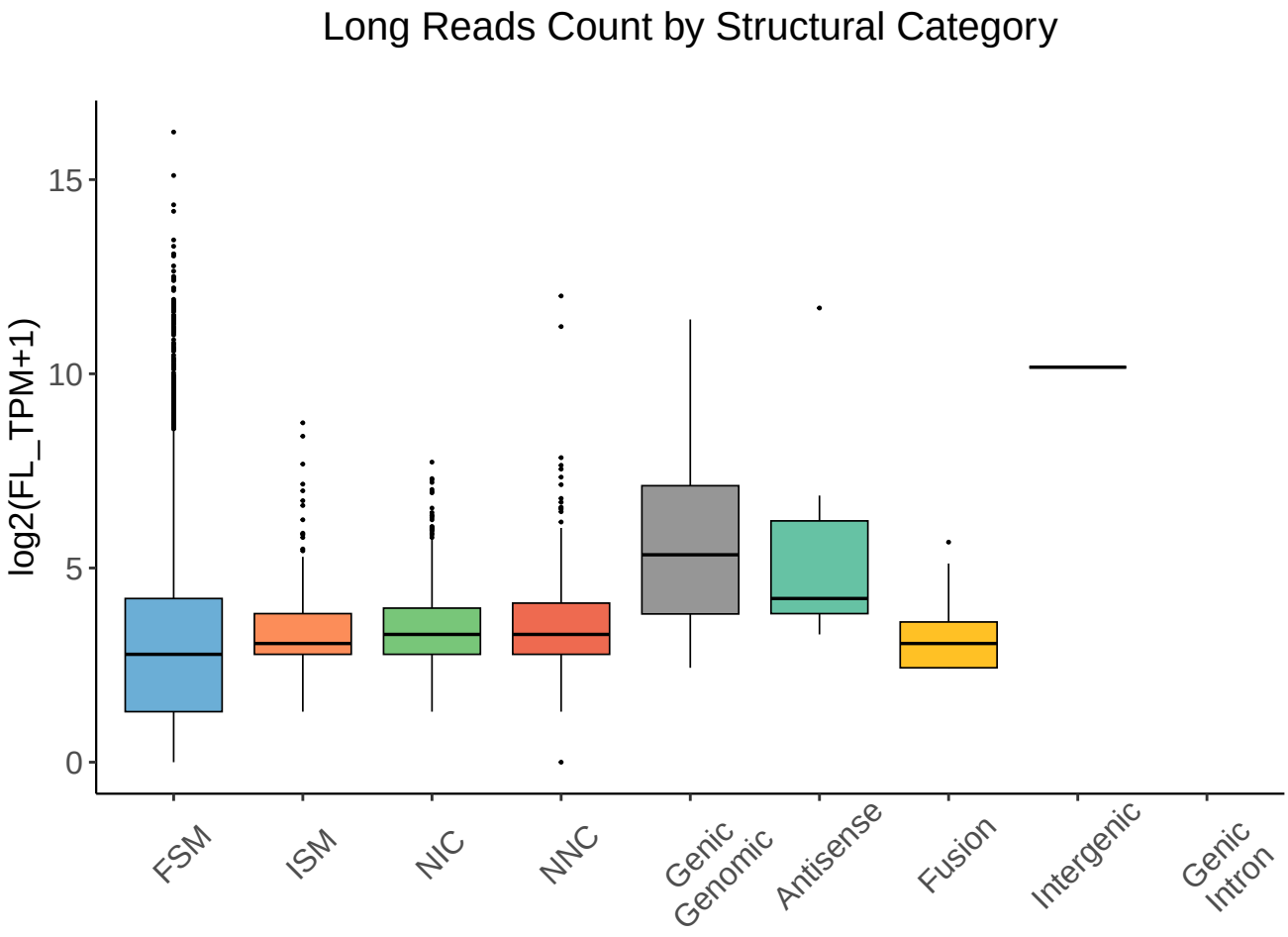

Isoform Distribution Across Structural Categories
Transcripts, %
100%
75%
50%
25%
0%
FSM
ISM
NIC
NNC
Supported Transcript Model
Fully supported
Not fully supported

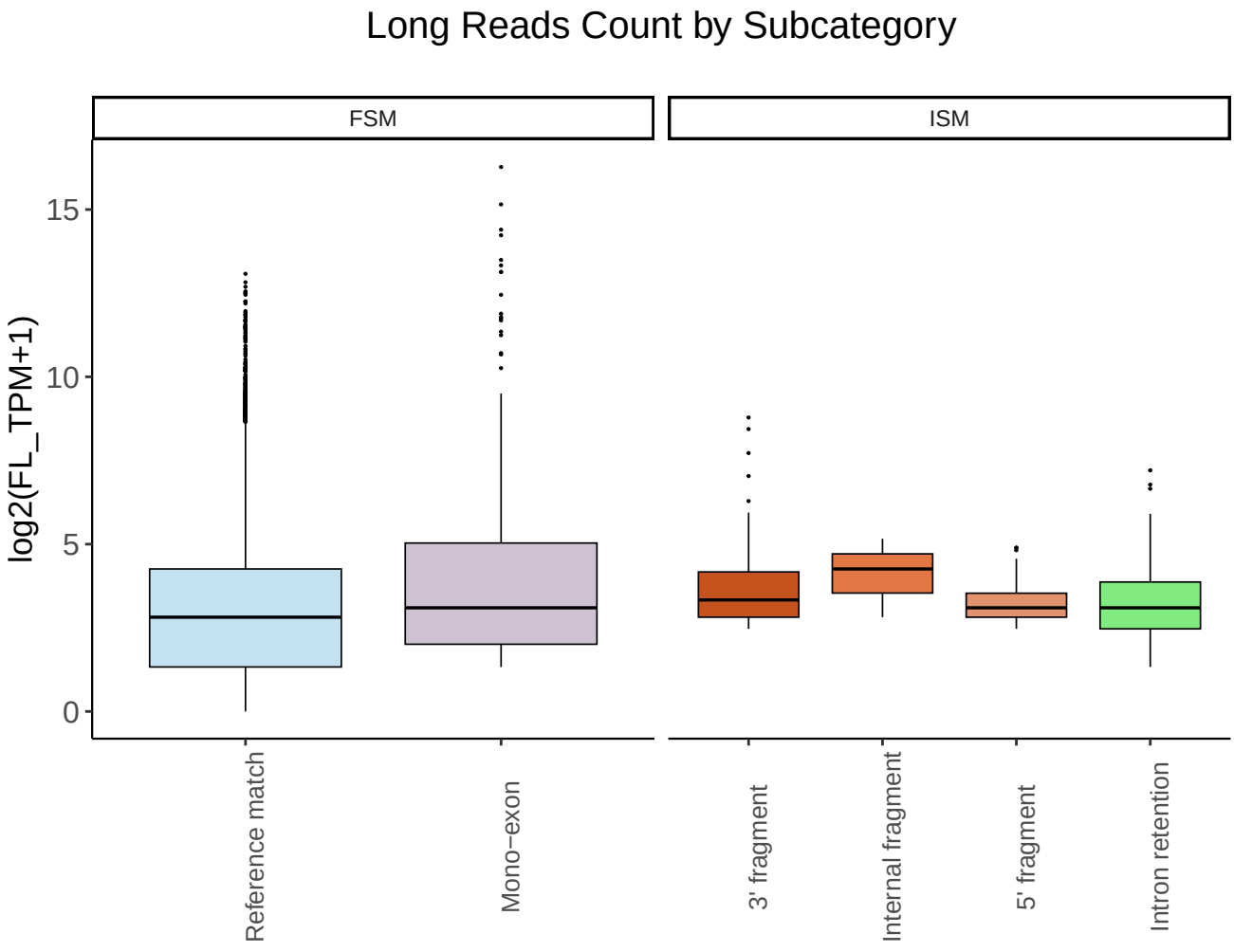

Isoform Distribution Across Structural Subcategories
FSM and ISM
FSM
ISM
Transcripts, count
25000
20000
15000
10000
5000
0
Reference match
Mono−exon
3' fragment
Internal fragment
5' fragment
Intron retention
Supported Transcript Model
Fully supported
Not fully supported

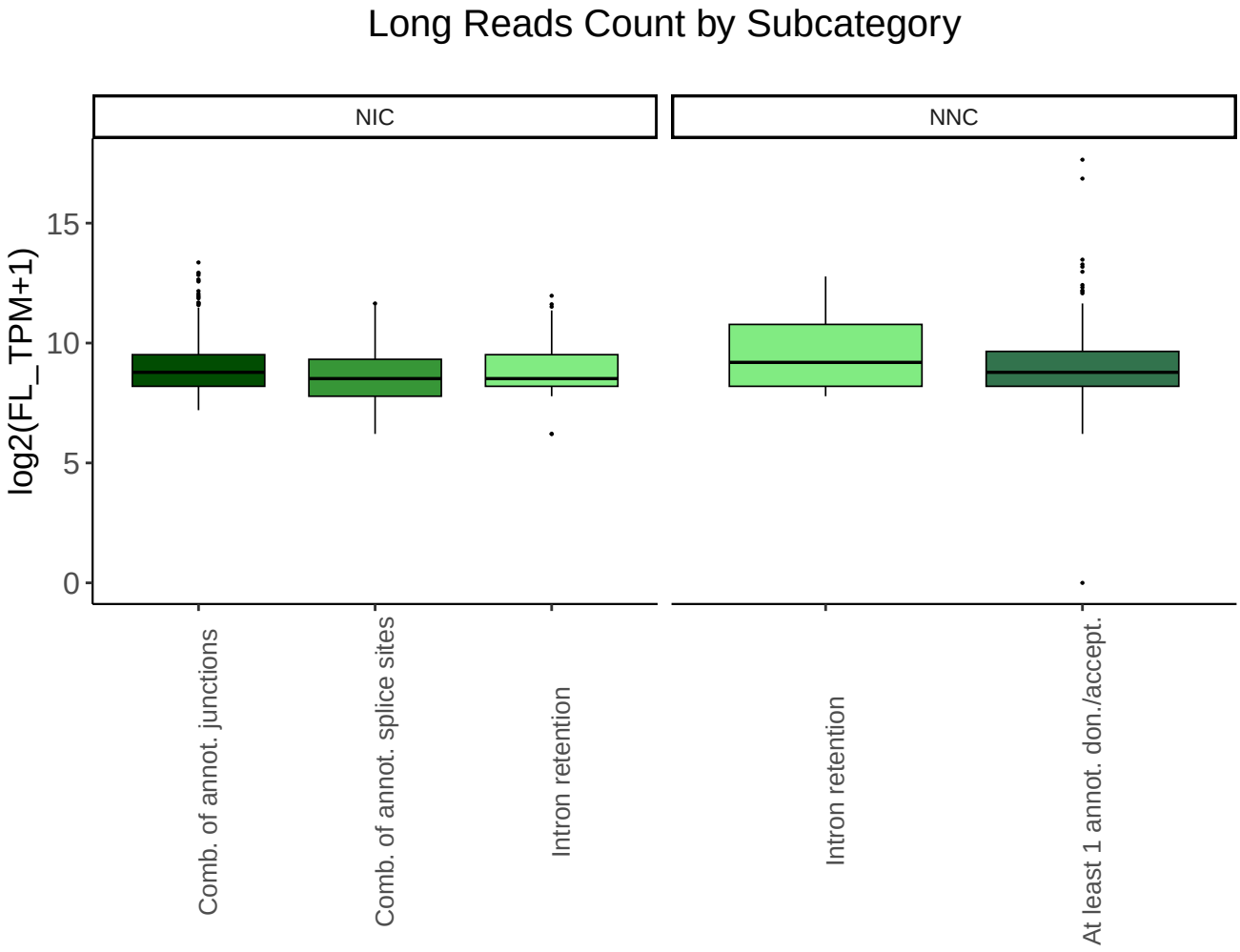

Isoform Distribution Across Structural Subcategories
FSM and ISM
FSM
ISM
100%
75%
50%
25%
0%
Transcripts, %
Reference match
Mono−exon
3' fragment
Internal fragment
5' fragment
Intron retention
Supported Transcript Model
Fully supported
Not fully supported

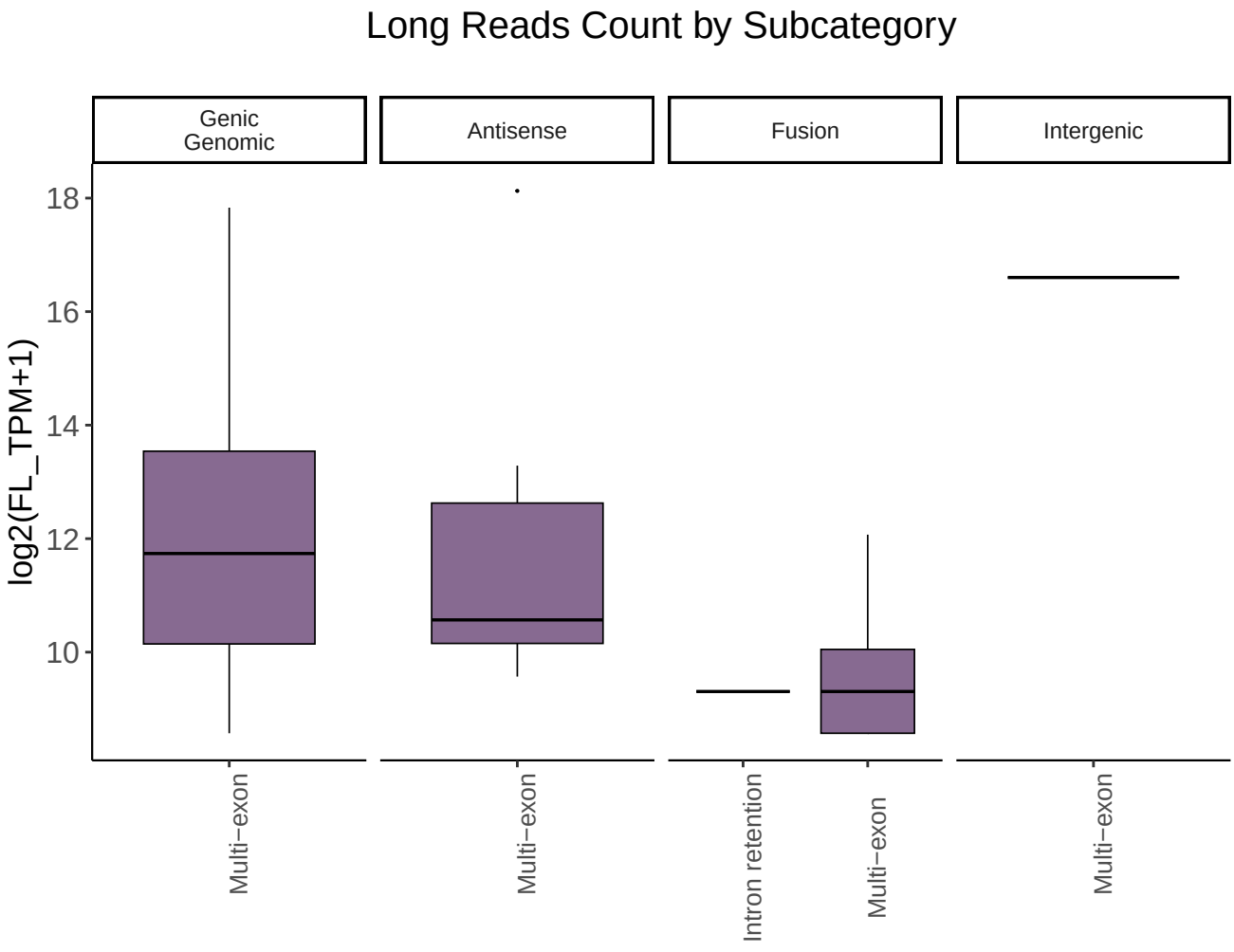

Isoform Distribution Across Structural Subcategories
NIC and NNC
NIC
NNC
Transcripts, count
500
400
300
200
100
0
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.
Supported Transcript Model
Not fully supported

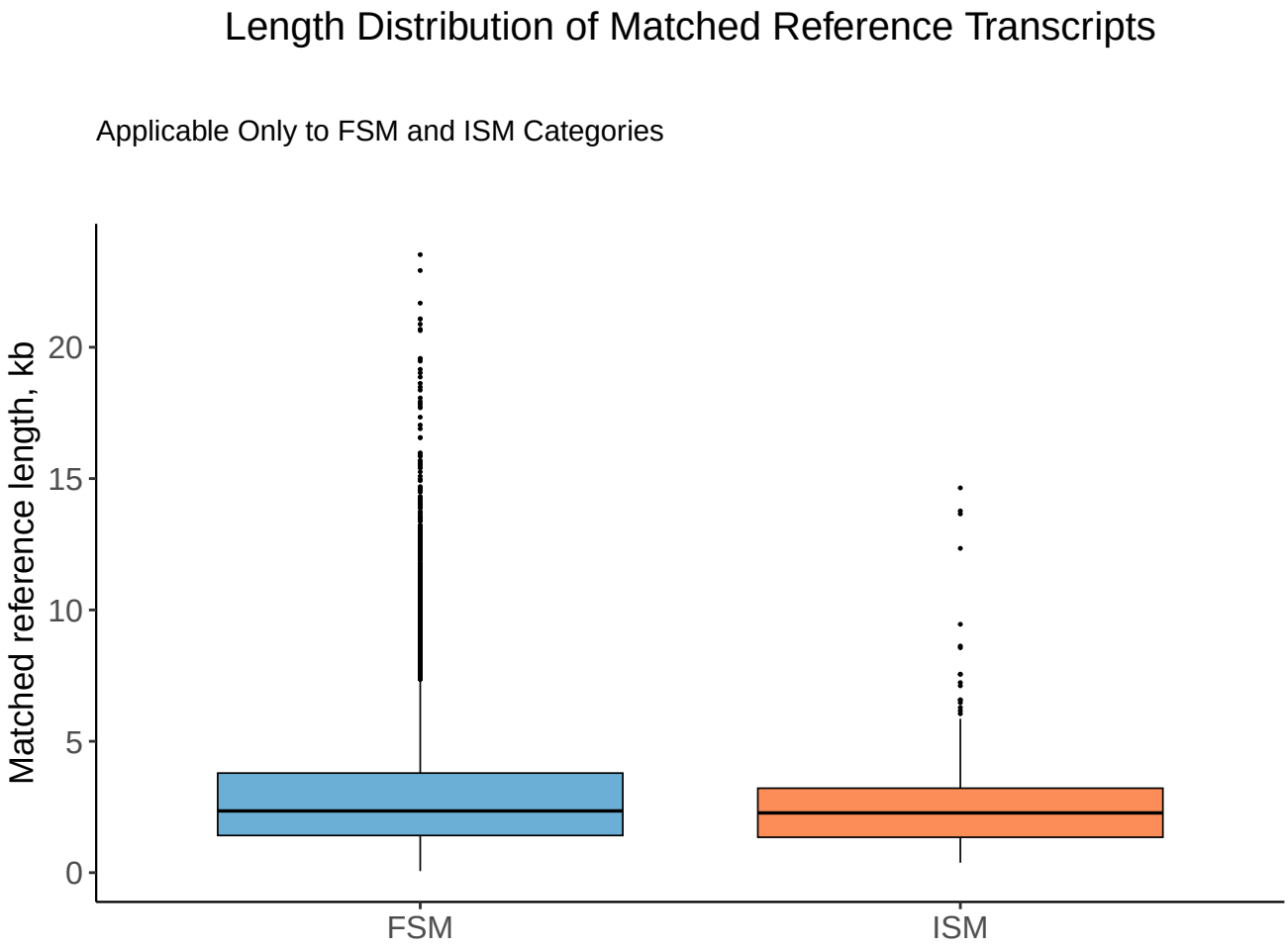

Isoform Distribution Across Structural Subcategories
NIC and NNC
NIC
NNC
Transcripts, %
100%
75%
50%
25%
0%
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.
Supported Transcript Model
Not fully supported

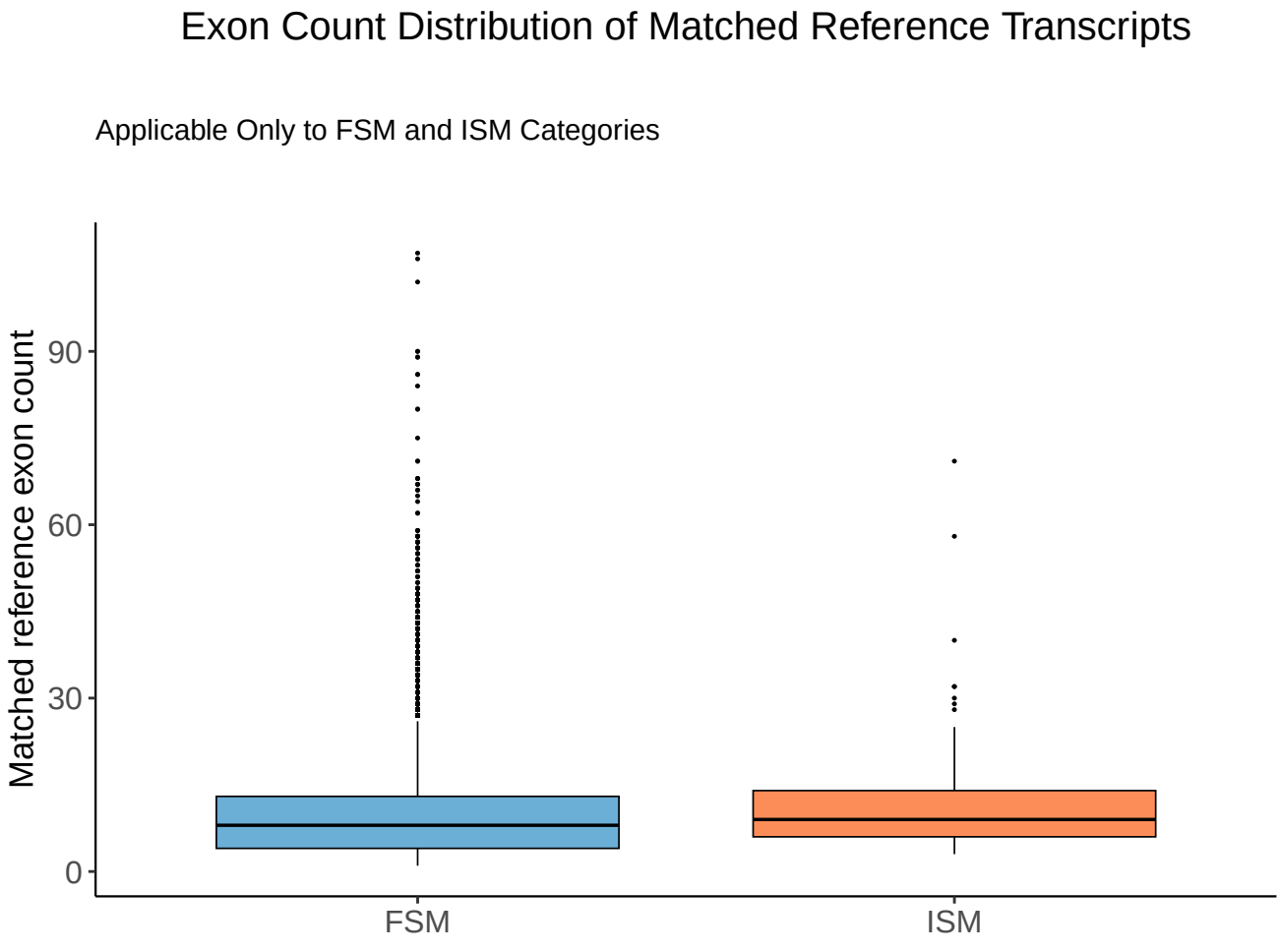

Long Reads Count by Structural Category
log2(FL_TPM+1)
15
10
5
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron


Long Reads Count by Subcategory
FSM
ISM
log2(FL_TPM+1)
15
10
5
0
Reference match
Mono−exon
3' fragment
Internal fragment
5' fragment
Intron retention

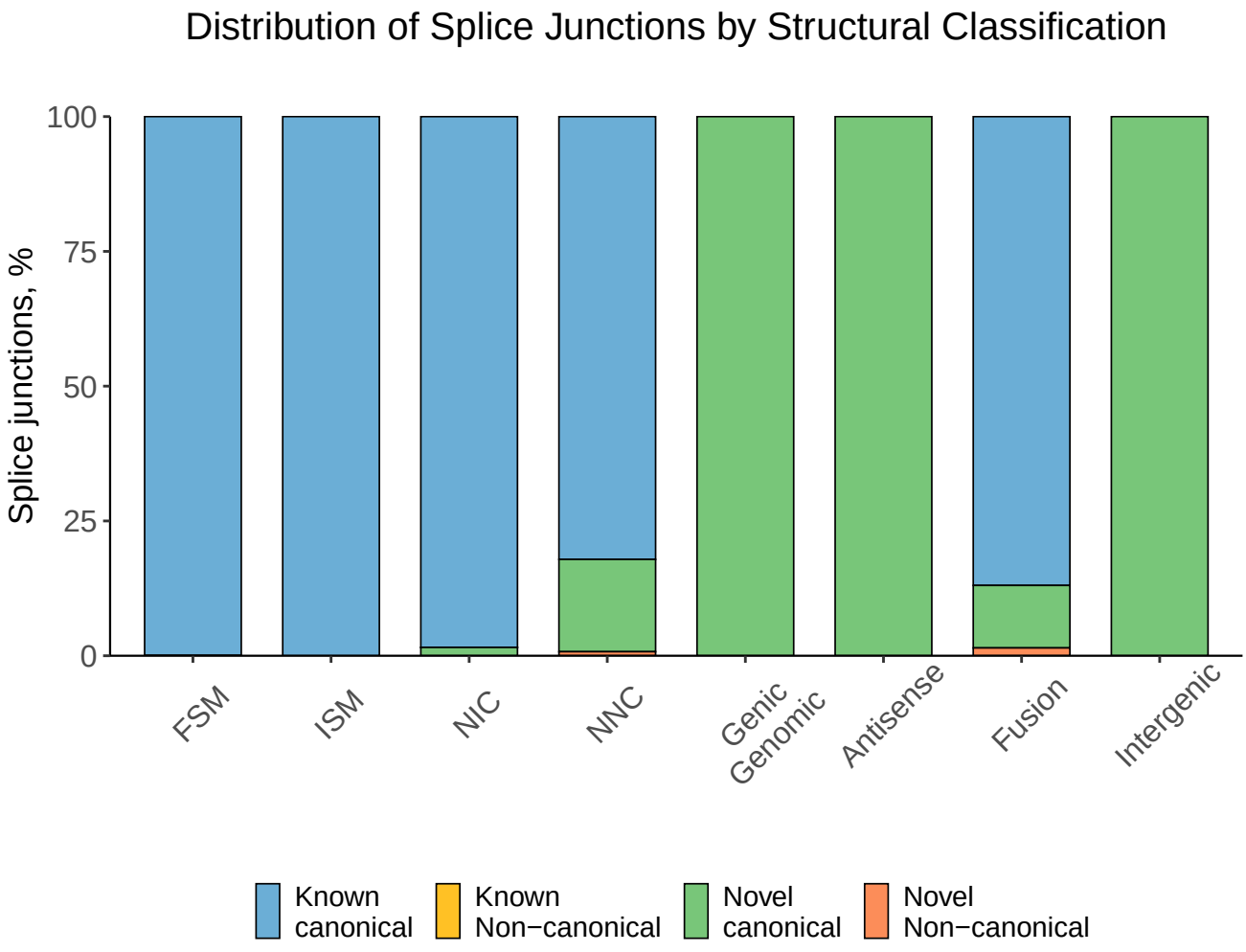

Long Reads Count by Subcategory
NIC
NNC
log2(FL_TPM+1)
15
10
5
0
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.

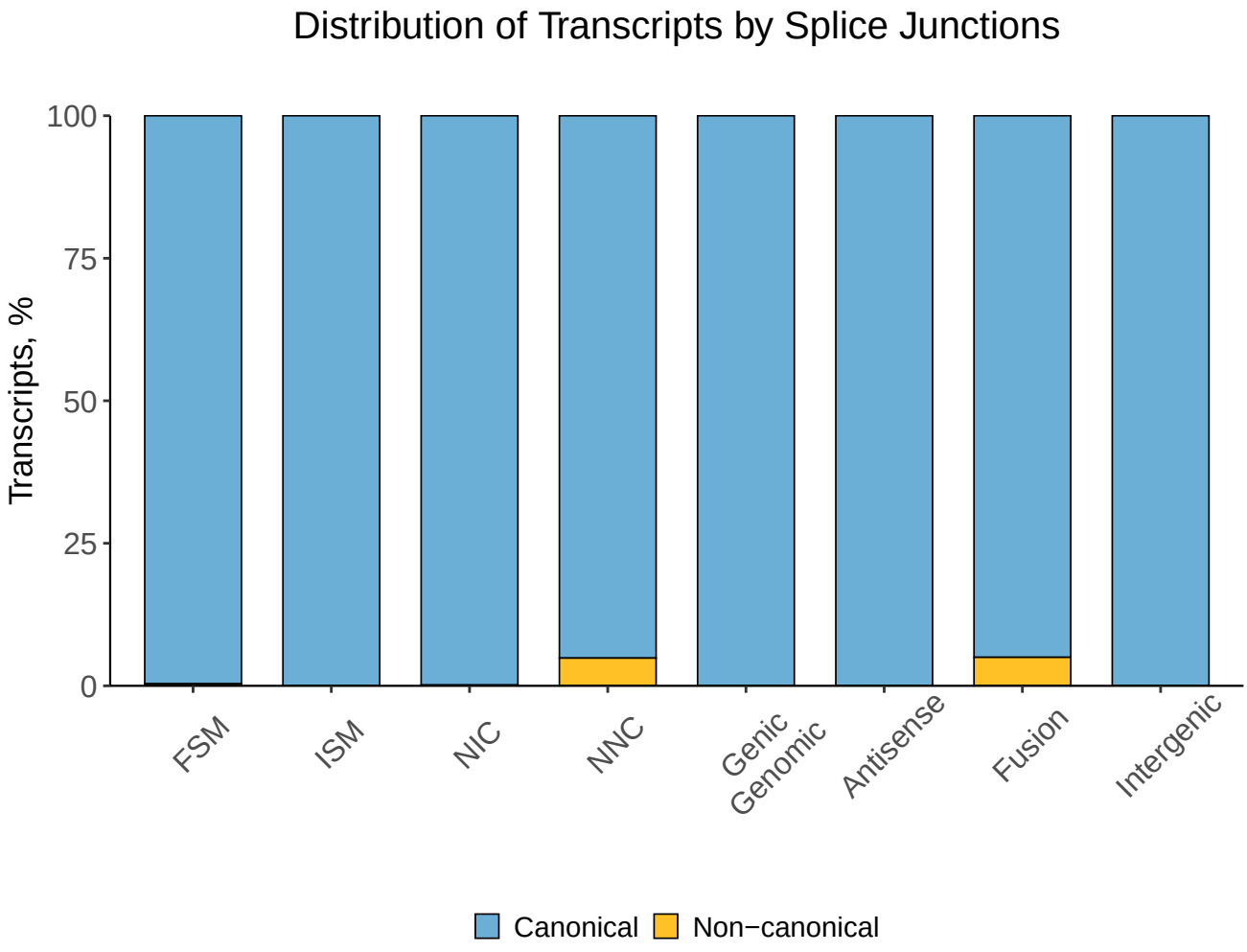

Long Reads Count by Subcategory
Genic Genomic
Antisense
Fusion
Intergenic
log2(FL_TPM+1)
18
16
14
12
10
Multi−exon
Multi−exon
Intron retention
Multi−exon
Multi−exon

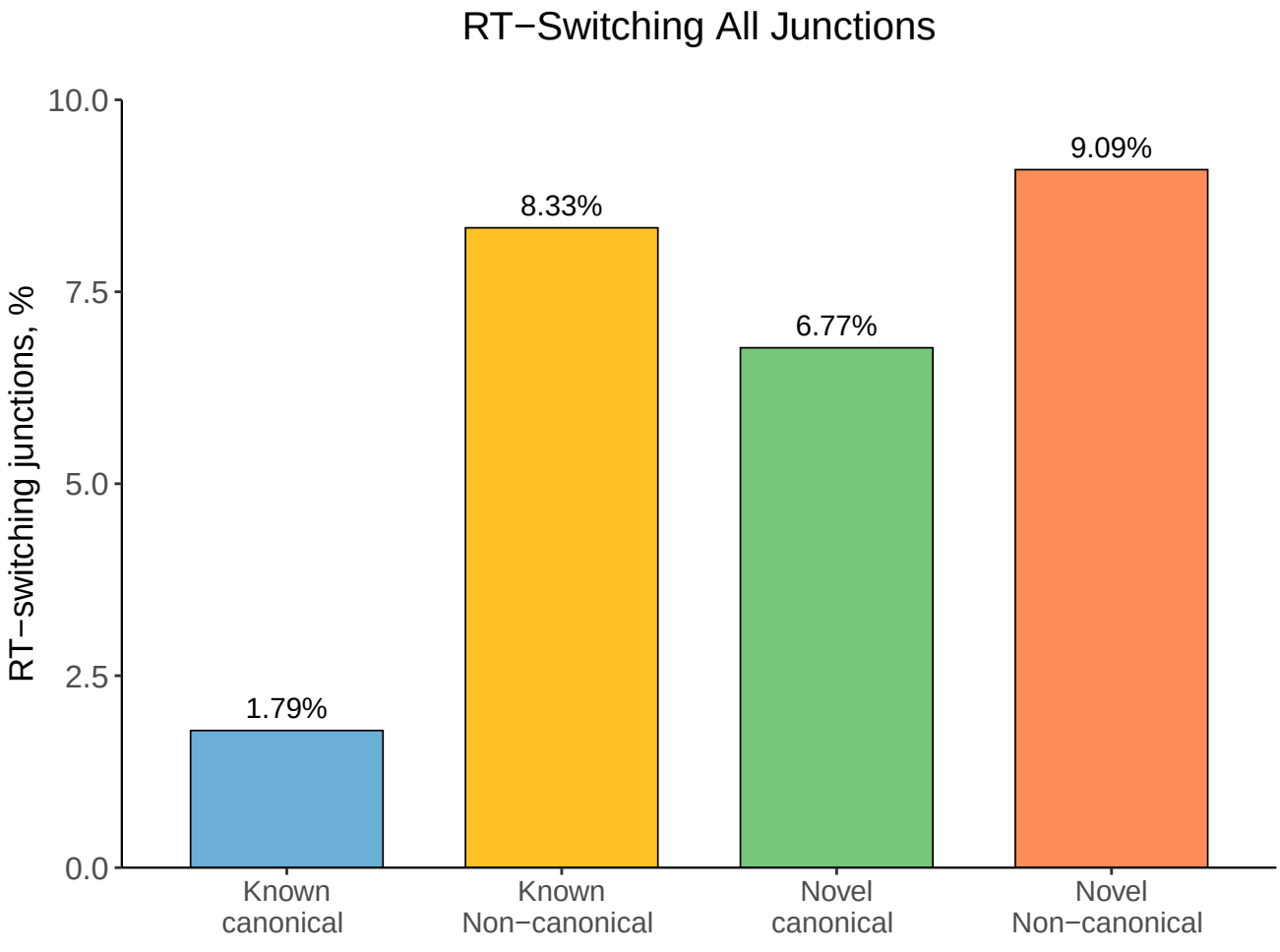

Length Distribution of Matched Reference Transcripts
Applicable Only to FSM and ISM Categories
Matched reference length, kb
0
5
10
15
20
FSM
ISM

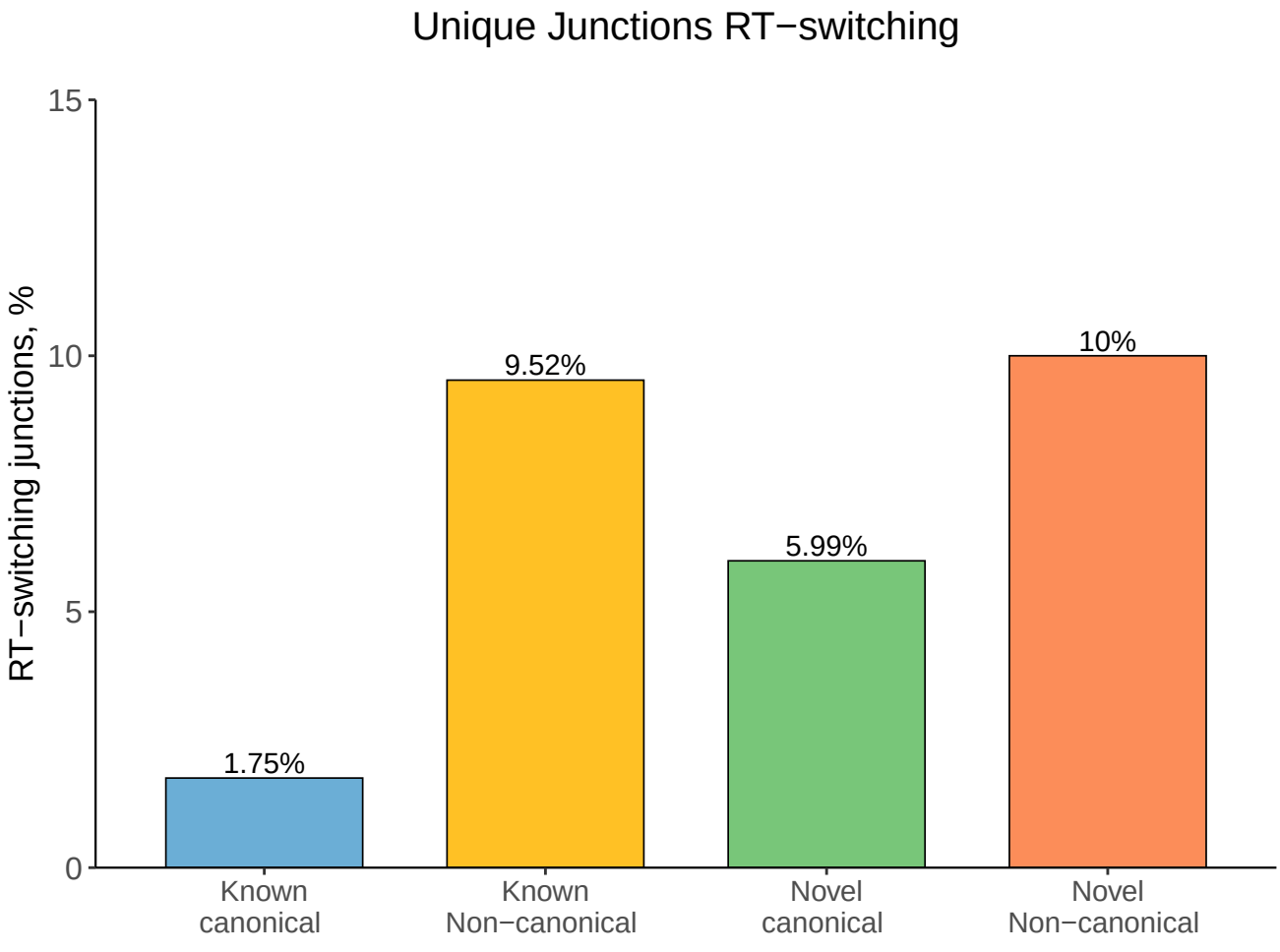

Exon Count Distribution of Matched Reference Transcripts
Applicable Only to FSM and ISM Categories
Matched reference exon count
0
30
60
90
FSM
ISM

Splice Junction Characterization

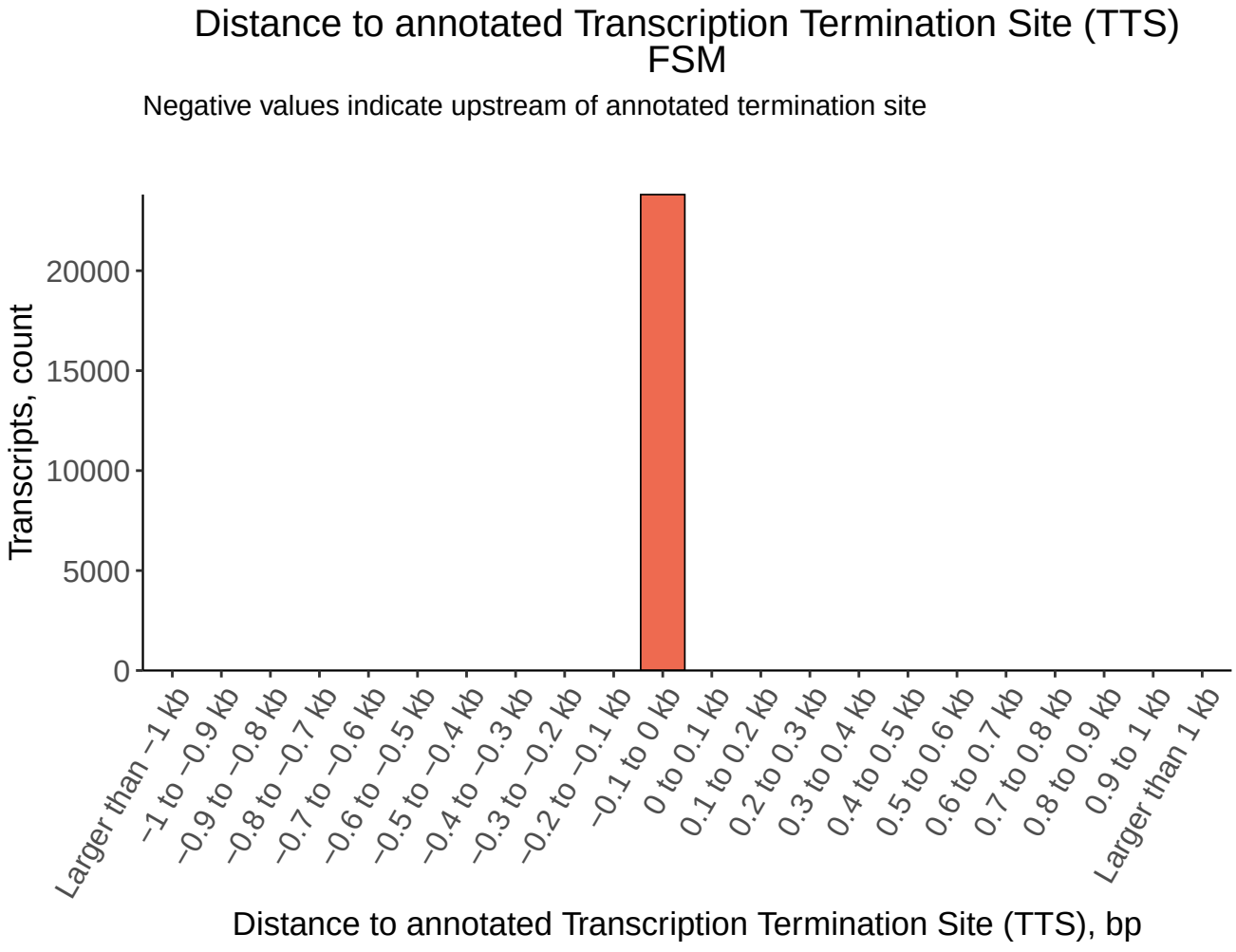

Distribution of Splice Junctions by Structural Classification
Splice junctions, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

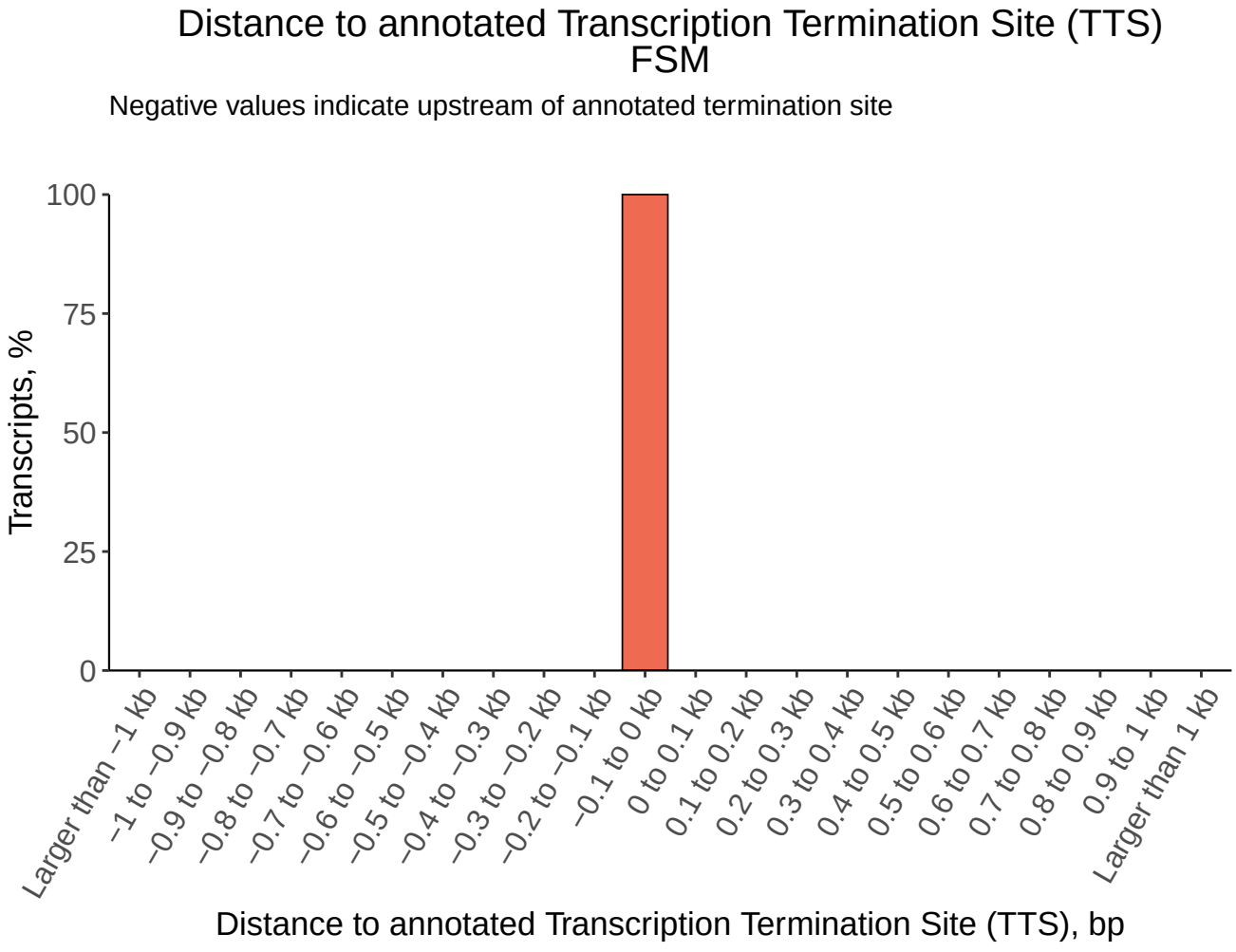

Distribution of Transcripts by Splice Junctions
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Canonical
Non−canonical

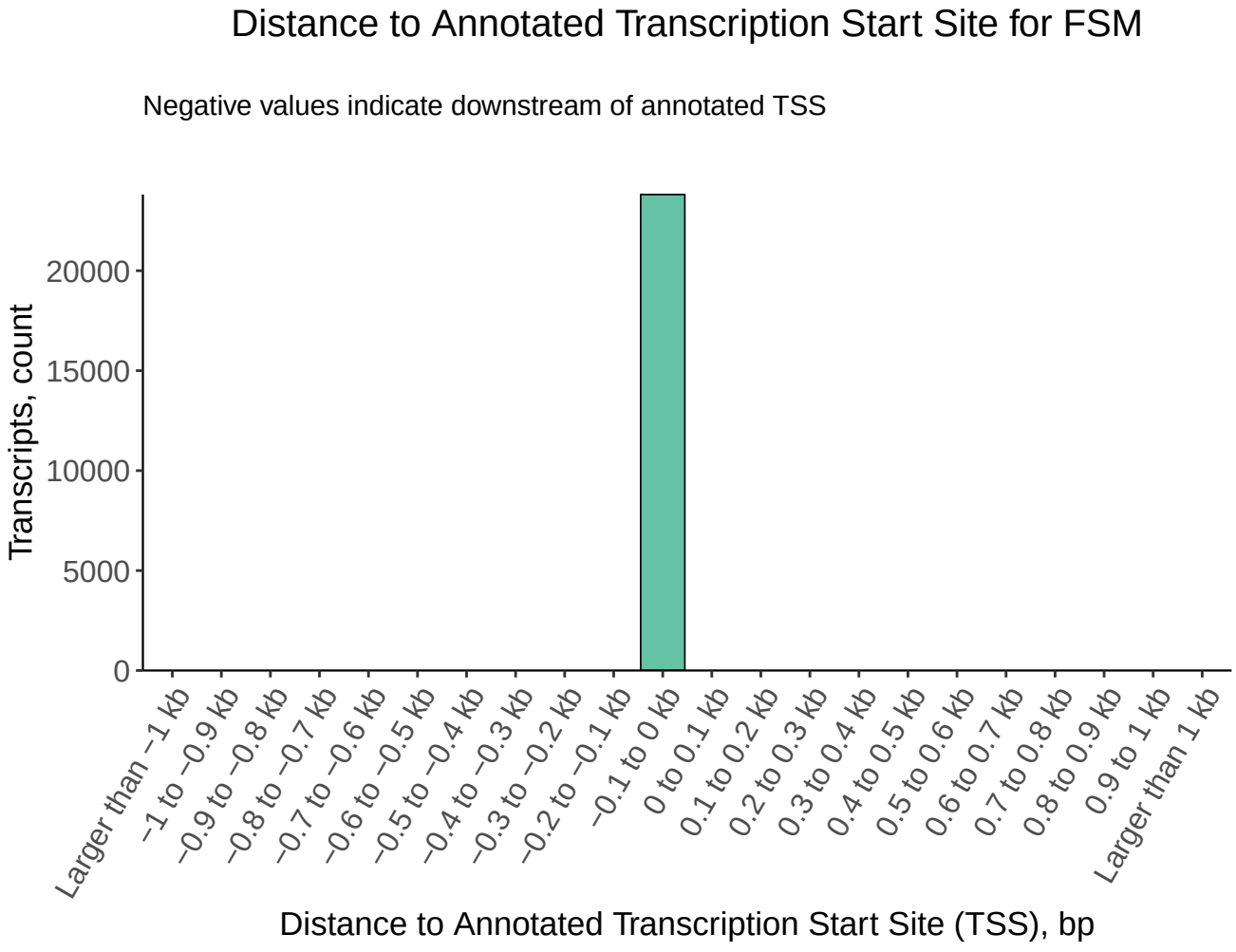

RT−Switching All Junctions
RT−switching junctions, %
10.0
7.5
5.0
2.5
0.0
1.79%
8.33%
6.77%
9.09%
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

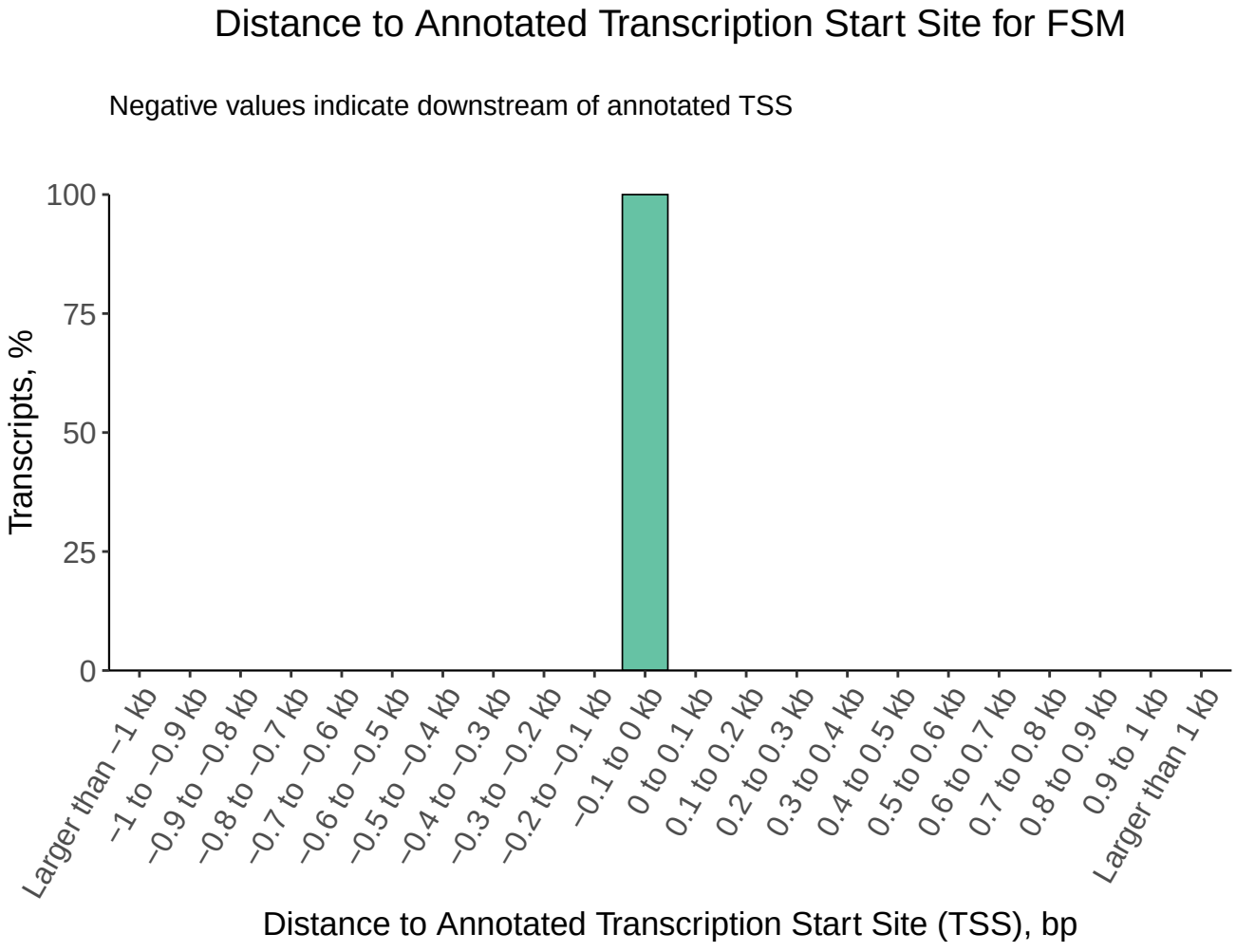

Unique Junctions RT−switching
RT−switching junctions, %
15
10
5
0
1.75%
9.52%
5.99%
10%
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

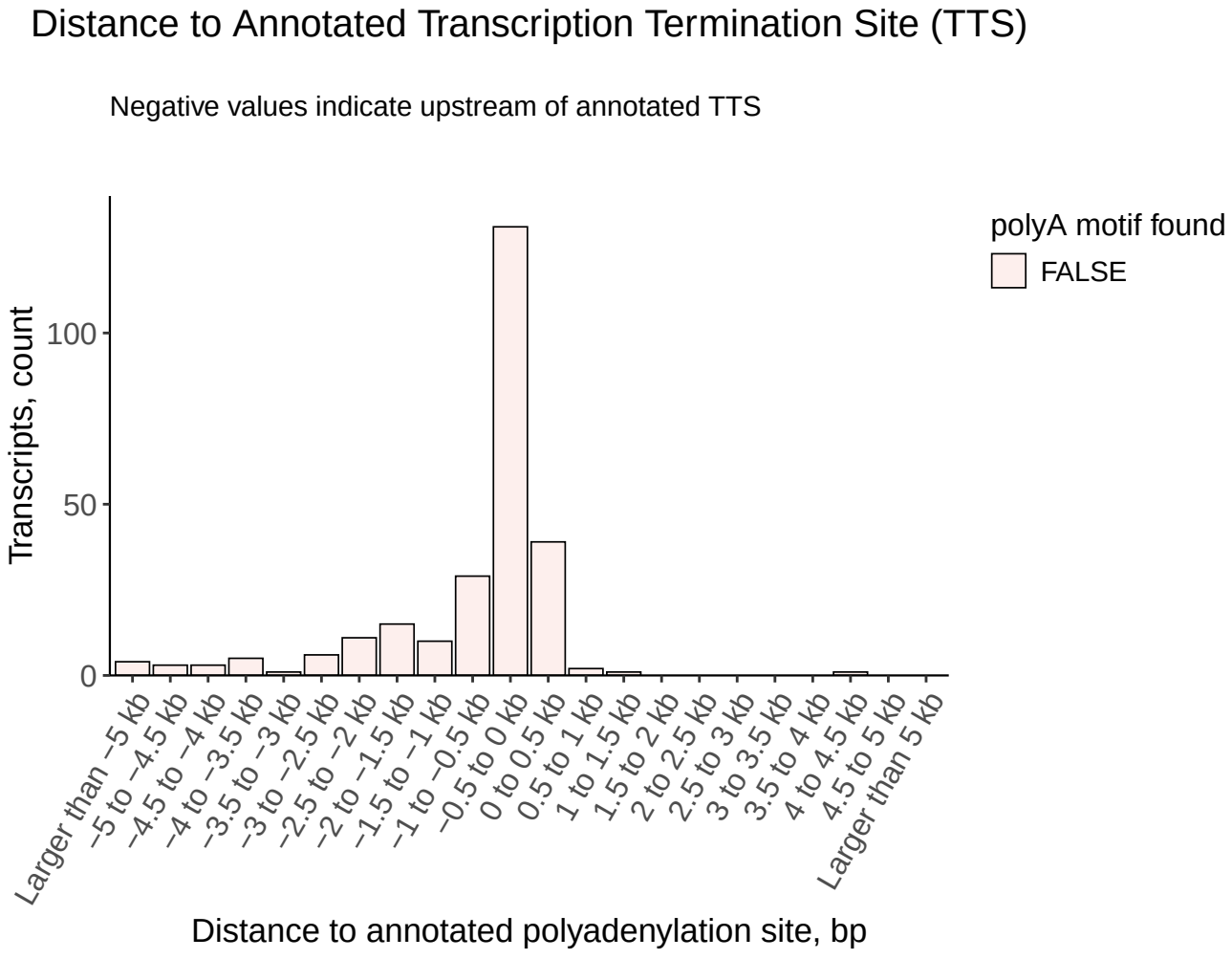
Comparison With Annotated TSS and TTS

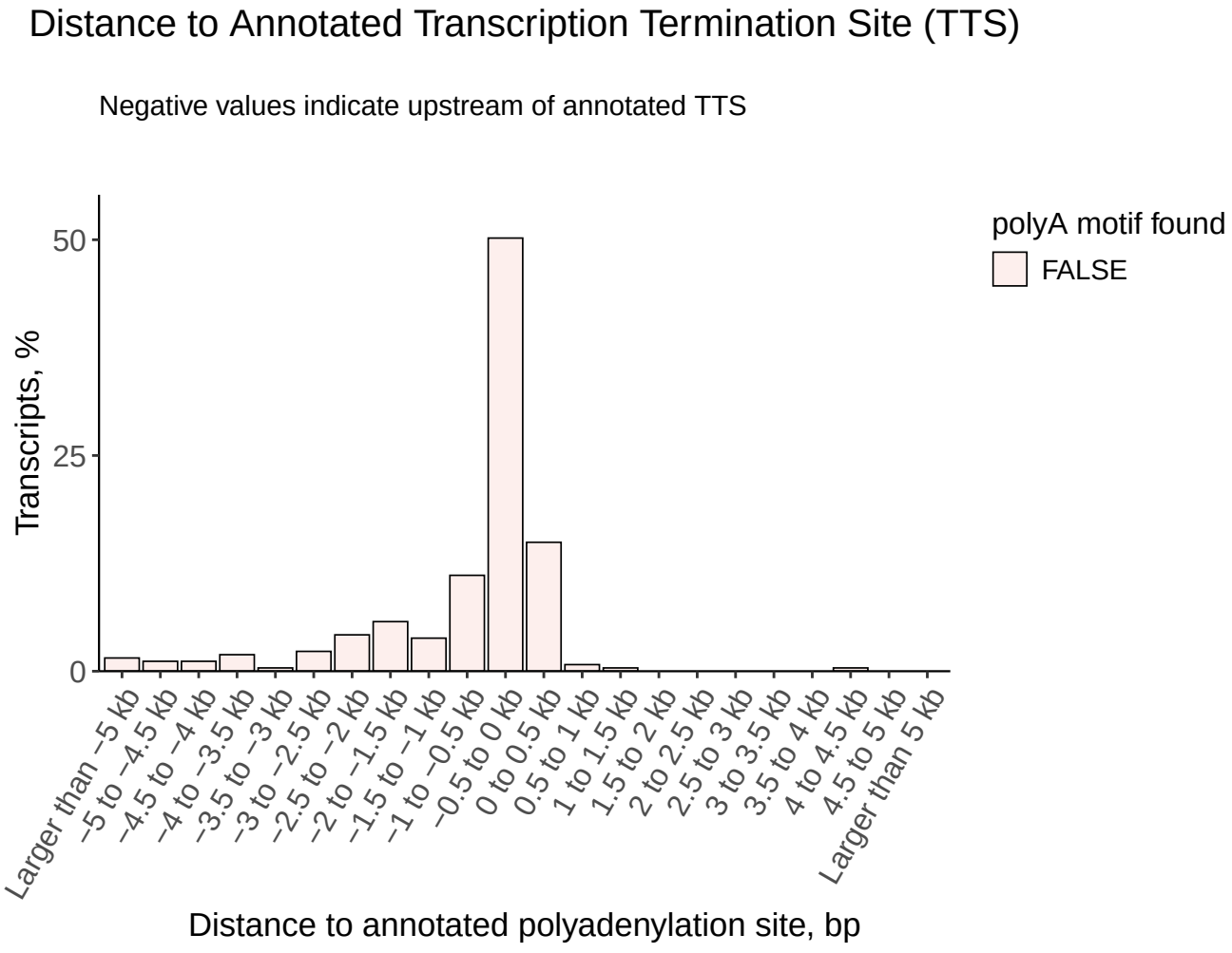

Distance to annotated Transcription Termination Site (TTS)
FSM
Negative values indicate upstream of annotated termination site
Transcripts, count
0
5000
10000
15000
20000
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to annotated Transcription Termination Site (TTS), bp

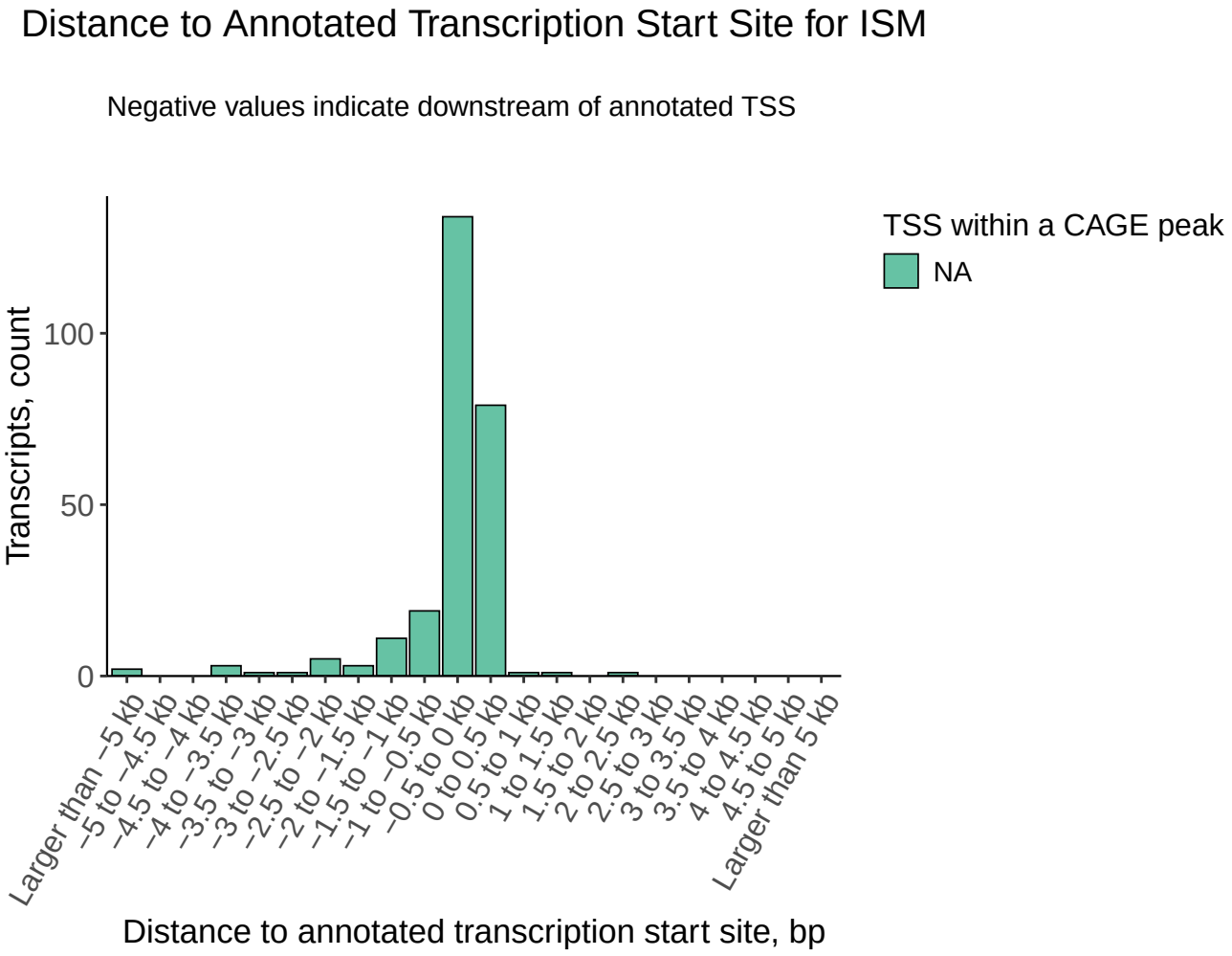

Distance to annotated Transcription Termination Site (TTS)
FSM
Negative values indicate upstream of annotated termination site
Transcripts, %
100
75
50
25
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to annotated Transcription Termination Site (TTS), bp

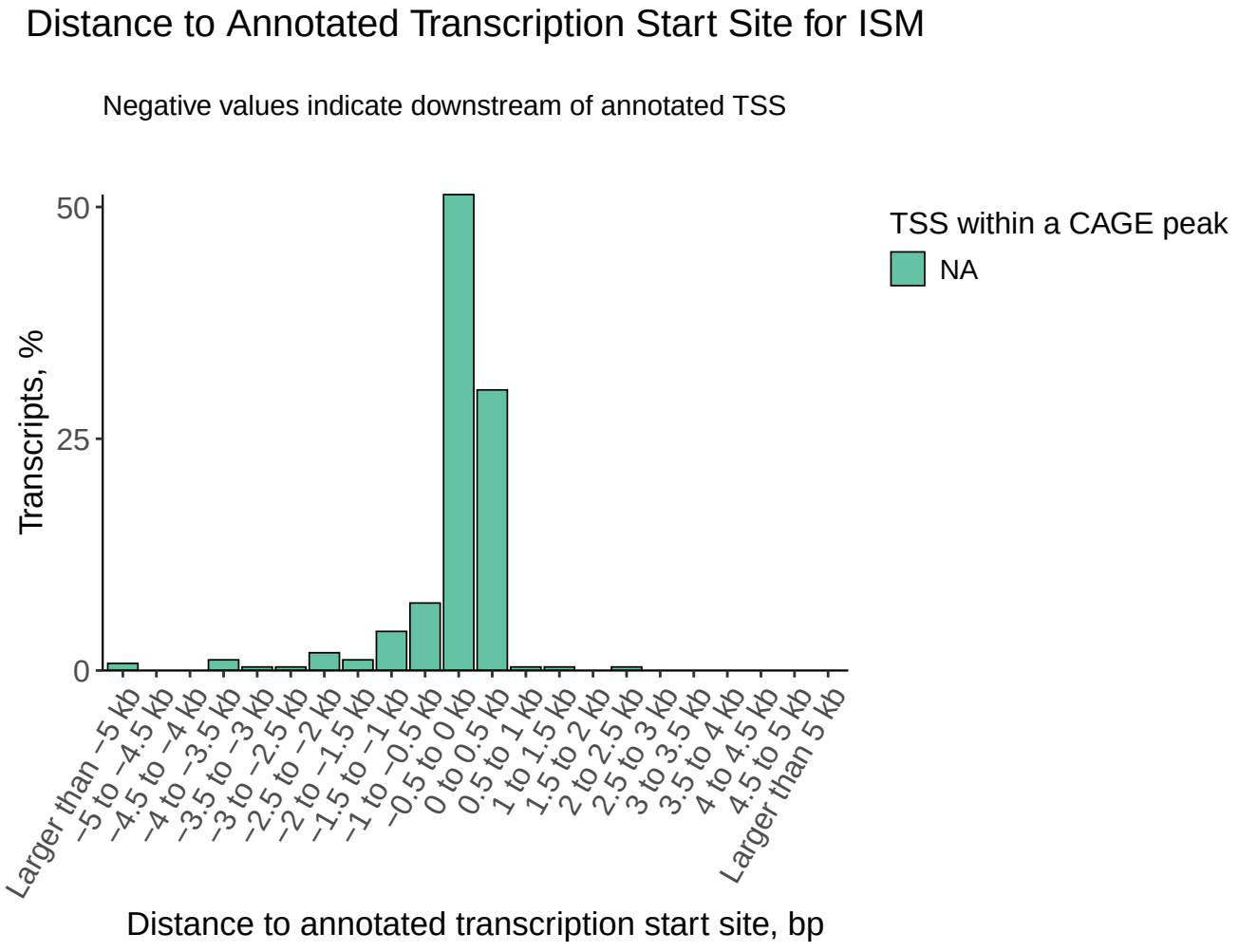

Distance to Annotated Transcription Start Site for FSM
Negative values indicate downstream of annotated TSS
Transcripts, count
20000
15000
10000
5000
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to Annotated Transcription Start Site (TSS), bp


Distance to Annotated Transcription Start Site for FSM
Negative values indicate downstream of annotated TSS
Transcripts, %
100
75
50
25
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to Annotated Transcription Start Site (TSS), bp


Distance to Annotated Transcription Termination Site (TTS)
Negative values indicate upstream of annotated TTS
polyA motif found
FALSE
Transcripts, count
0
50
100
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated polyadenylation site, bp


Distance to Annotated Transcription Termination Site (TTS)
Negative values indicate upstream of annotated TTS
Transcripts, %
50
25
0
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated polyadenylation site, bp
polyA motif found
FALSE

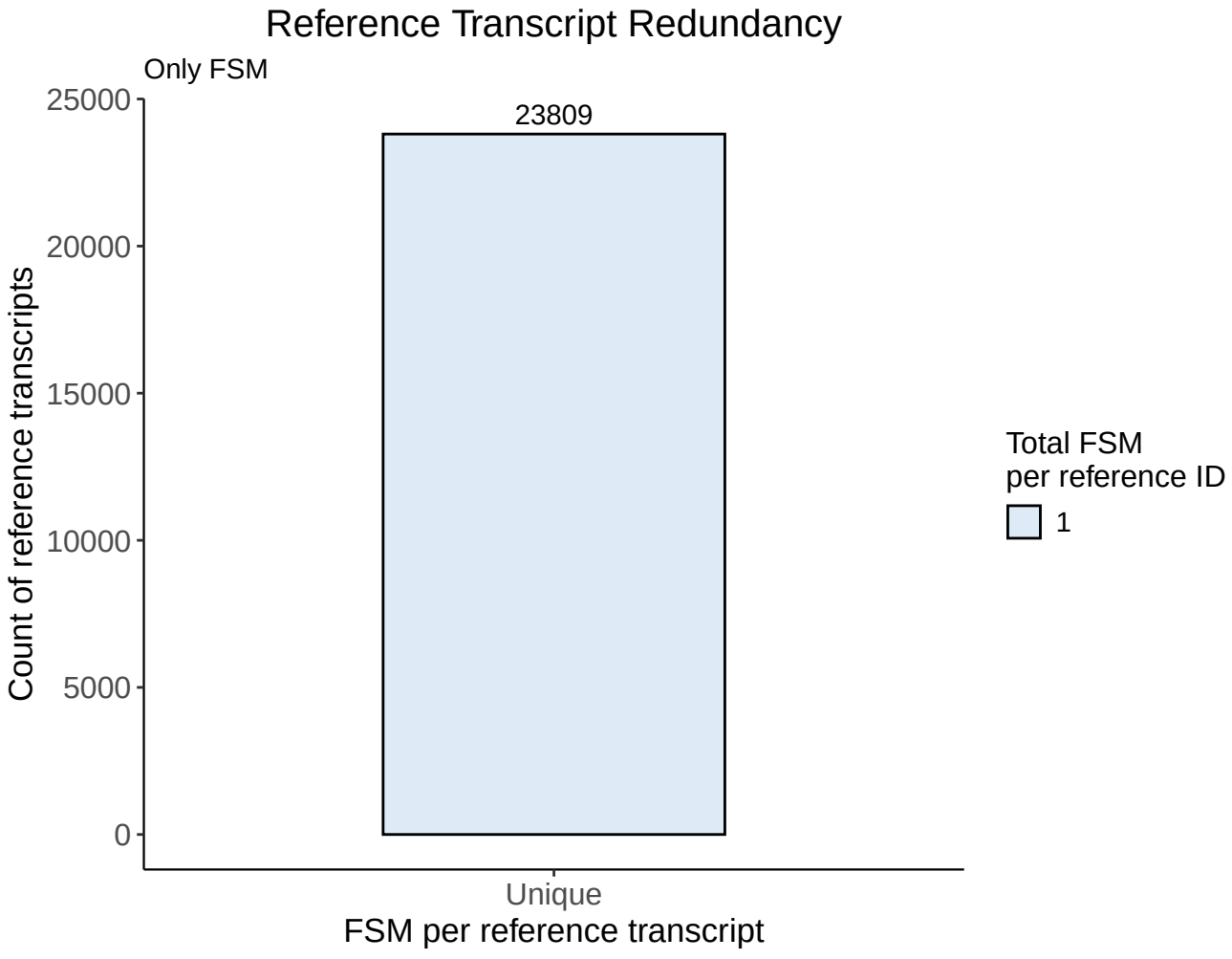

Distance to Annotated Transcription Start Site for ISM
Negative values indicate downstream of annotated TSS
Transcripts, count
0
50
100
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated transcription start site, bp
TSS within a CAGE peak
NA

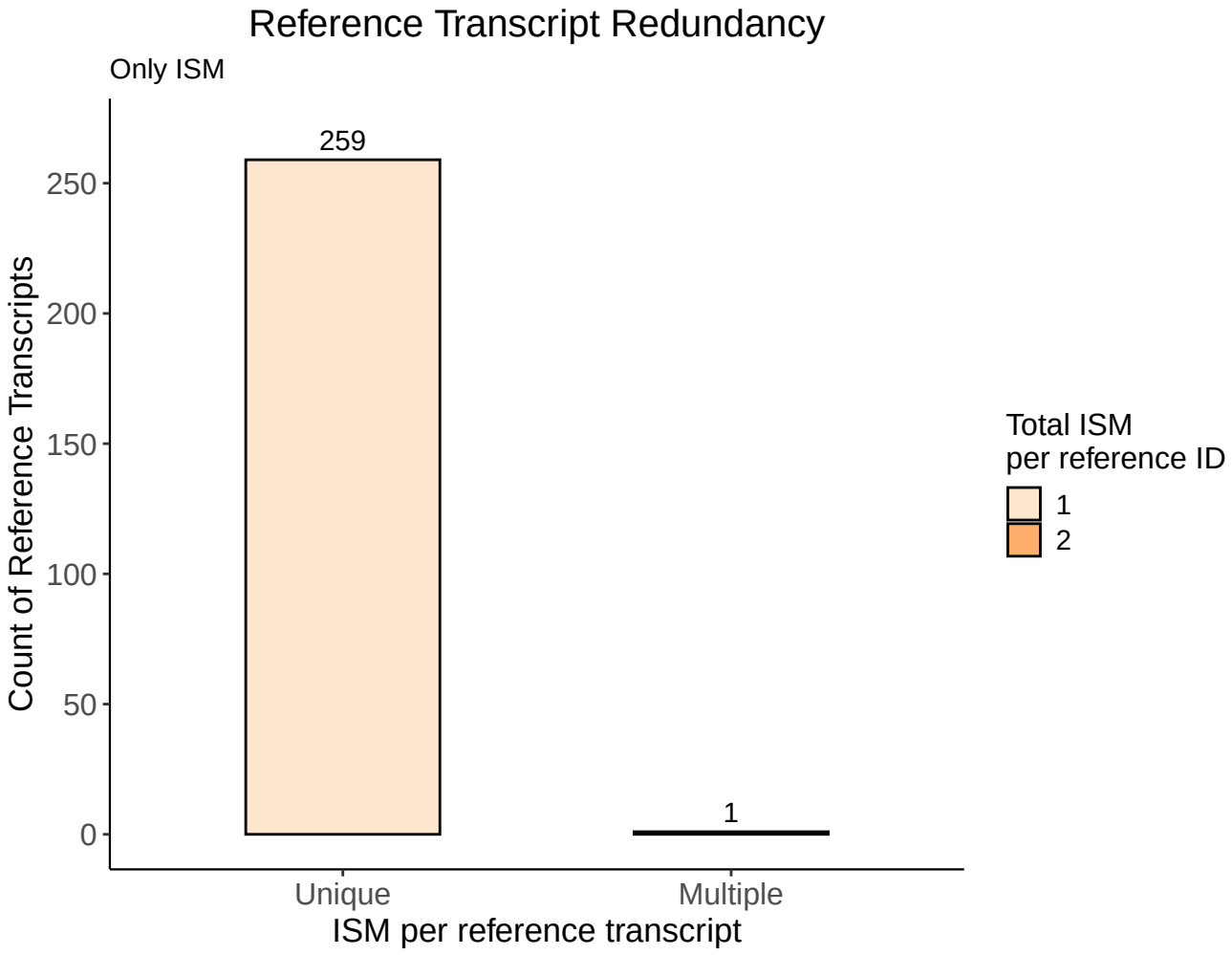

Distance to Annotated Transcription Start Site for ISM
Negative values indicate downstream of annotated TSS
Transcripts, %
50
25
0
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated transcription start site, bp
TSS within a CAGE peak
NA

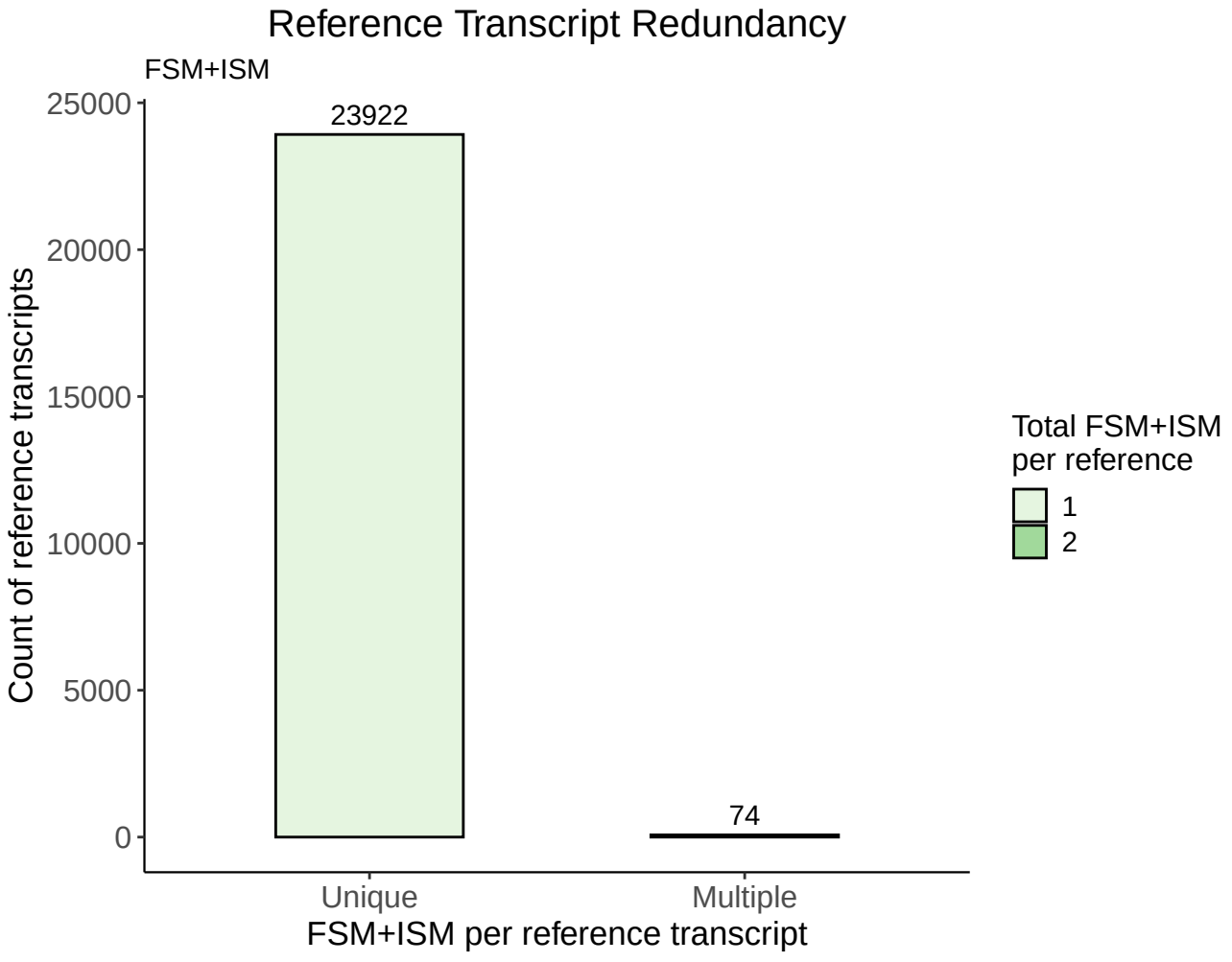
Comparison With Annotated TSS and TTS by Subcategories

PolyA Distance Analysis

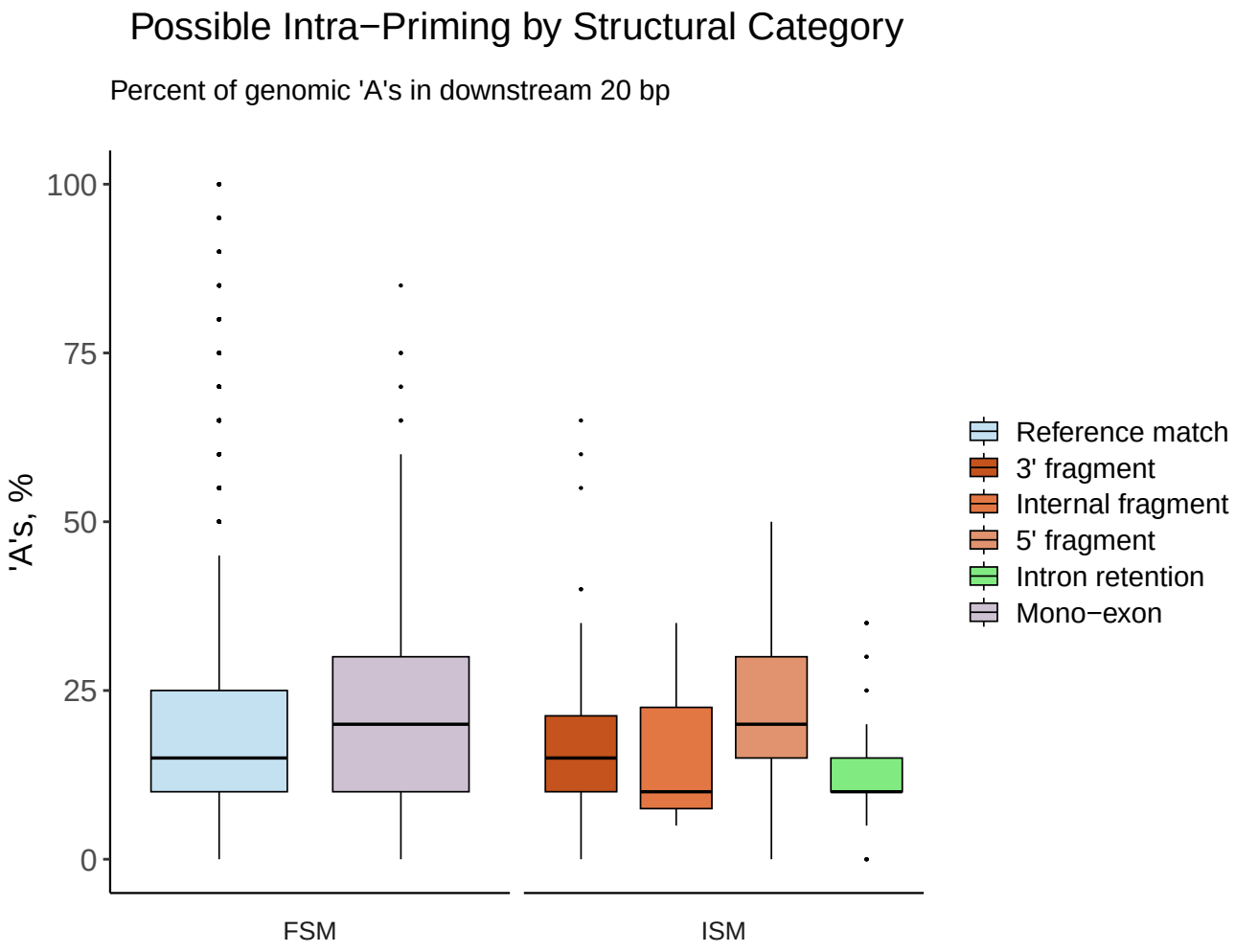
Redundancy Analysis

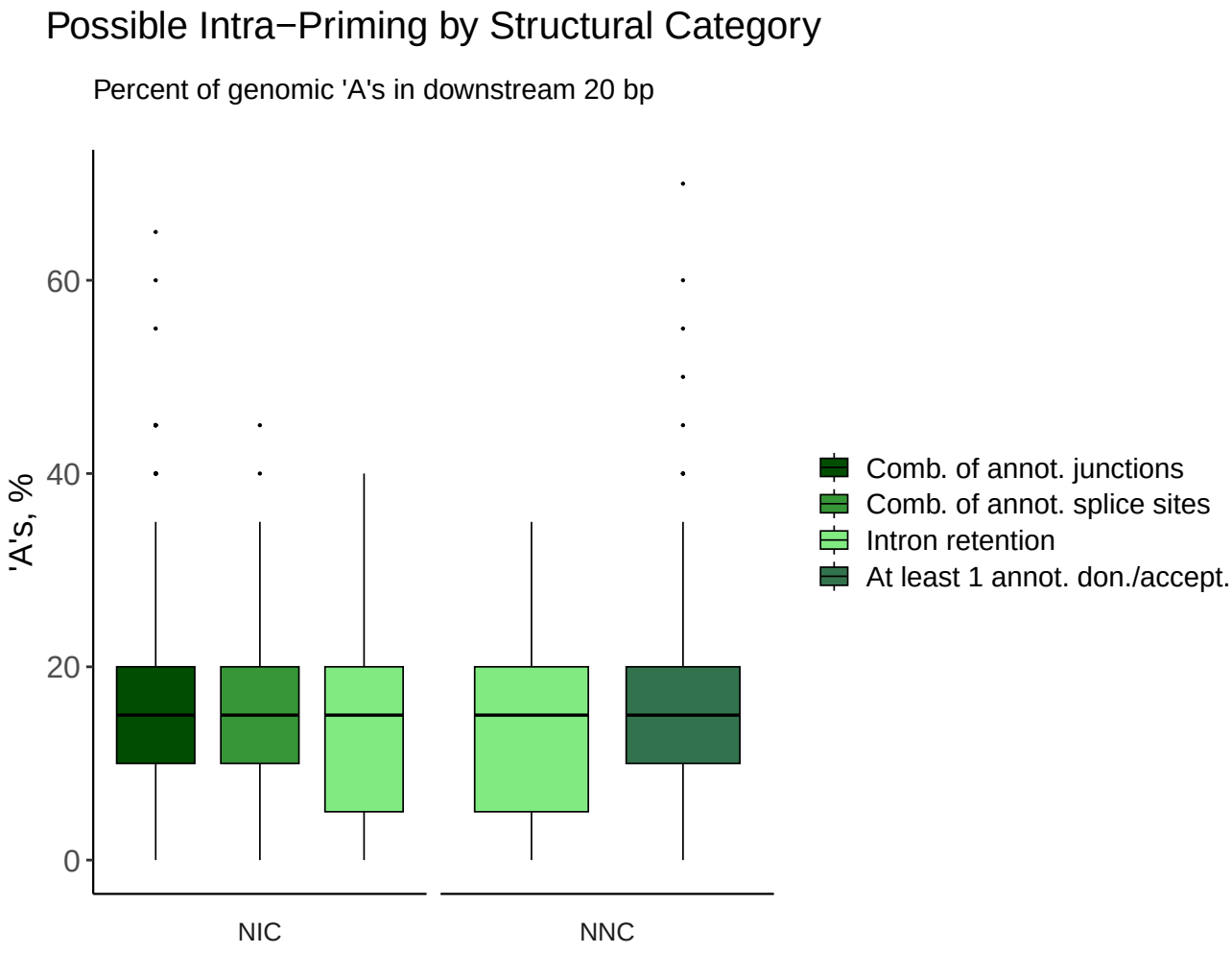

Reference Transcript Redundancy
Only FSM
23809
25000
20000
15000
10000
5000
0
Count of reference transcripts
Unique
FSM per reference transcript
Total FSM
per reference ID
1

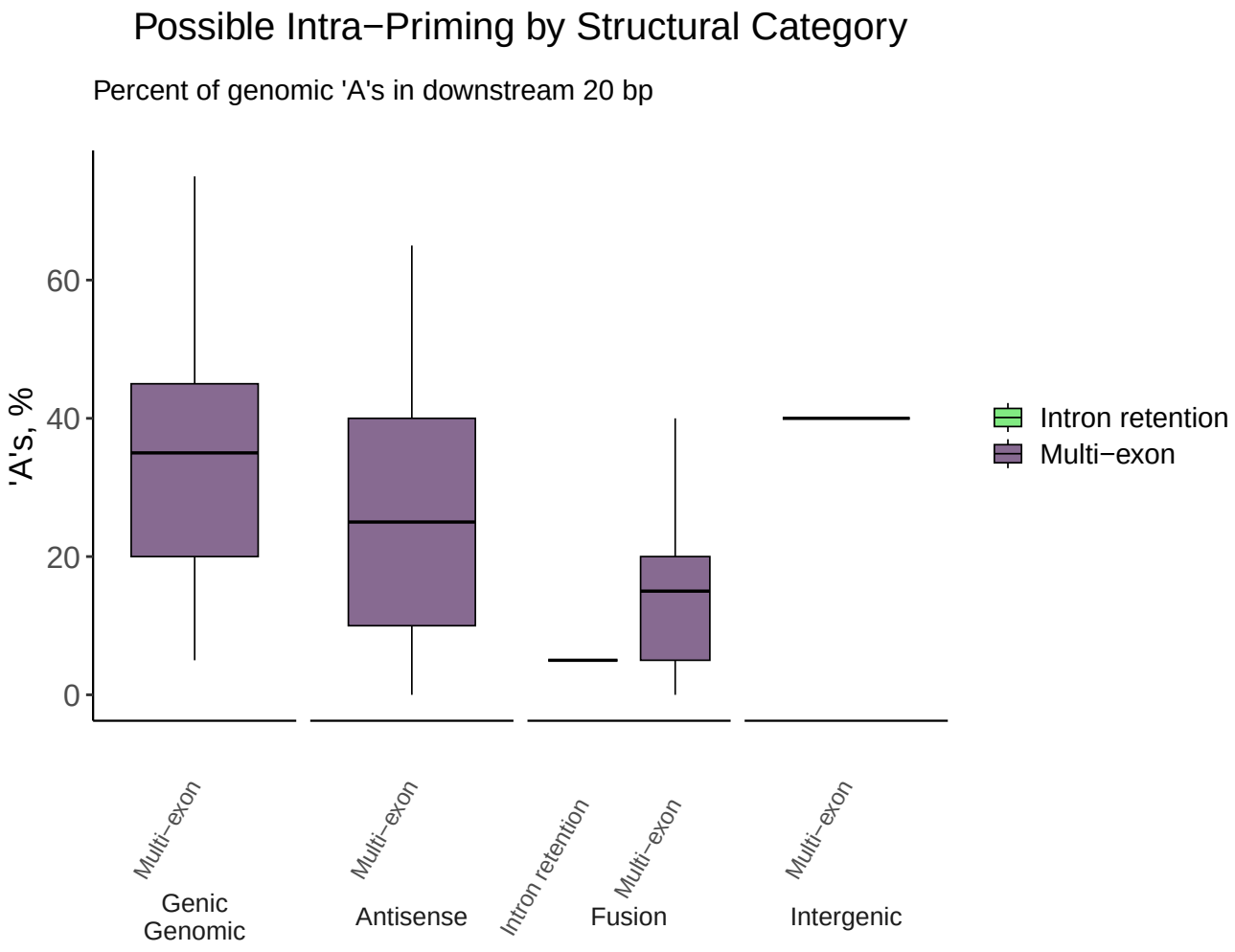

Reference Transcript Redundancy
Only ISM
259
1
250
200
150
100
50
0
Count of Reference Transcripts
Unique
Multiple
ISM per reference transcript
Total ISM
per reference ID
1
2

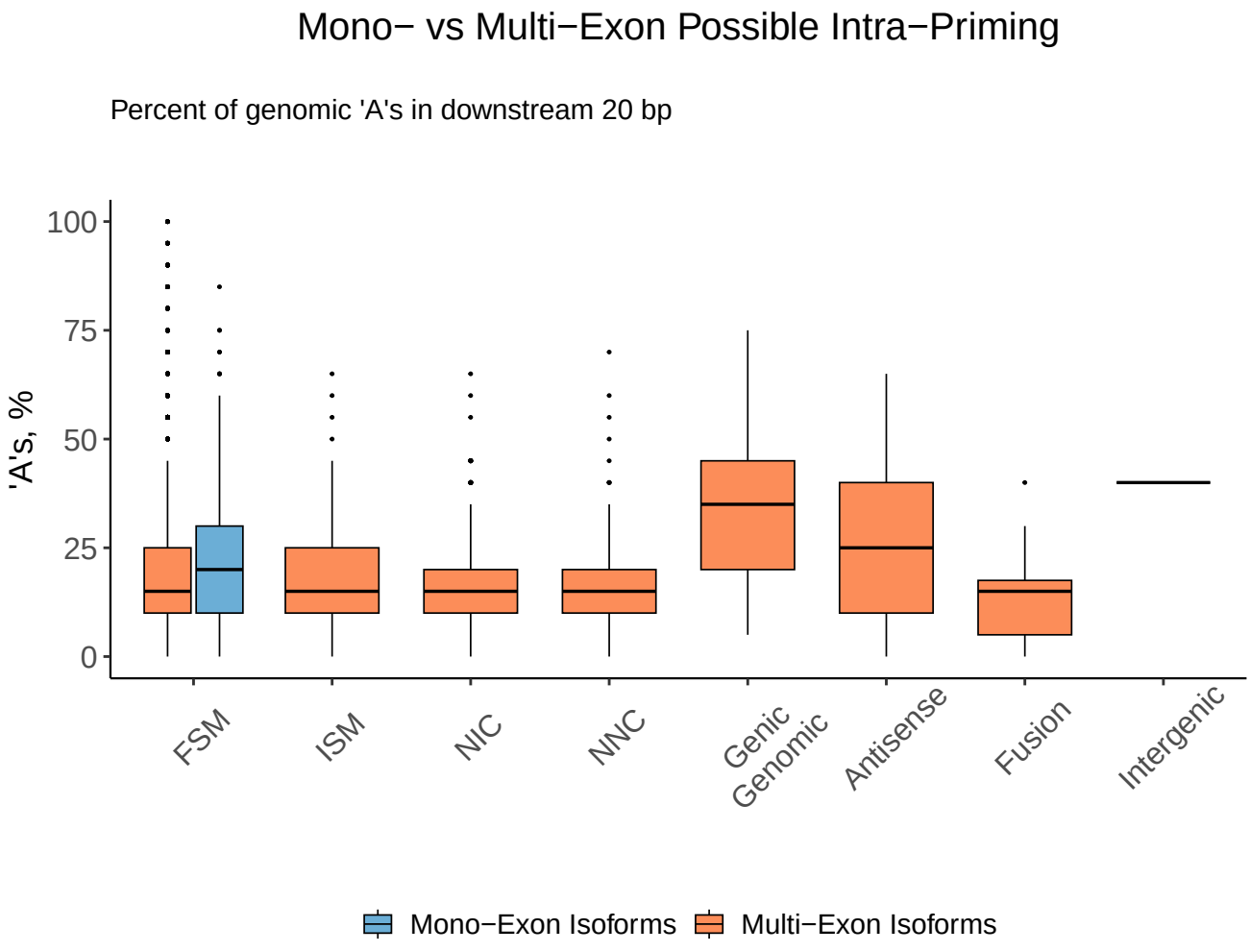

Reference Transcript Redundancy
FSM+ISM
25000
20000
15000
10000
5000
0
23922
74
Count of reference transcripts
Unique
Multiple
FSM+ISM per reference transcript
Total FSM+ISM per reference
1
2

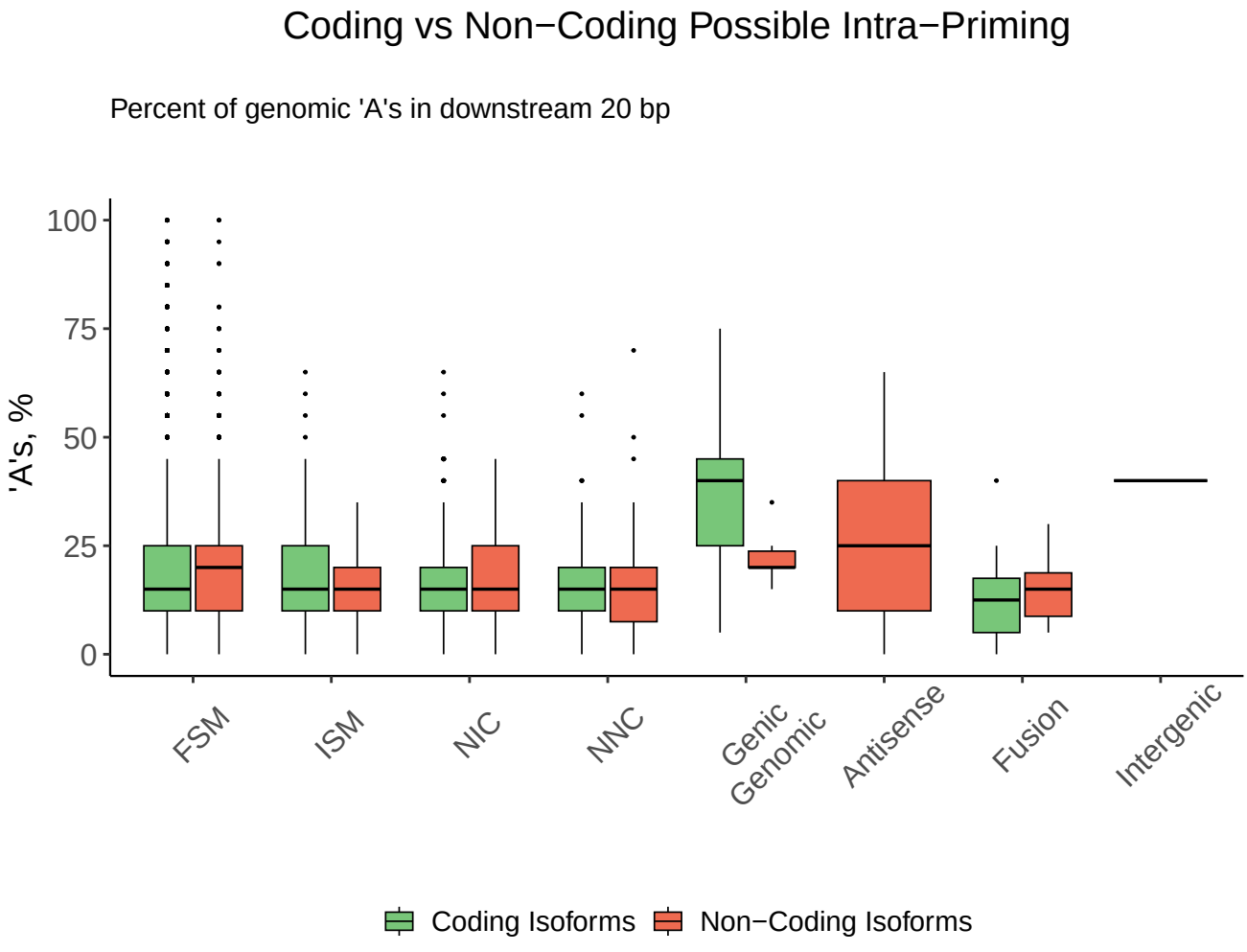
Intra−Priming Quality Check


Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon

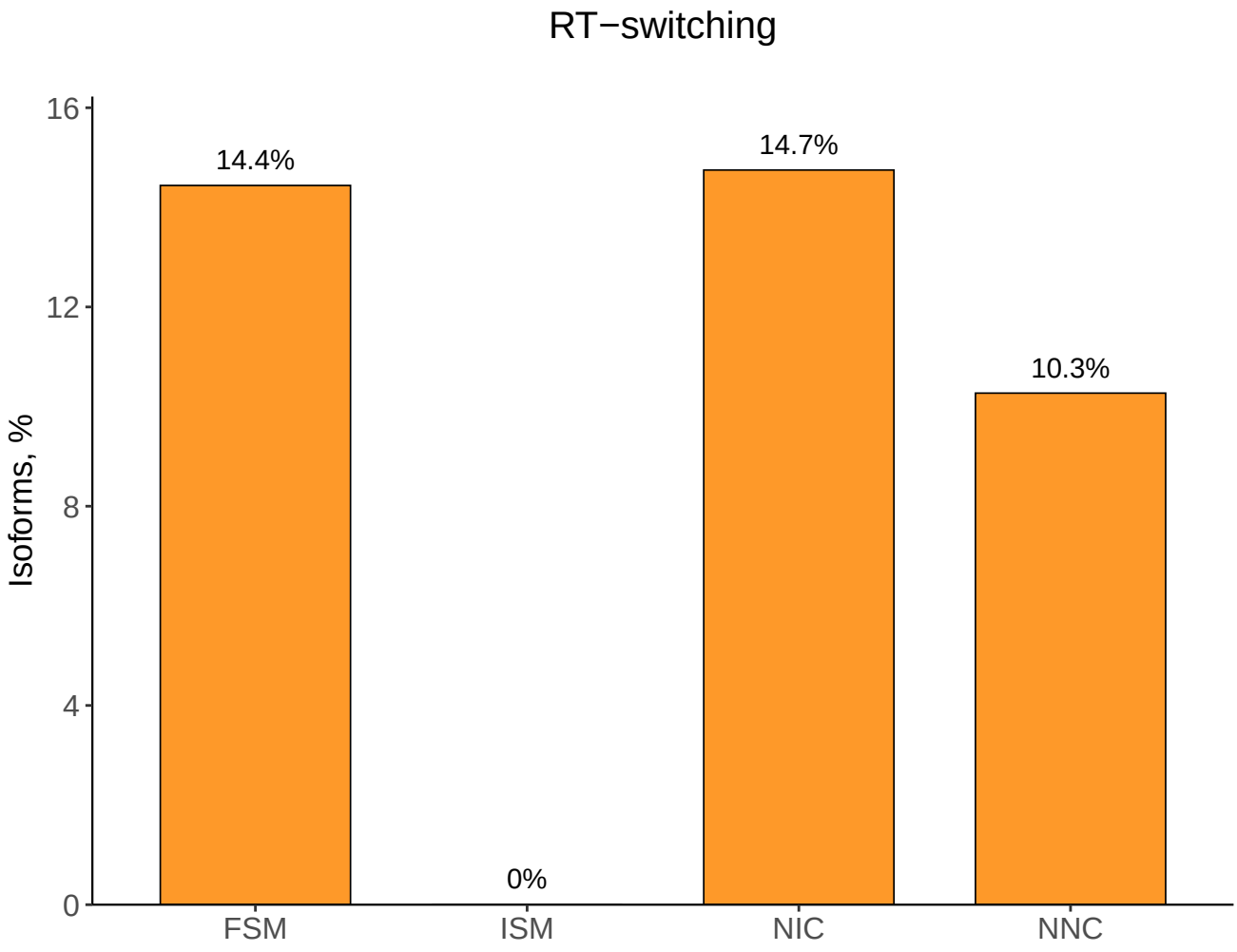

Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
0
20
40
60
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

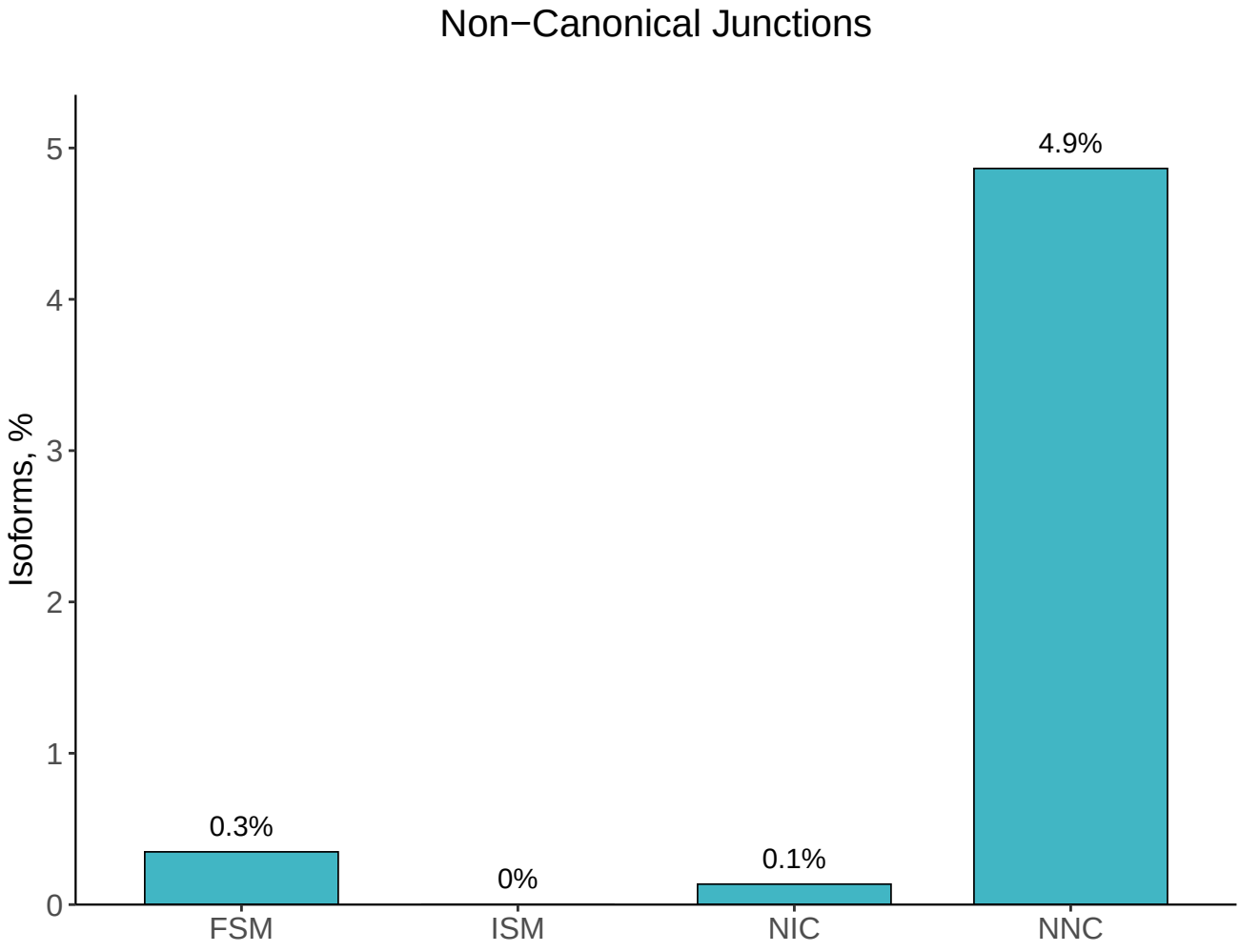

Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
0
20
40
60
Multi−exon
Genic Genomic
Multi−exon
Antisense
Intron retention
Multi−exon
Fusion
Multi−exon
Intergenic
Intron retention
Multi−exon

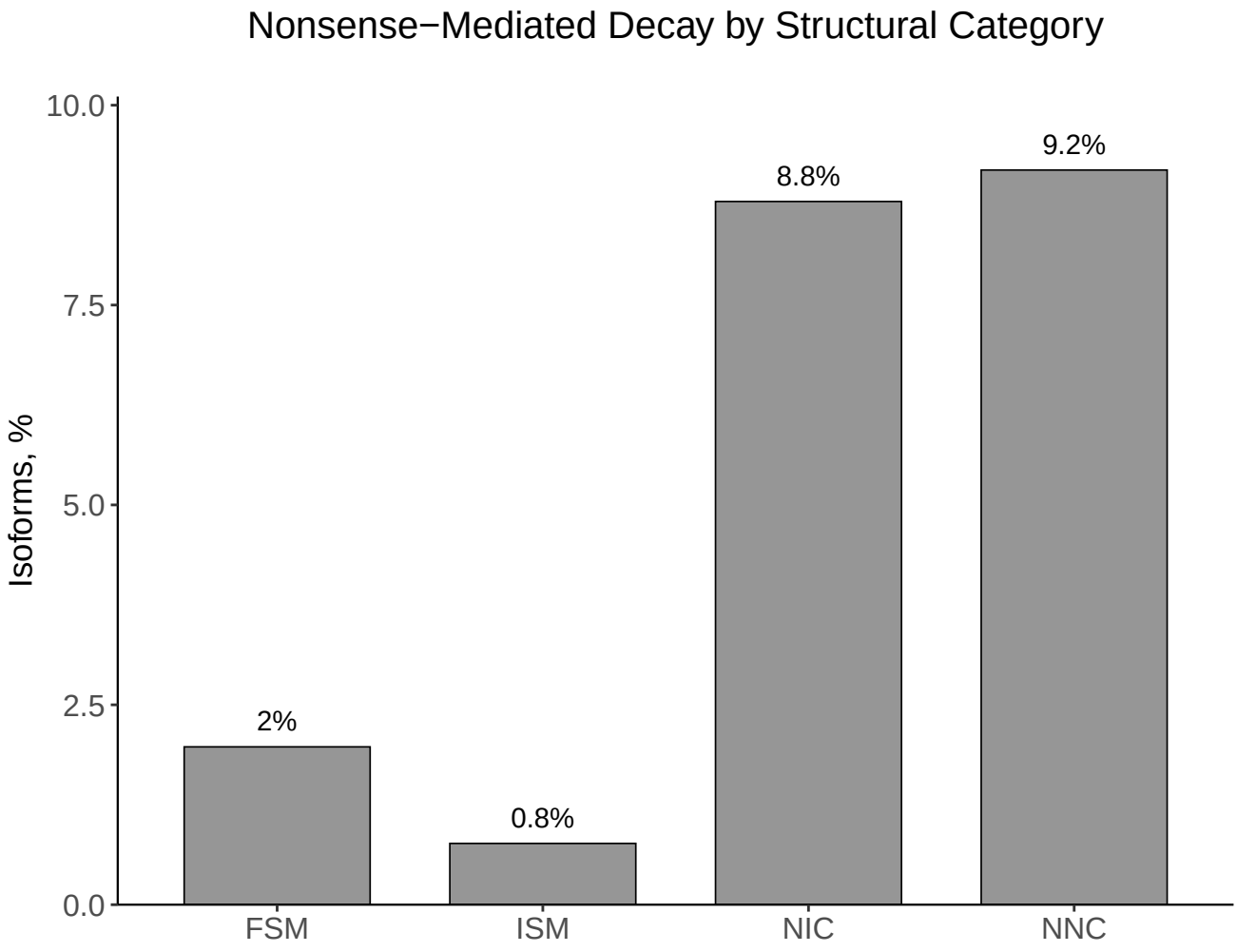

Mono− vs Multi−Exon Possible Intra−Priming
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Mono−Exon Isoforms
Multi−Exon Isoforms

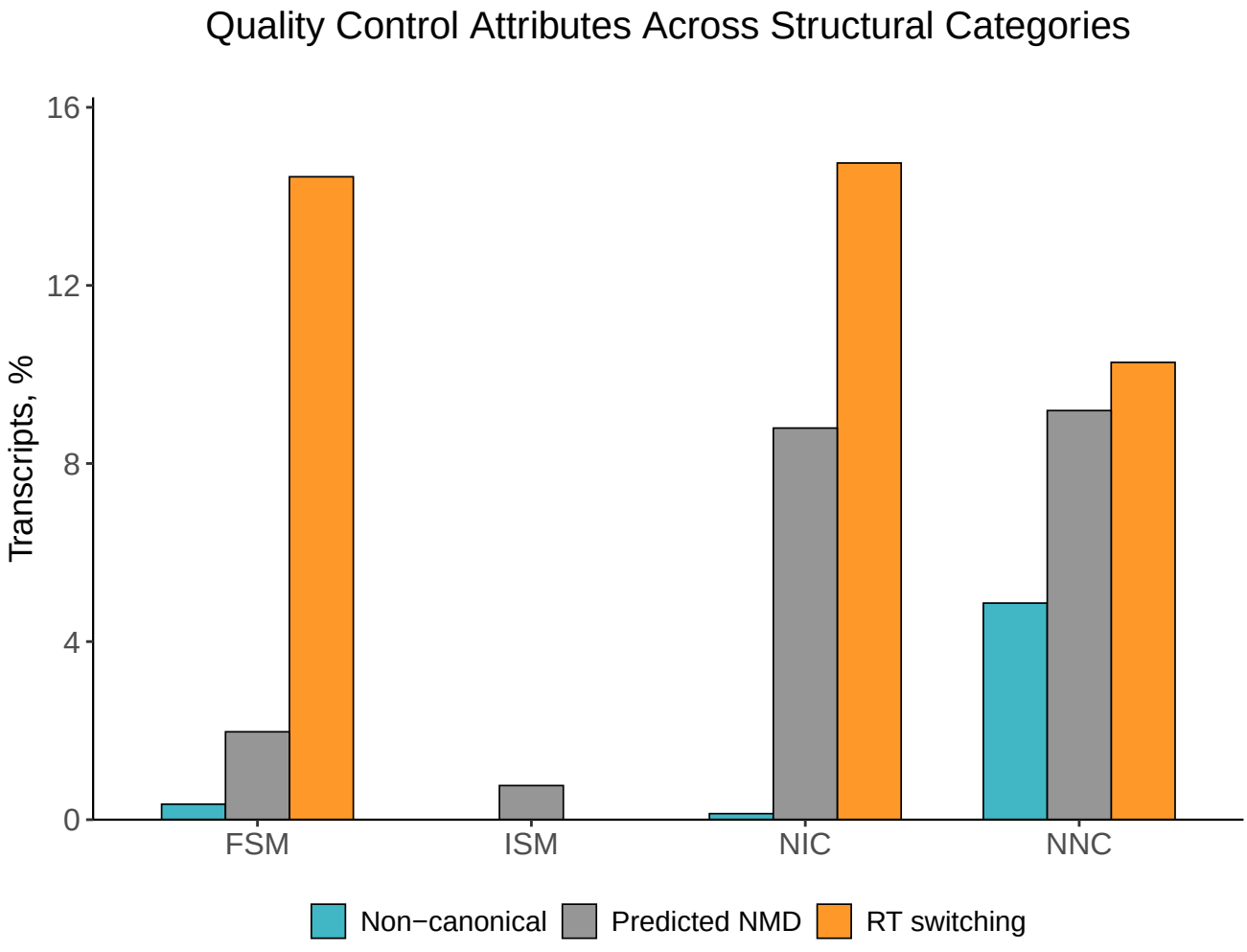

Coding vs Non−Coding Possible Intra−Priming
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Coding Isoforms
Non−Coding Isoforms

Features of Bad Quality

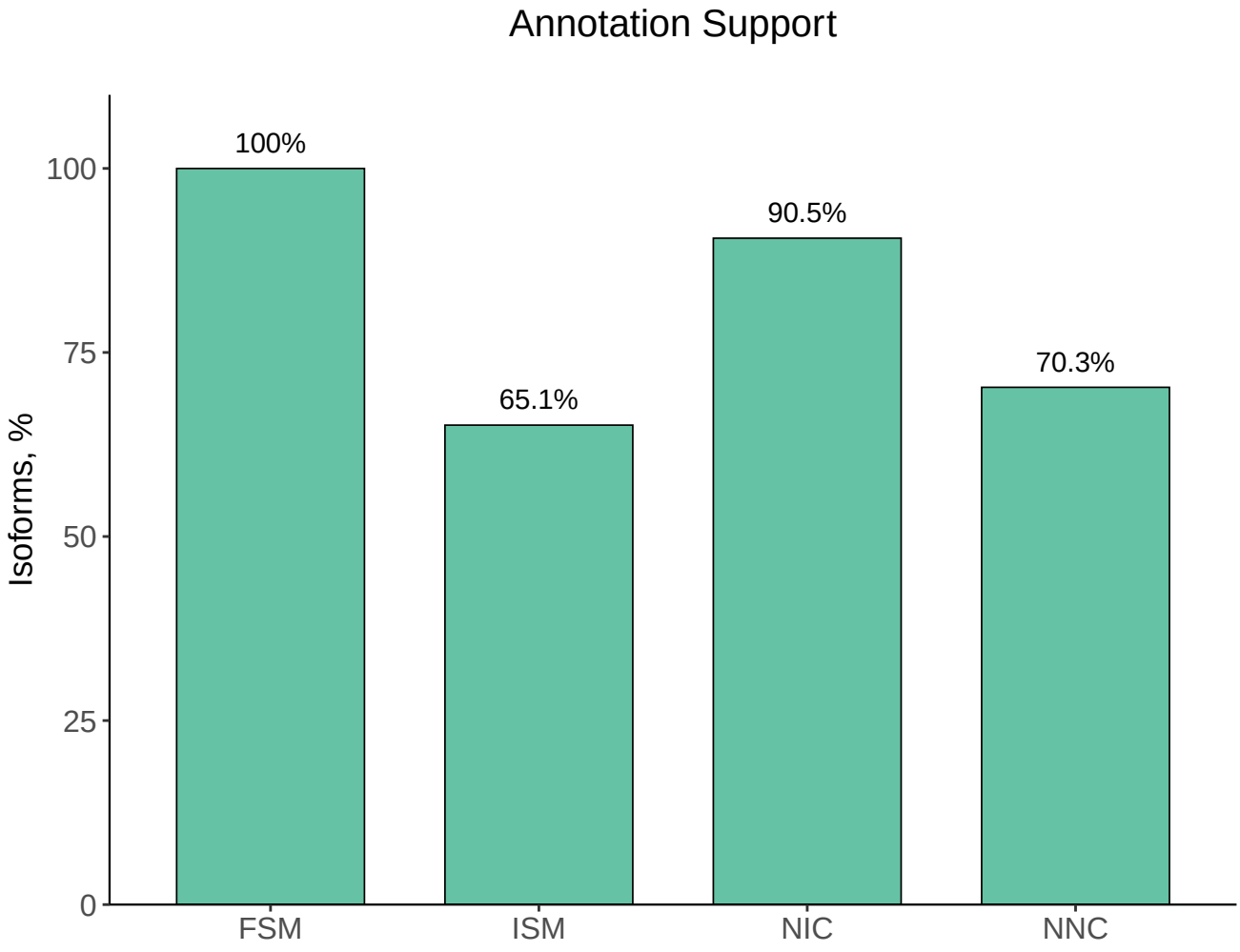

RT−switching
Isoforms, %
16
12
8
4
0
14.4%
0%
14.7%
10.3%
FSM
ISM
NIC
NNC

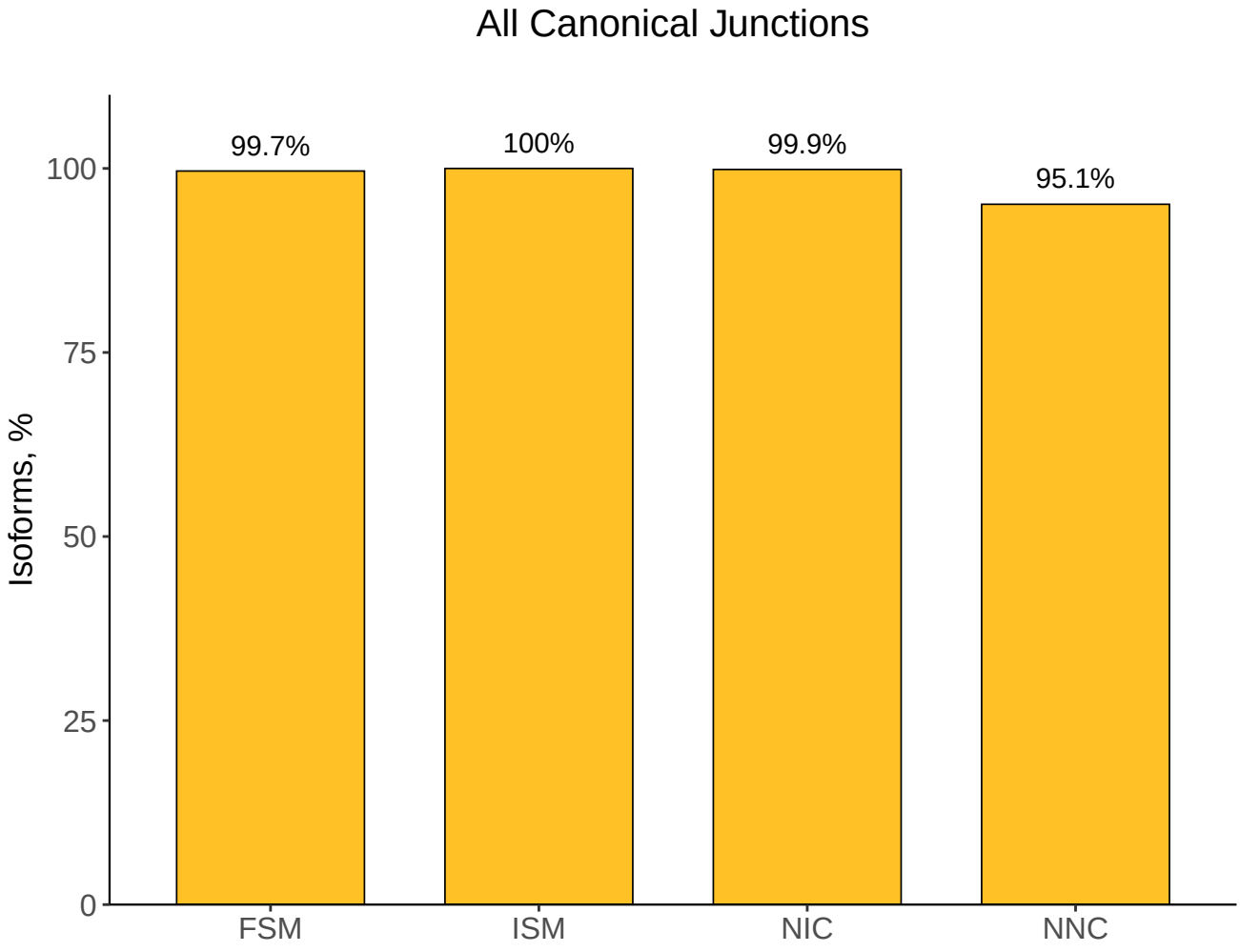

Non−Canonical Junctions
Isoforms, %
0
1
2
3
4
5
0.3%
0%
0.1%
4.9%
FSM
ISM
NIC
NNC

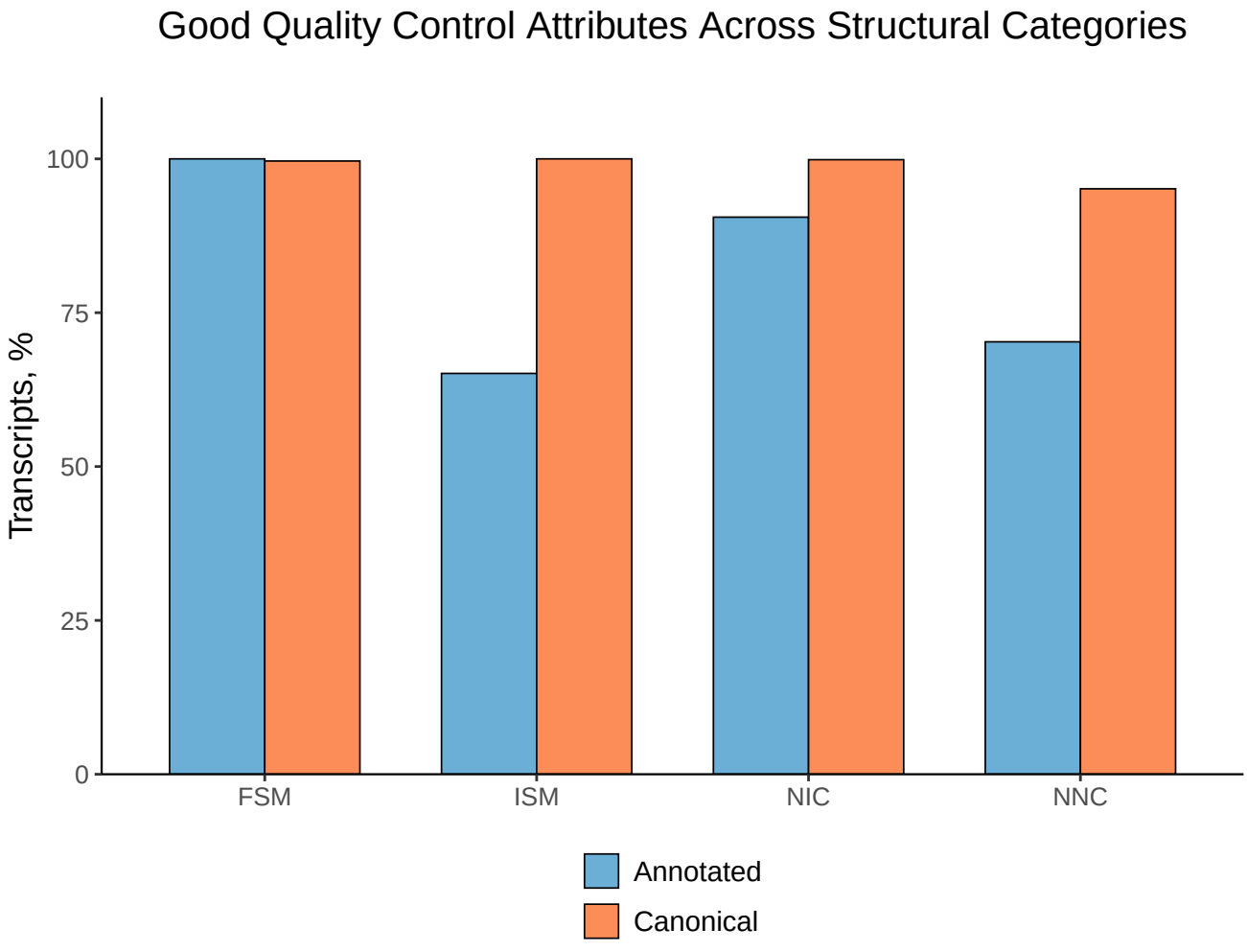

Nonsense−Mediated Decay by Structural Category
Isoforms, %
10.0
7.5
5.0
2.5
0.0
2%
0.8%
8.8%
9.2%
FSM
ISM
NIC
NNC


Quality Control Attributes Across Structural Categories
Transcripts, %
16
12
8
4
0
FSM
ISM
NIC
NNC
Non−canonical
Predicted NMD
RT switching

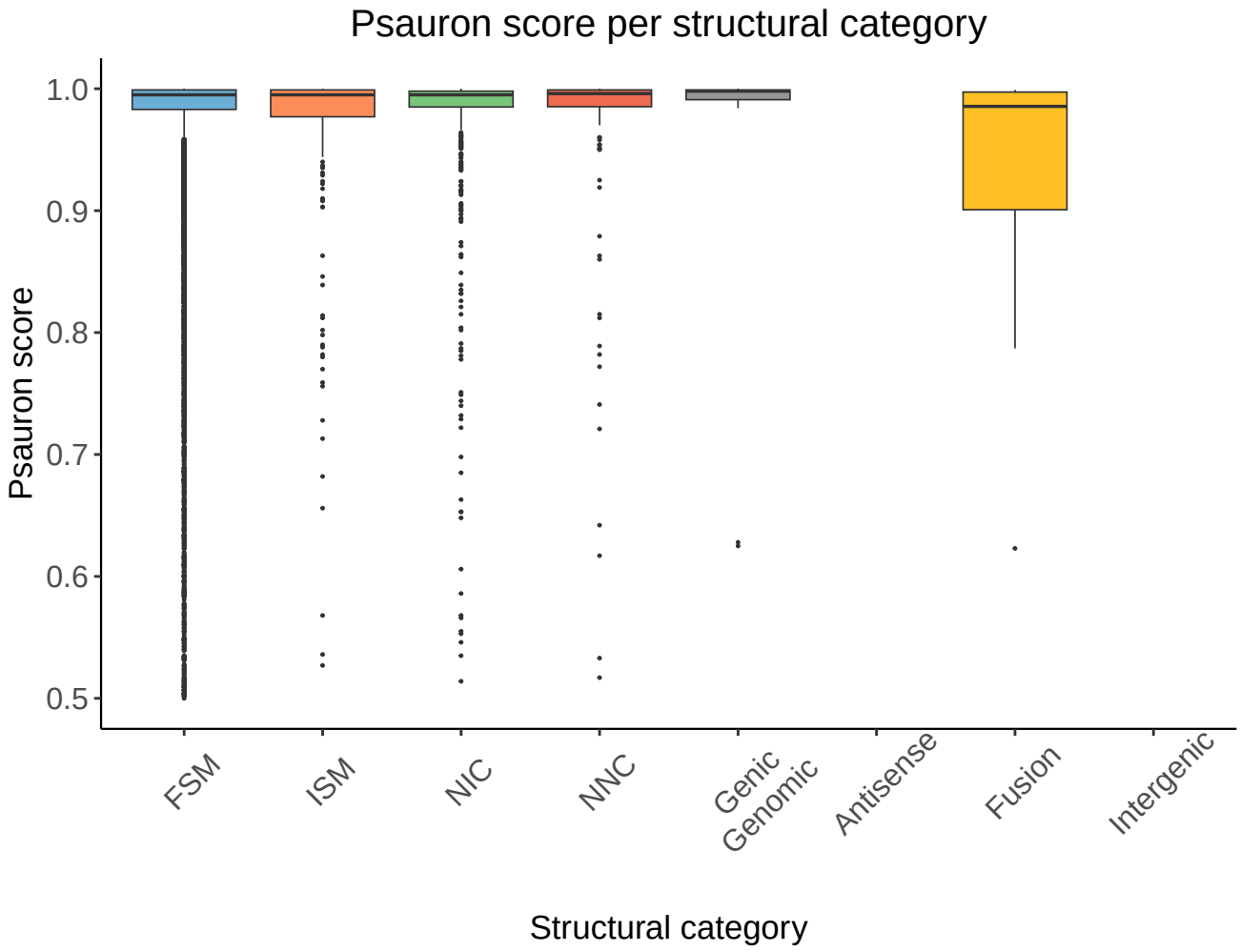
Features of Good Quality

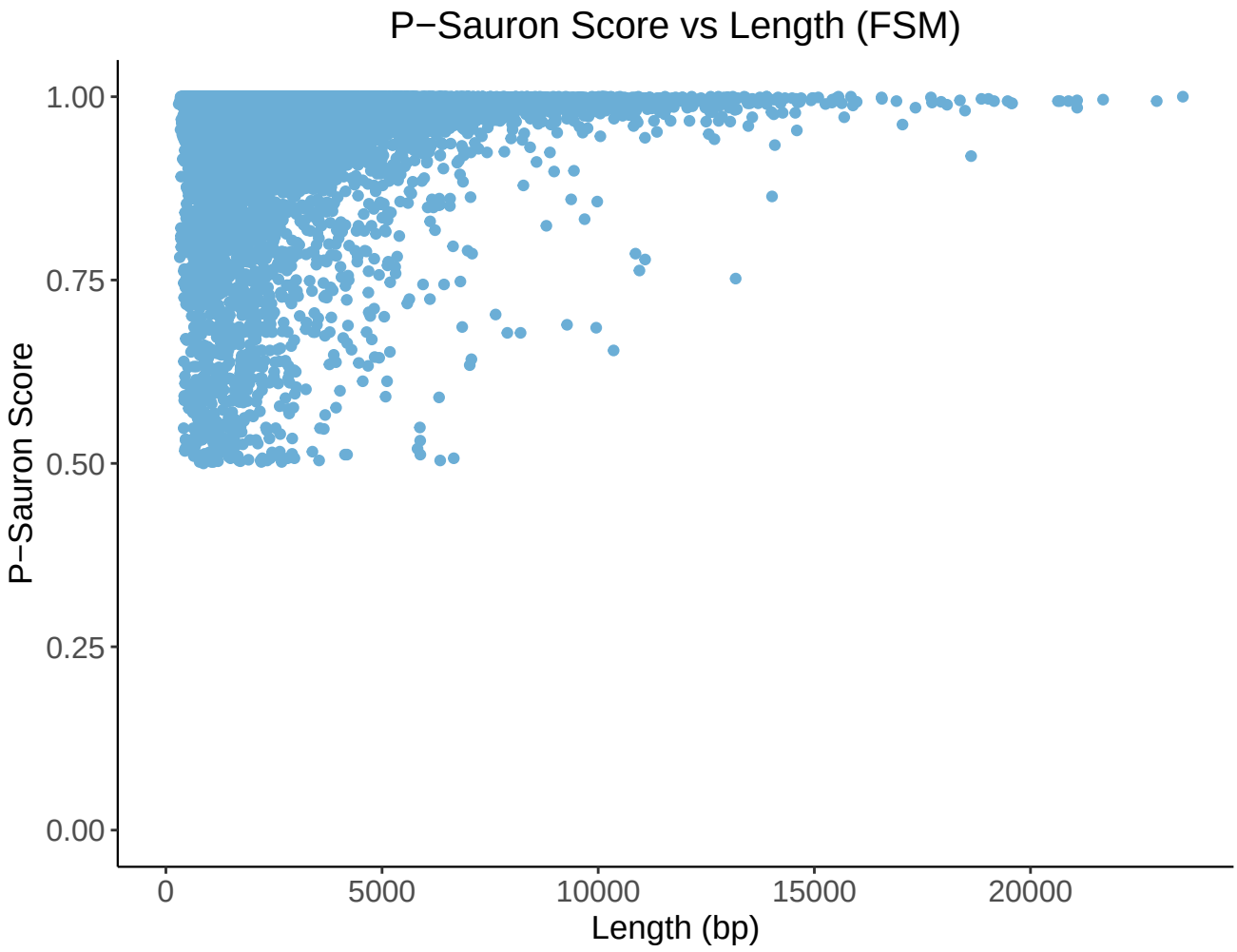

Annotation Support
Isoforms, %
100
75
50
25
0
100%
65.1%
90.5%
70.3%
FSM
ISM
NIC
NNC

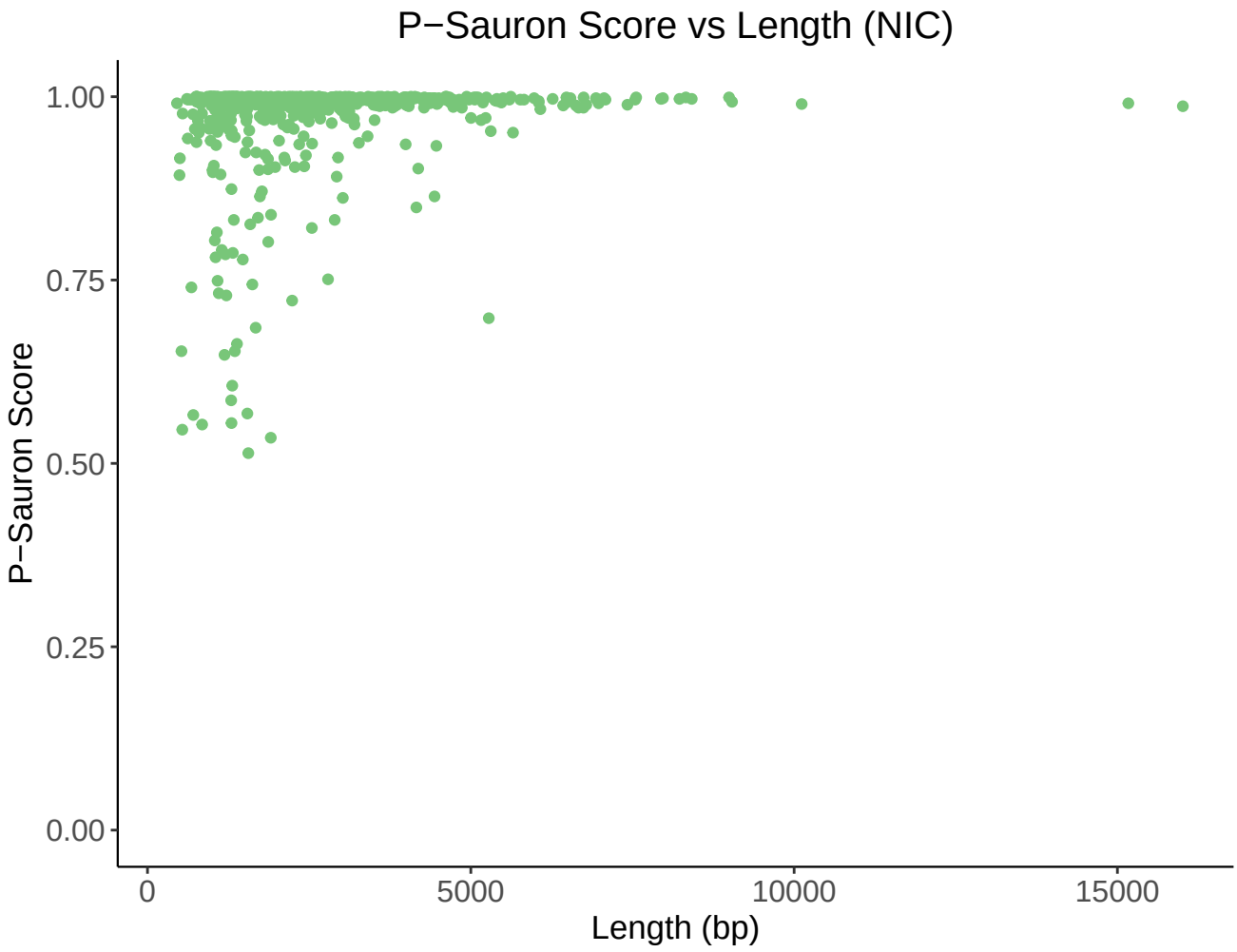

All Canonical Junctions
Isoforms, %
100
75
50
25
0
99.7%
100%
99.9%
95.1%
FSM
ISM
NIC
NNC

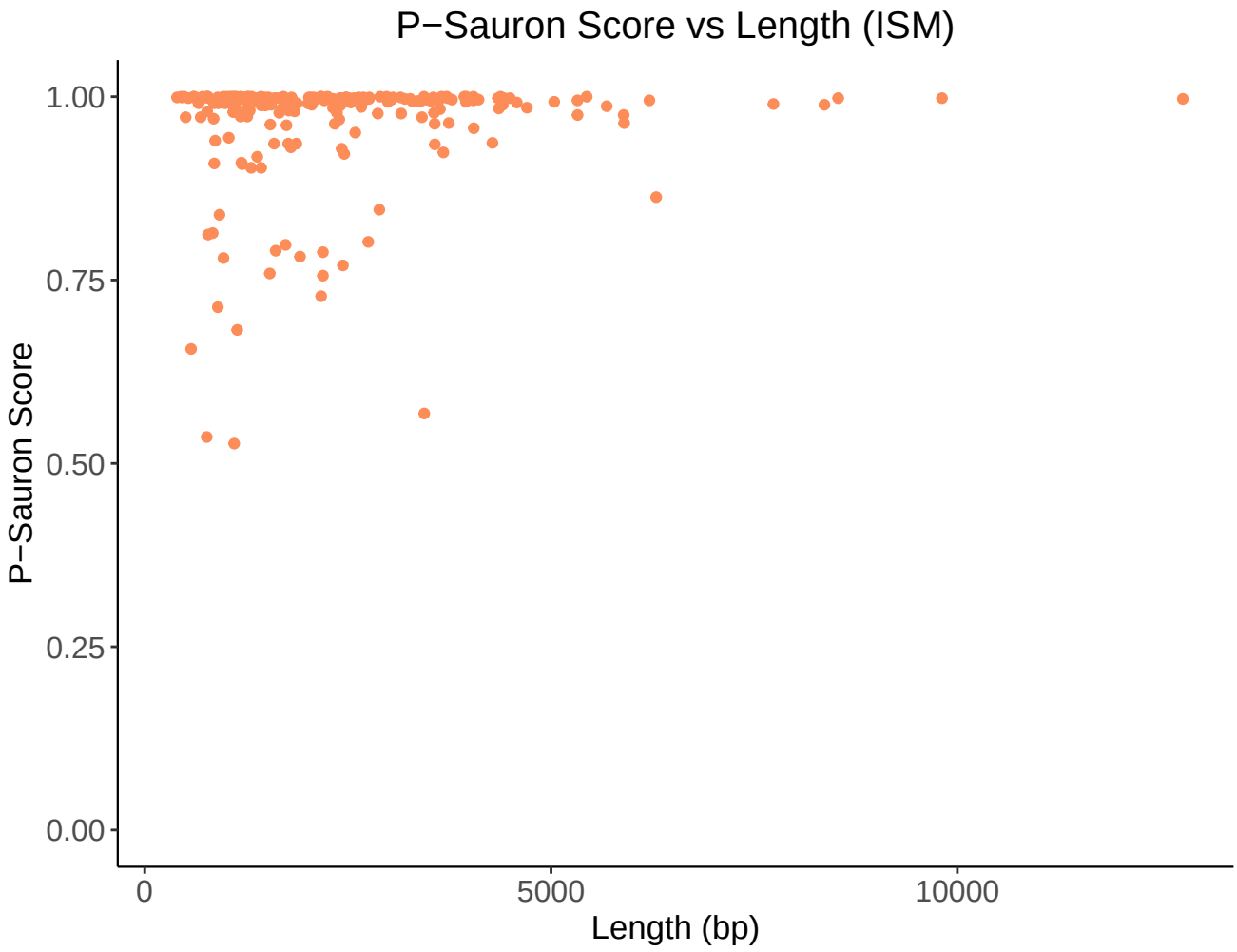

Good Quality Control Attributes Across Structural Categories
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Annotated
Canonical

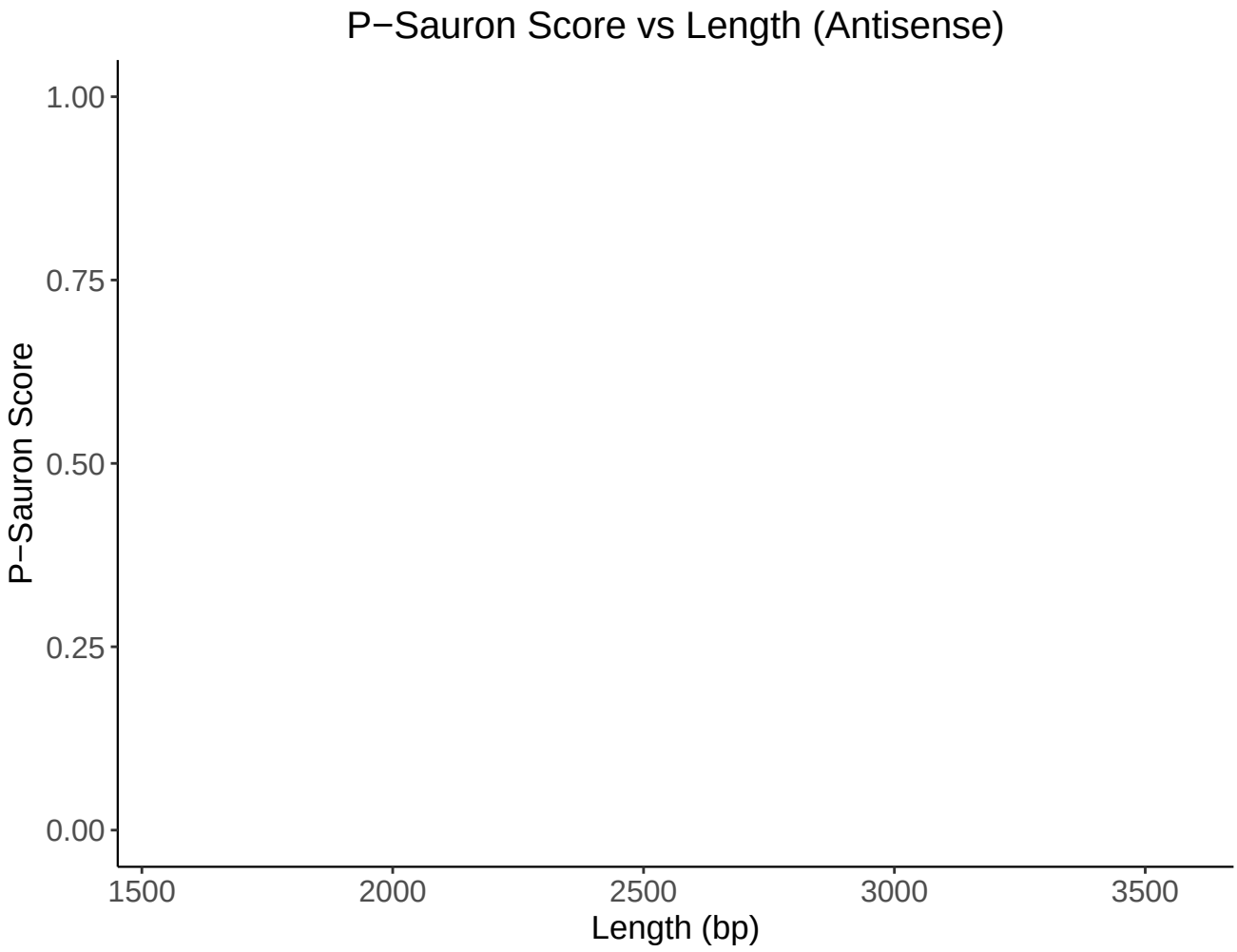
TransDecoder2 CDS prediction statistics

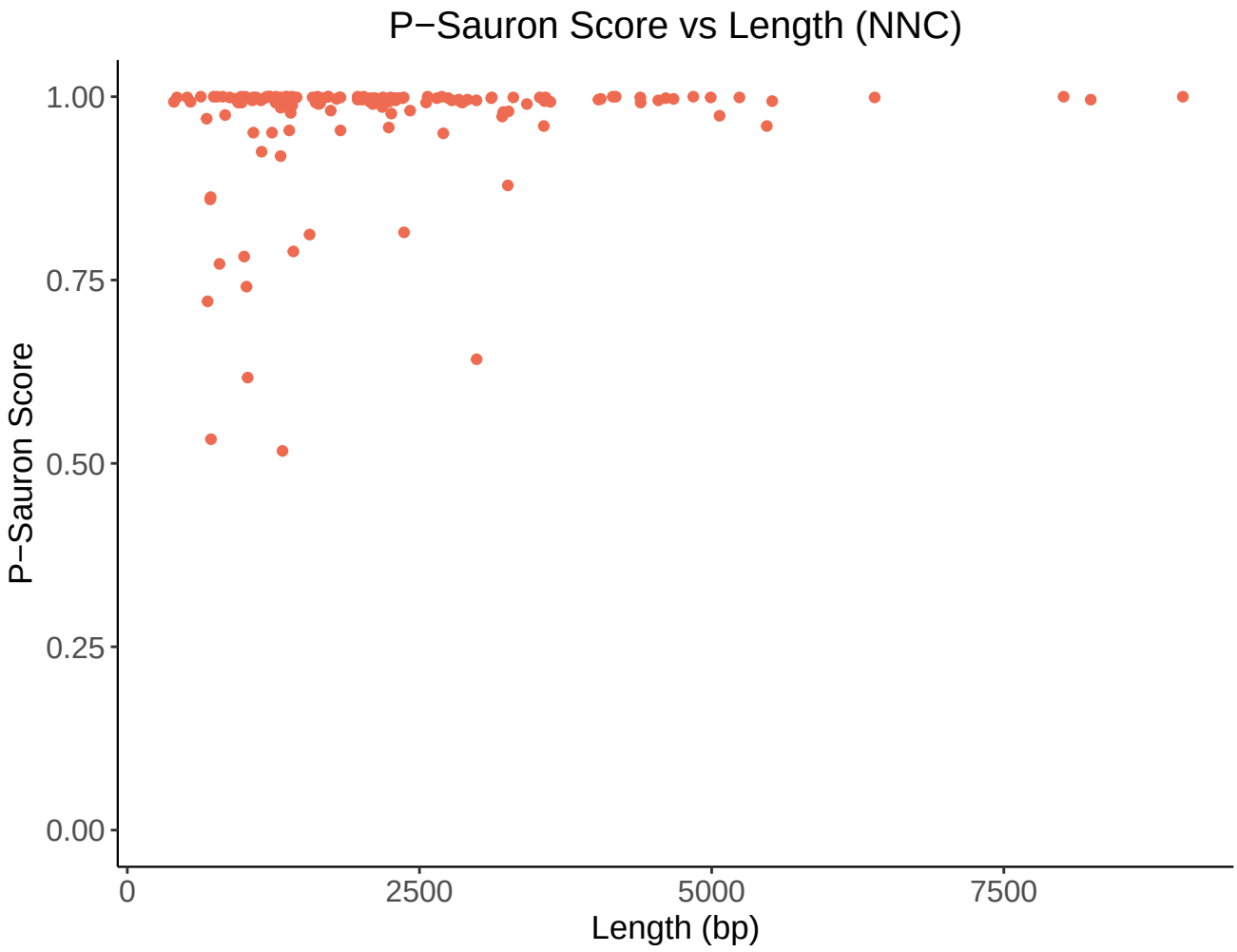

Psauron score per structural category
Psauron score
1.0
0.9
0.8
0.7
0.6
0.5
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Structural category

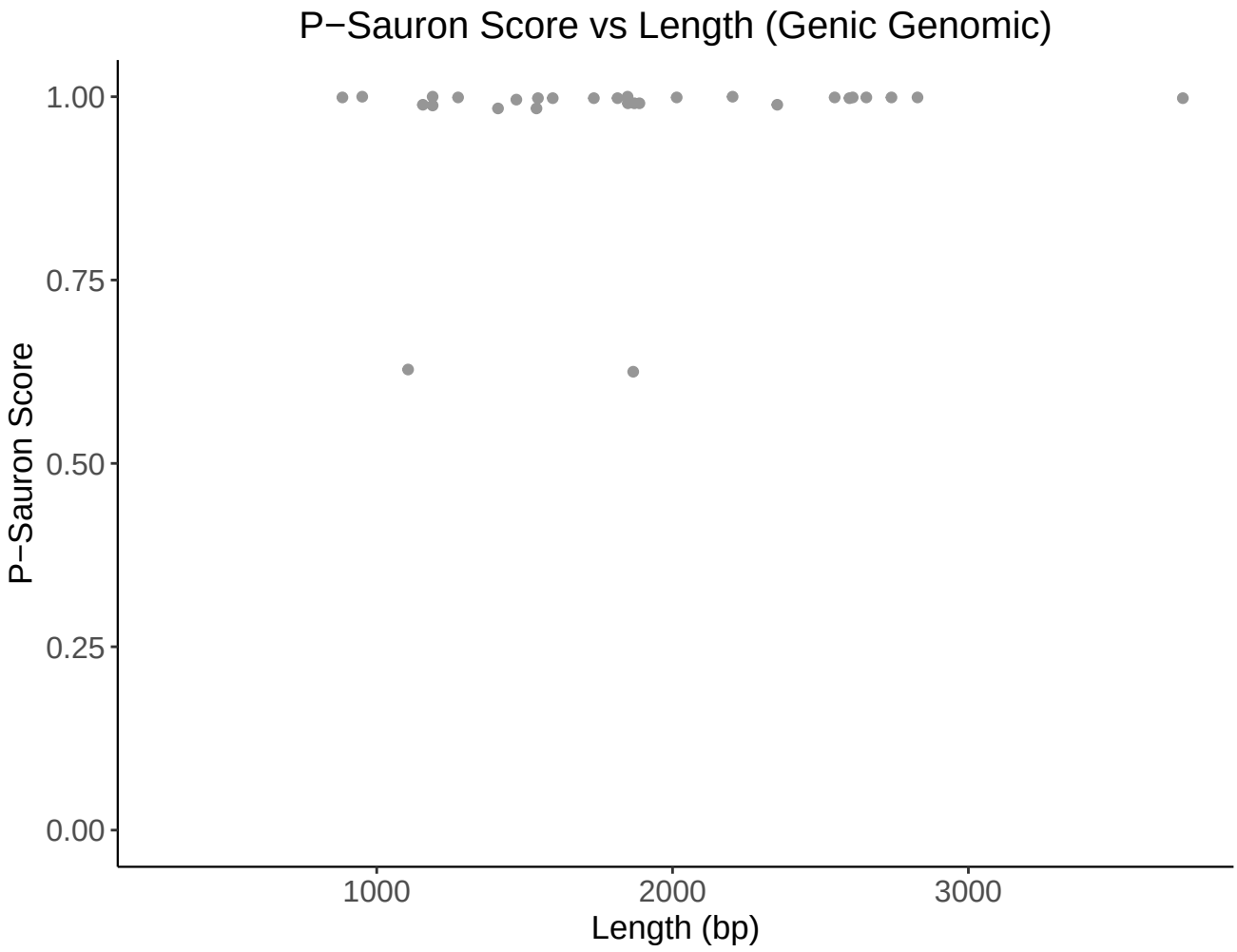

P−Sauron Score vs Length (FSM)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
0
5000
10000
15000
20000
Length (bp)

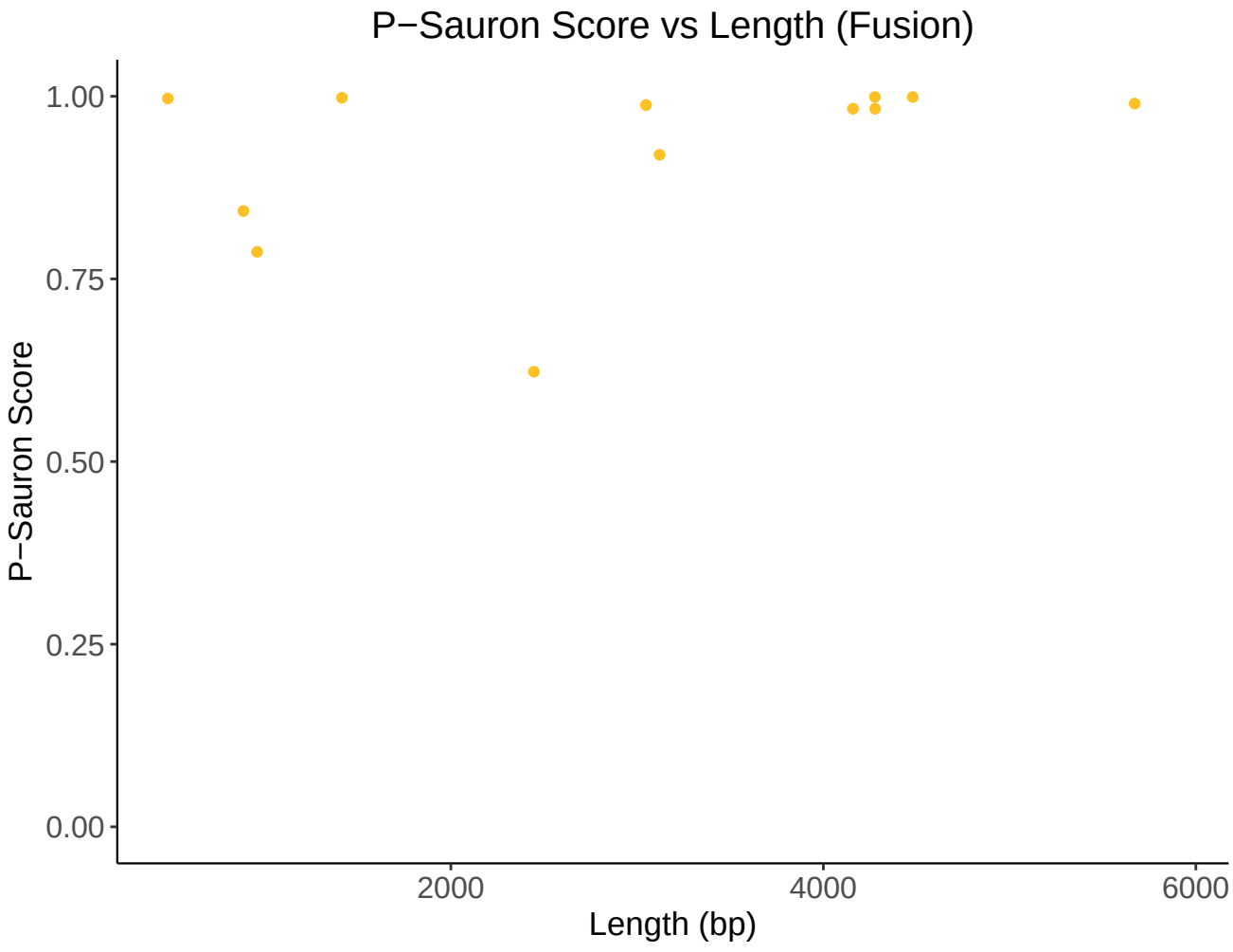

P−Sauron Score vs Length (NIC)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
0
5000
10000
15000
Length (bp)

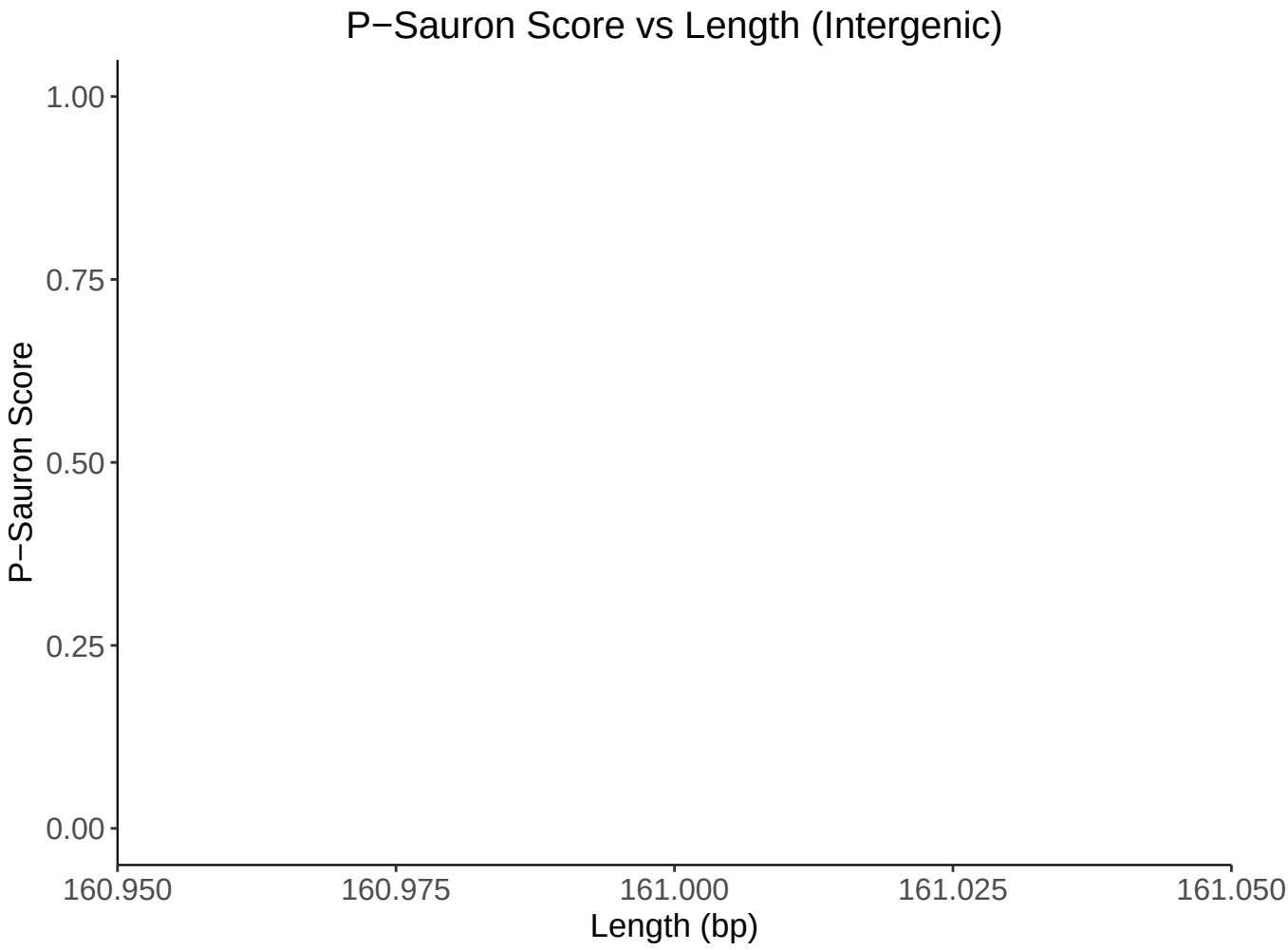

P−Sauron Score vs Length (ISM)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
0
5000
10000
Length (bp)

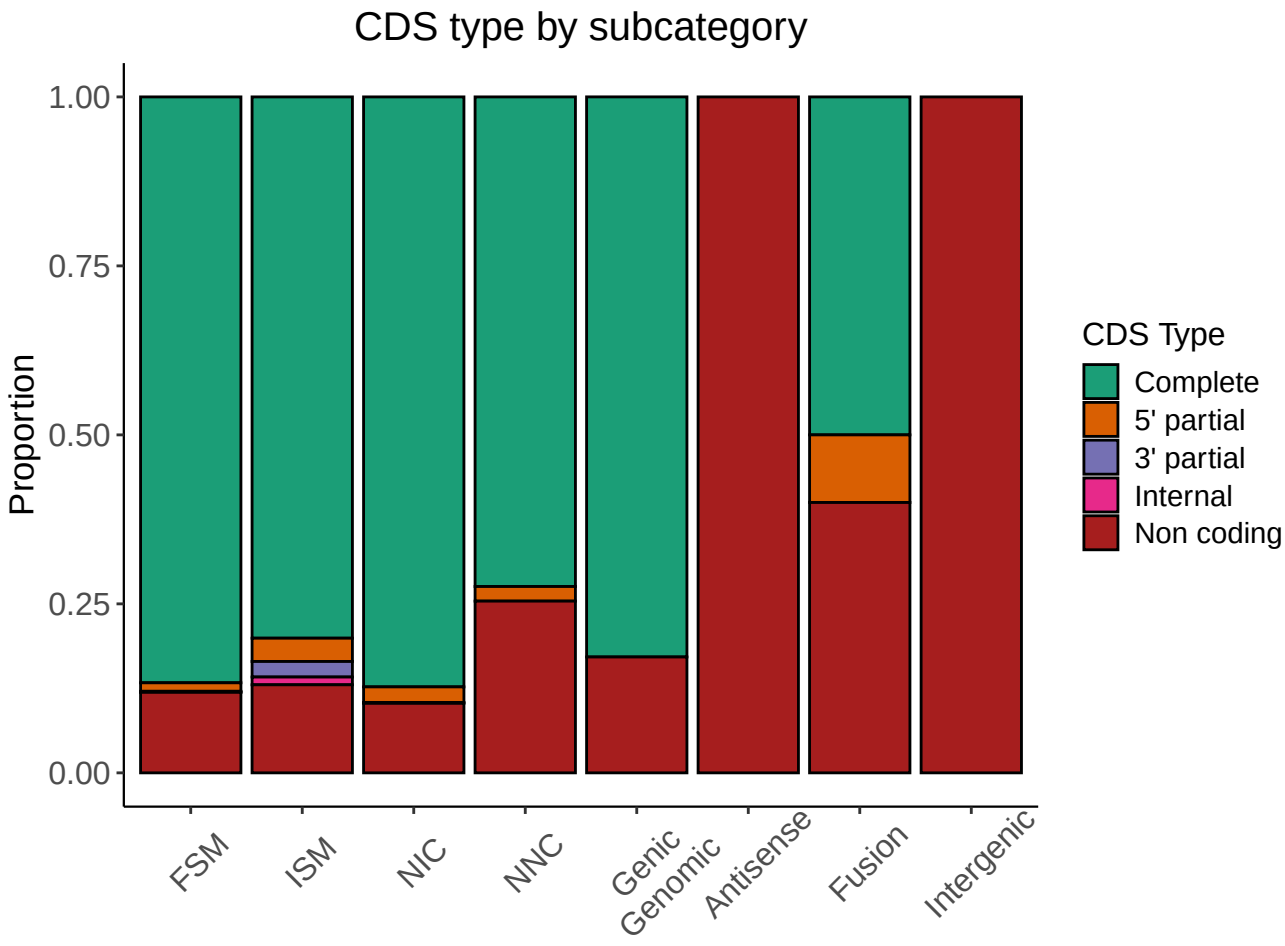

P−Sauron Score vs Length (Antisense)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
1500
2000
2500
3000
3500
Length (bp)

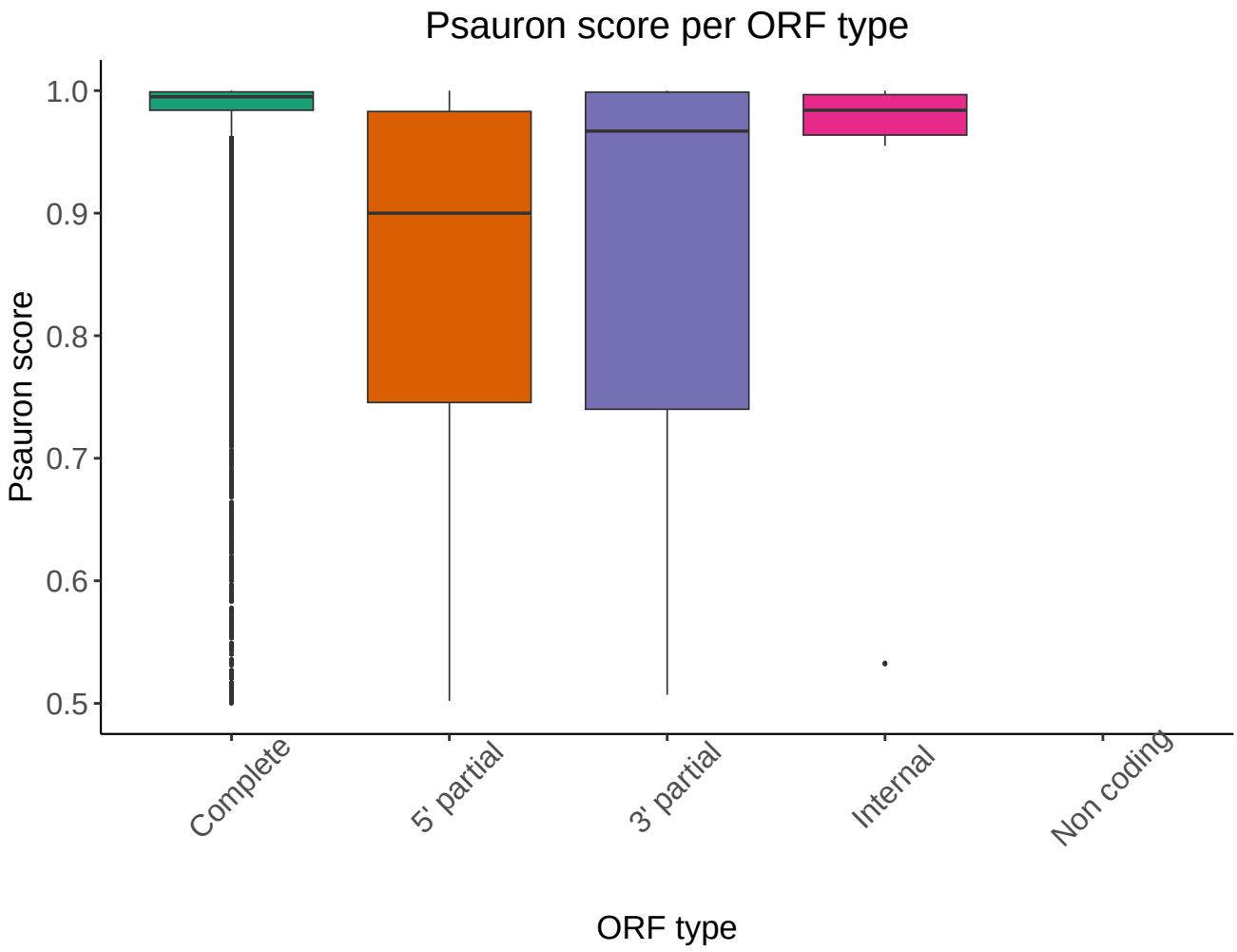

P−Sauron Score vs Length (NNC)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
0
2500
5000
7500
Length (bp)


P−Sauron Score vs Length (Genic Genomic)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
1000
2000
3000
Length (bp)


P−Sauron Score vs Length (Fusion)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
2000
4000
6000
Length (bp)


P−Sauron Score vs Length (Intergenic)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
160.950
160.975
161.000
161.025
161.050
Length (bp)


CDS type by subcategory
Proportion
1.00
0.75
0.50
0.25
0.00
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
CDS Type
Complete
5' partial
3' partial
Internal
Non coding


Psauron score per ORF type
Psauron score
1.0
0.9
0.8
0.7
0.6
0.5
Complete
5' partial
3' partial
Internal
Non coding
ORF type